# SUSTAINABLE DESIGN OF STUDENTS' ACCOMMODATION BLOCK

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Sustainable Environmental Design

MSc. Climate Resilience and Environmental Sustainability in Architecture



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# Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design
Afua Ampomah-Amoako



## 1.0 PROJECT LOCATION : SEATTLE, WASHINGTON STATE, U.S.A.

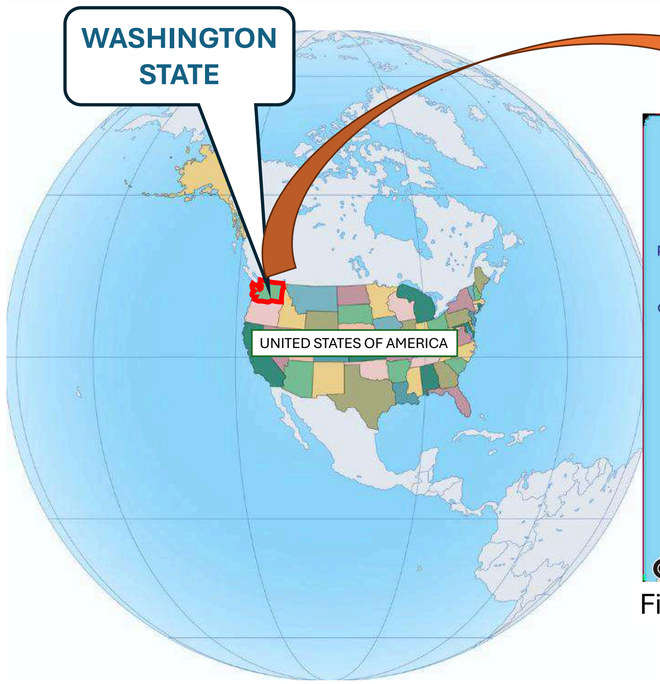



Figure 1:Location of United States of America (VectorStock, 2025)




Figure 2 : Map of Washington State (Tripsmap, 2018)

## 1.1 KOPPEN CLIMATE CLASSIFICATION

Six different climate types and Seattle falls within the Cold-summer mediterranean.

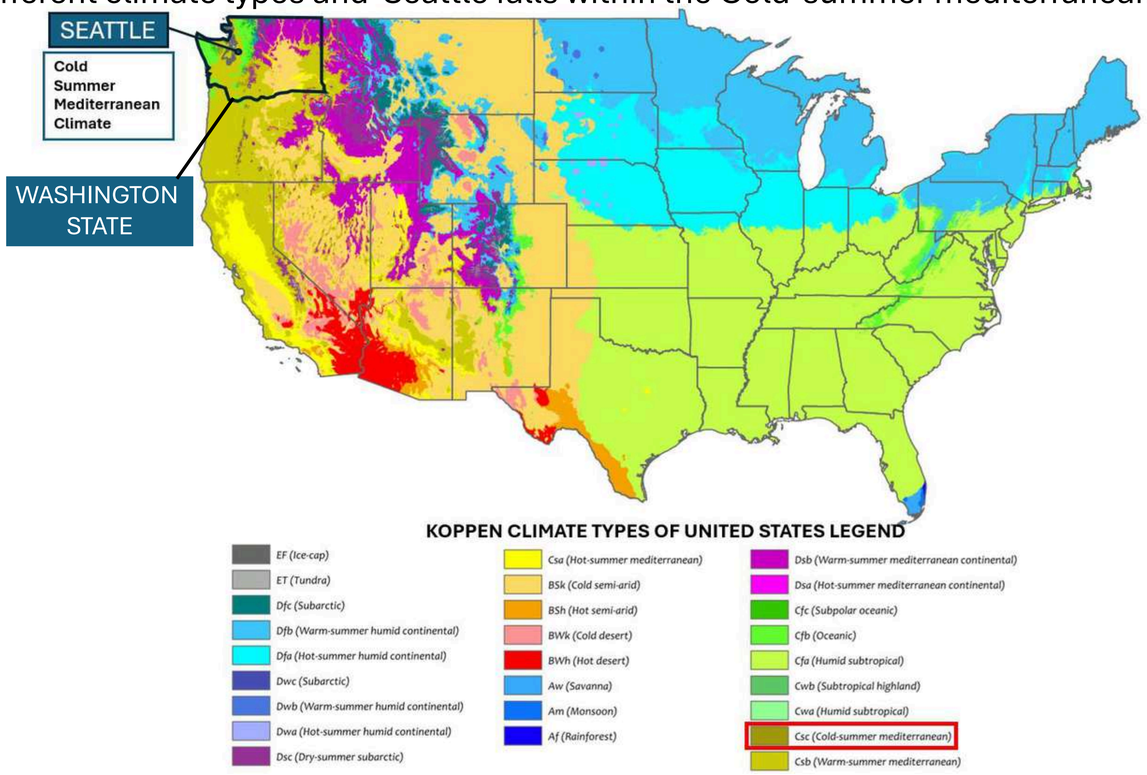



Figure 3 : Koppen Climate Classification of United States of America (Beck et al.,2023 p.3)

## 1.2 LOCAL GEOGRAPHIC CHARACTERISTICS OF SEATTLE

Seattle is an urban coastal city surrounded by **water, mountains, and parks**.




Figure 4 : Aerial view of Seattle City (Google Earth, 2025 with modifications by author)

## 1.3 TEMPERATURE RANGE OF SEATTLE

Temperature at Seattle range between -4 degrees Celsius and 31 degrees Celsius as seen in Figure 3. This indicates that at most times of the year (83.5%) Temperatures fall below the comfort zone requiring heating.

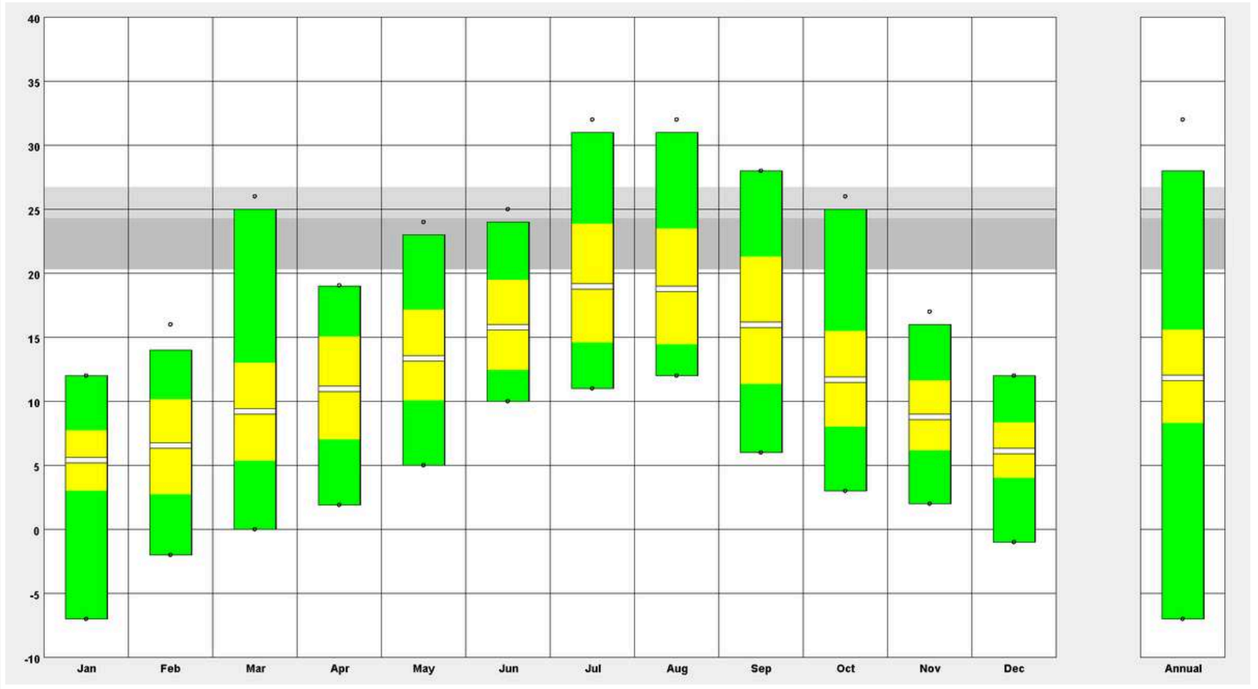



Figure 5 : Temperature range chart of climate (UCLA Energy Design Group, 2021)

## 1.3.1 RELATIVE HUMIDITY

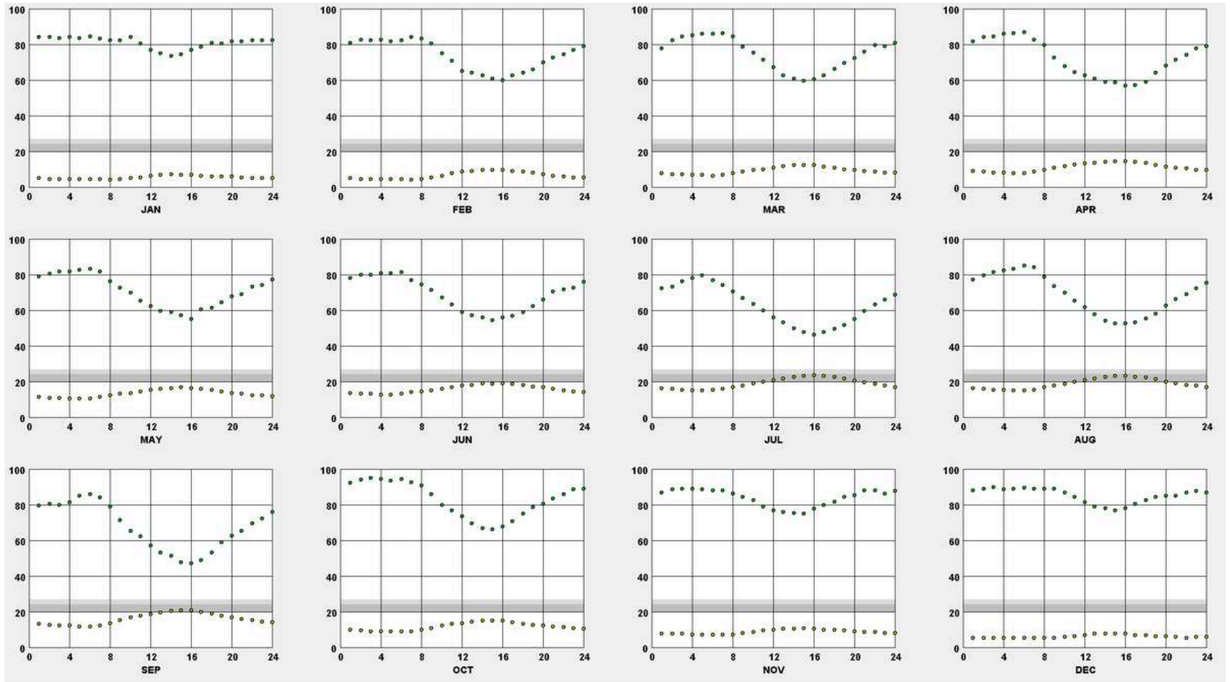
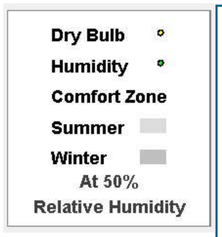



Figure 6: Dry Bulb and Relative Humidity (UCLA Energy Design Group, 2021)




Dry bulb temperatures below the comfort zone except during summer months which falls within the comfort range.
Average relative humidity of 70% but rises during the winter months above 80%.
Seattle had an average precipitation of 32.7 inches in 2024 which was below the yearly average of 39.34 inches indicating a drop in rainfall.

## 1.3.2 PRECIPITATION PATTERN OF SEATTLE

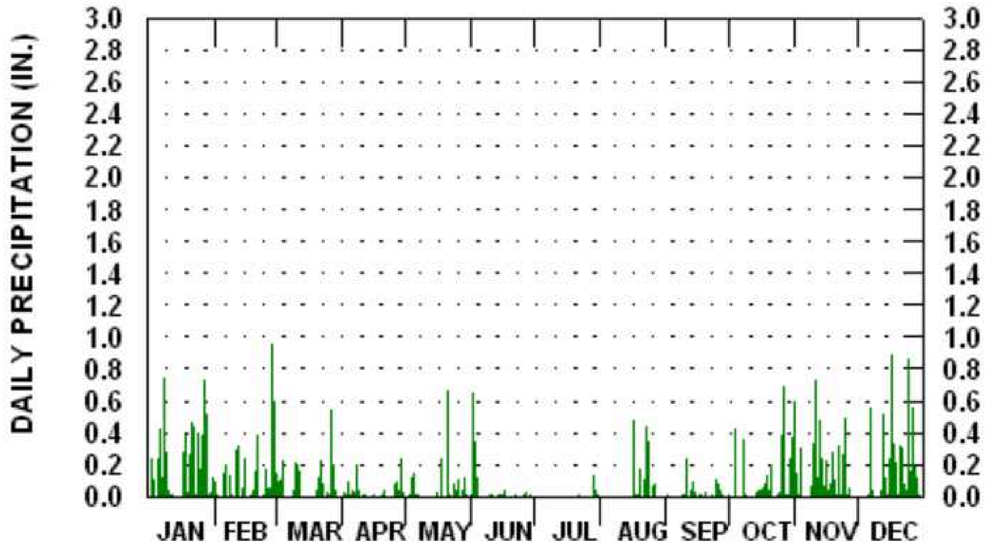



Figure 7: Monthly precipitation for 2024 year (OceanWP, 2025)

## 1.3.3 SUN SHADING CHART

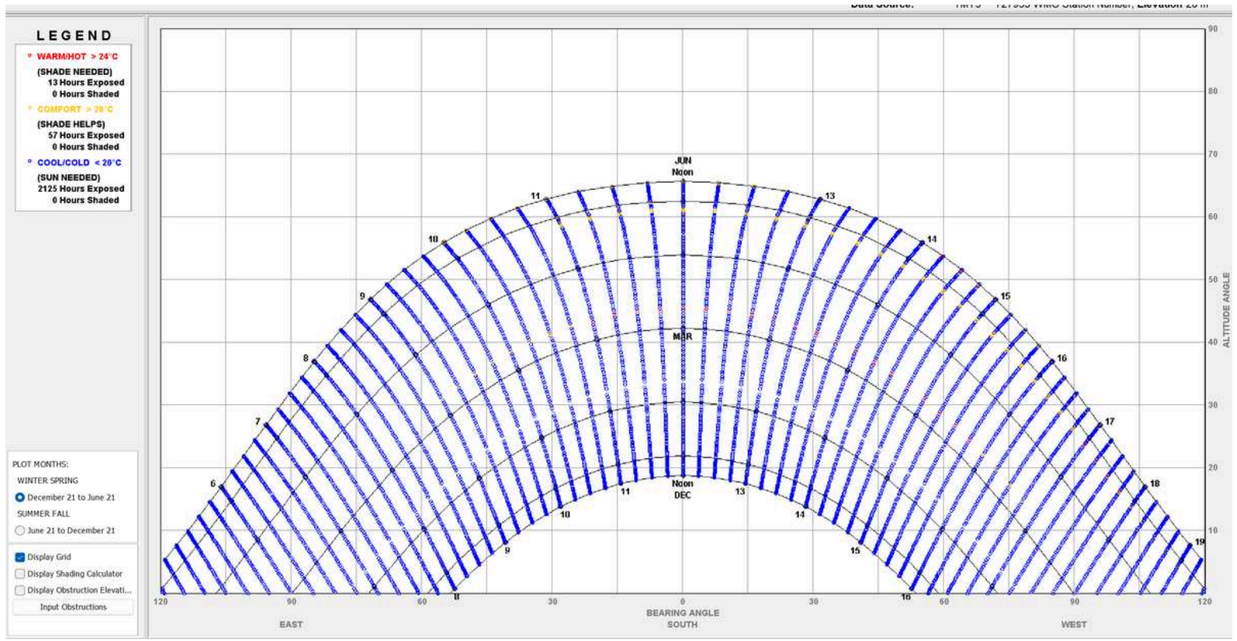



Figure 8: Sun shading chart from December to June
(UCLA Energy Design Group, 2021)

Shading is required from **June to September** because of the high exposure to sun specifically on the Southeast, South, Southwest and West facades. This will require both horizontal and vertical sun shading devices along the named facades.

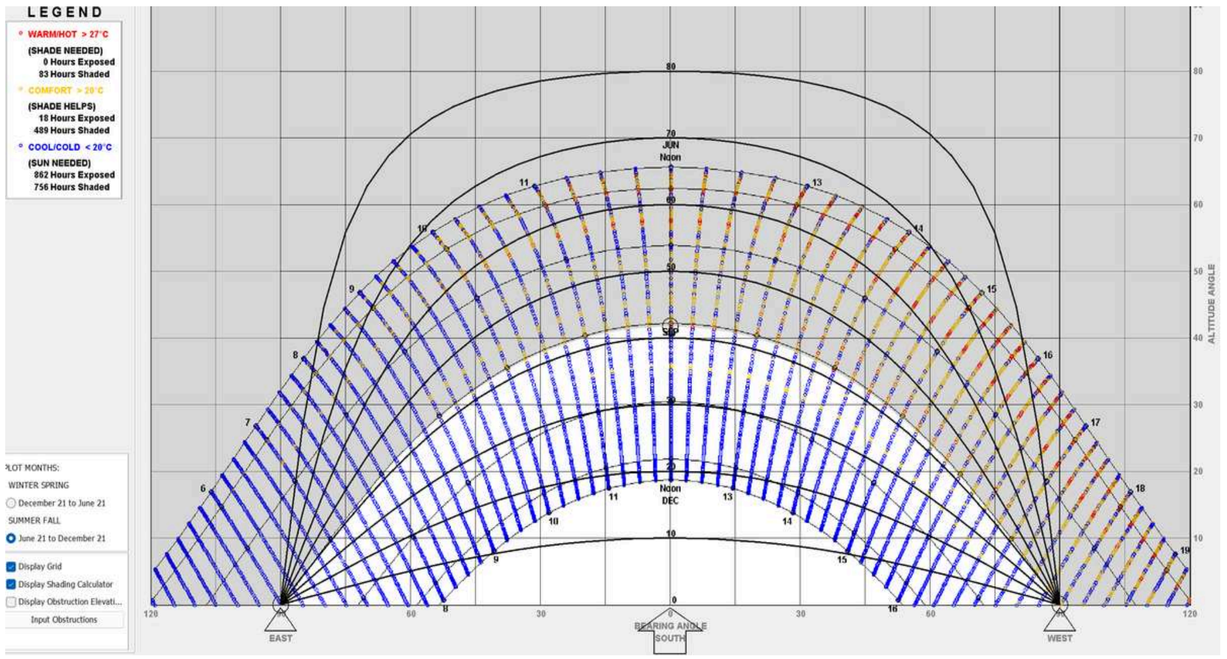



Figure 9: Sun shading chart from June to December
(UCLA Energy Design Group, 2021)

## 1.3.4 PSYCHROMETRIC CHART

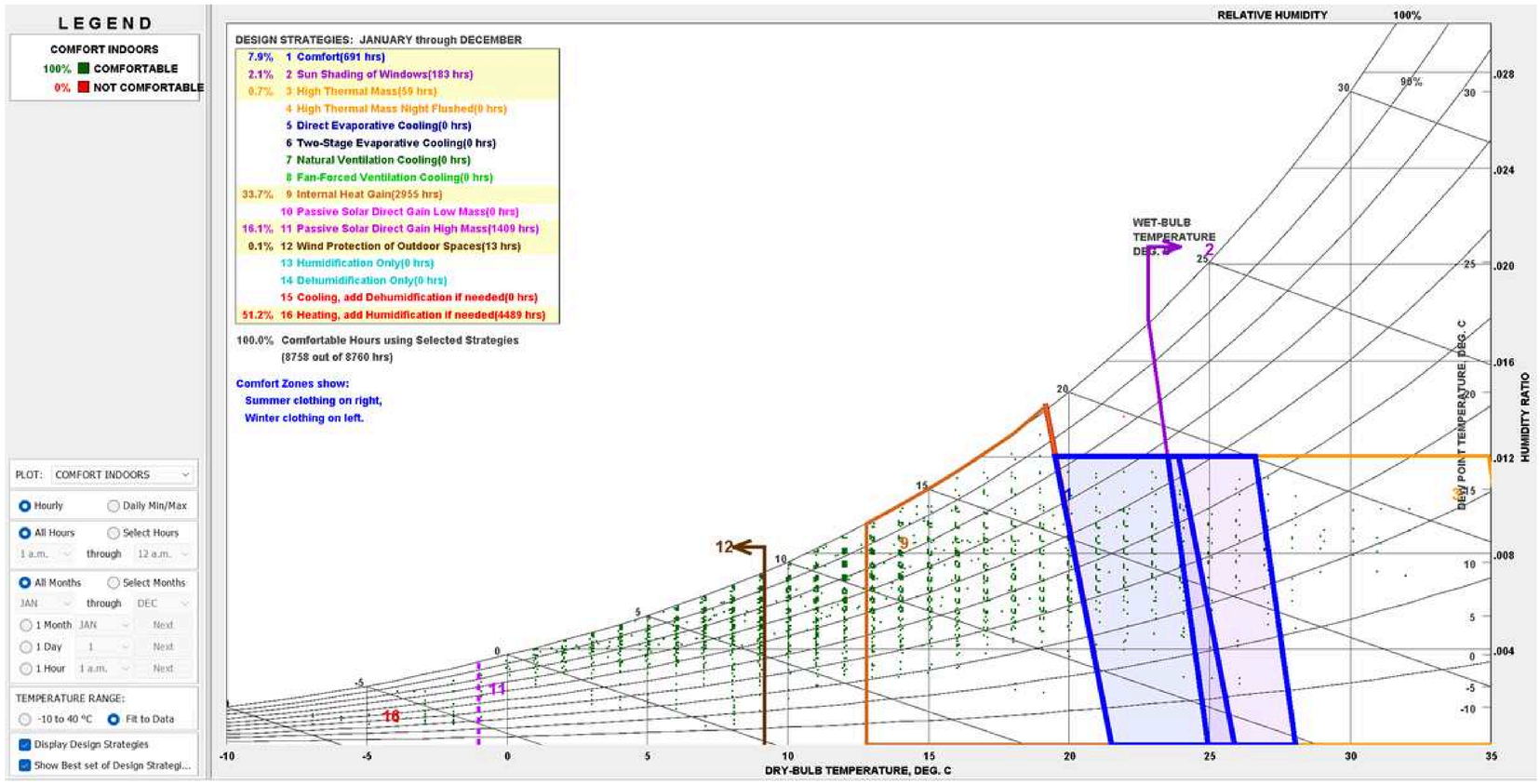



Figure 10: Psychrometric chart (UCLA Energy Design Group, 2021)

The psychrometric chart informs design strategies to bring optimum comfort as listed below:

- **Heating of spaces and internal heat gain as most months are cold below the comfort zone.**
- **Passive solar direct gain and incorporating high thermal mass.**
- **Sun shading of windows on South, Southeast and Southwest facades**
- **Wind protection of outdoor spaces**

The ground temperature chart (Figure 11) indicates a minimum temperature of 9 degrees and 15 degrees maximum at 4m depth.

The prevailing wind directions (Figure 12) are mainly from South, Southwest (10 m/s) and Northwest (8m/s).

## 1.3.5 GROUND TEMPERATURE (MONTHLY AVERAGE)

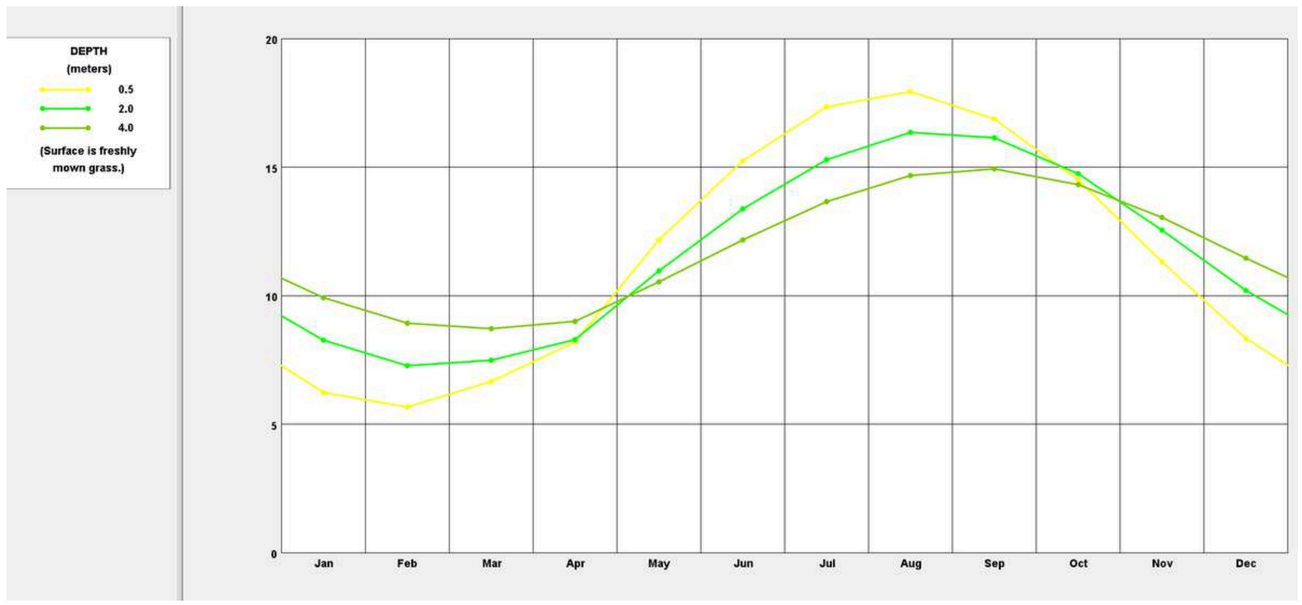



Figure 11: Ground Temperature (UCLA Energy Design Group, 2021)

## 1.3.6 WIND WHEEL

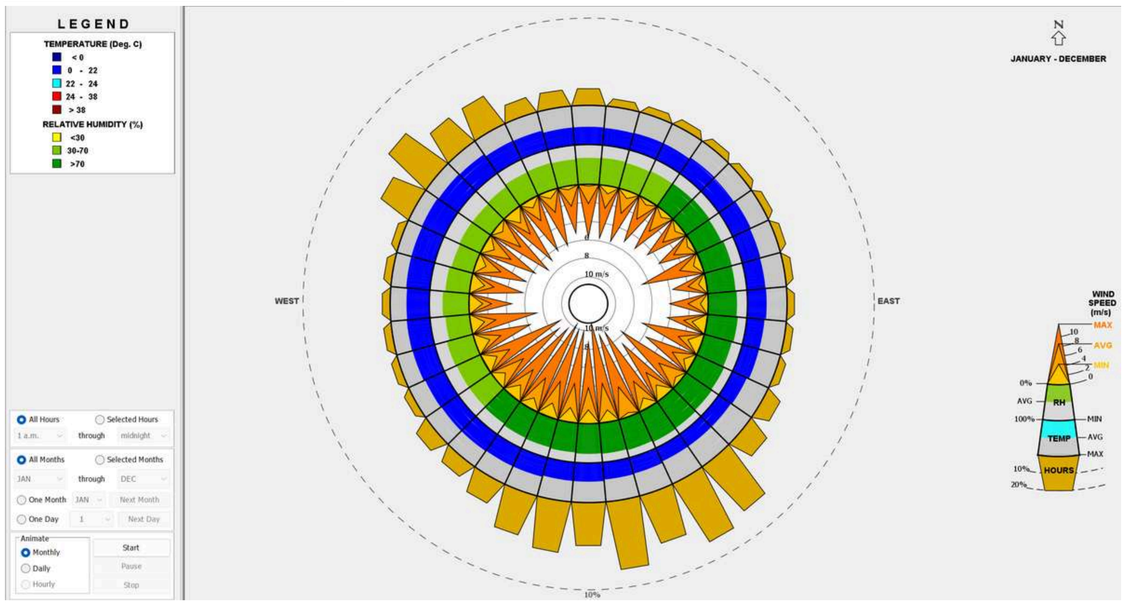



Figure 12: Wind Wheel (UCLA Energy Design Group, 2021)

## 1.4.0 CLIMATE CHANGE IMPACT – DROUGHT RISK

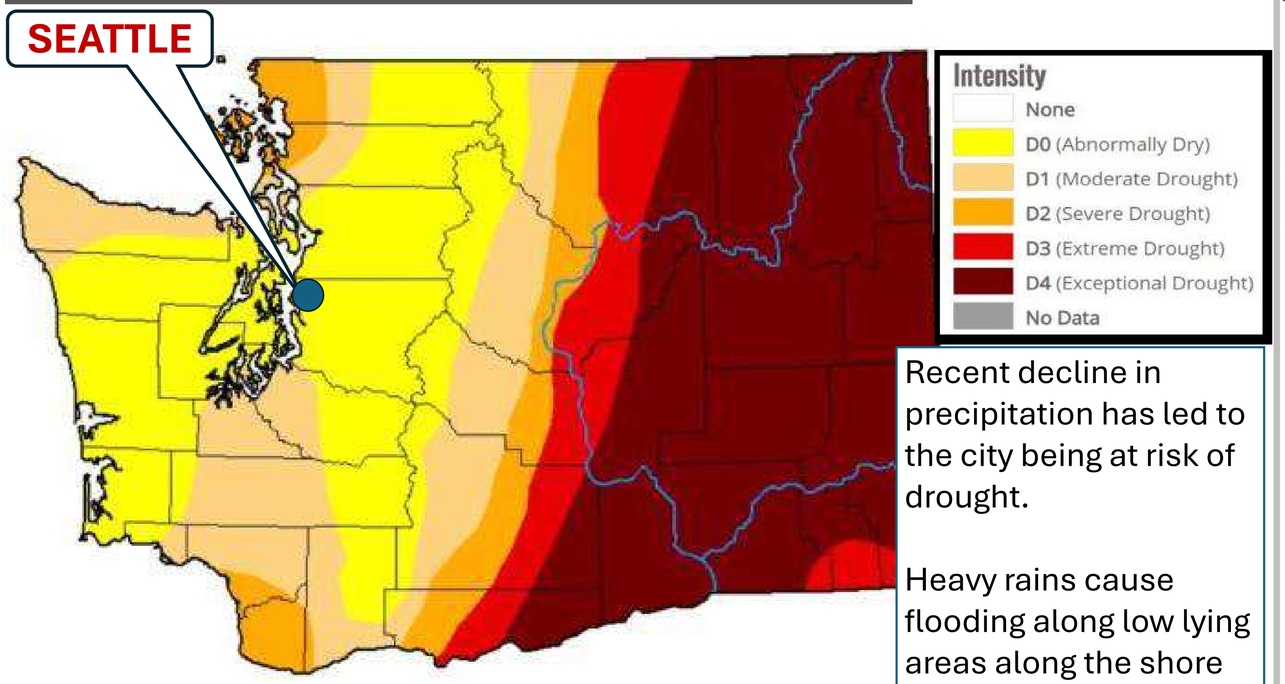



Figure 13: Drought risk
(Washington Water Trust, 2023)

Recent decline in precipitation has led to the city being at risk of drought.

Heavy rains cause flooding along low lying areas along the shore of River Duwamish.

## 1.4.1 FLOOD RISK

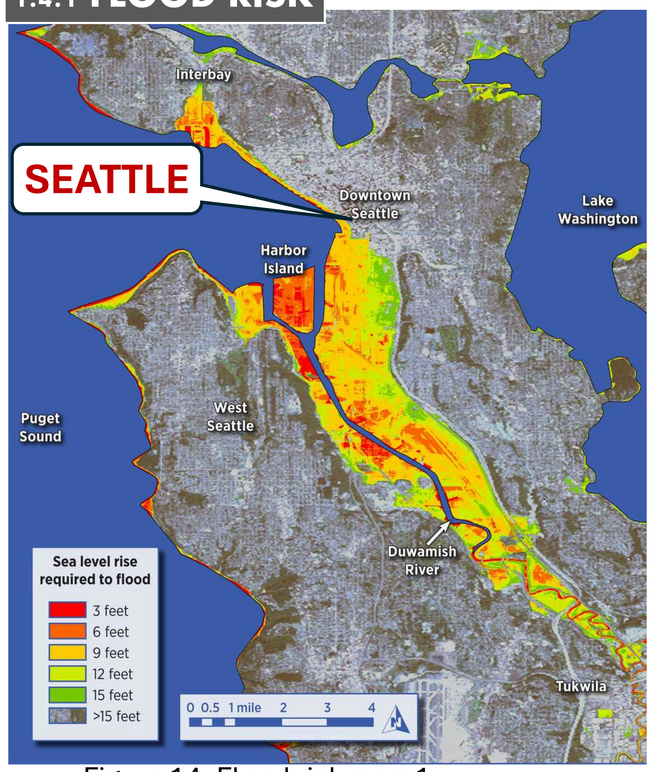
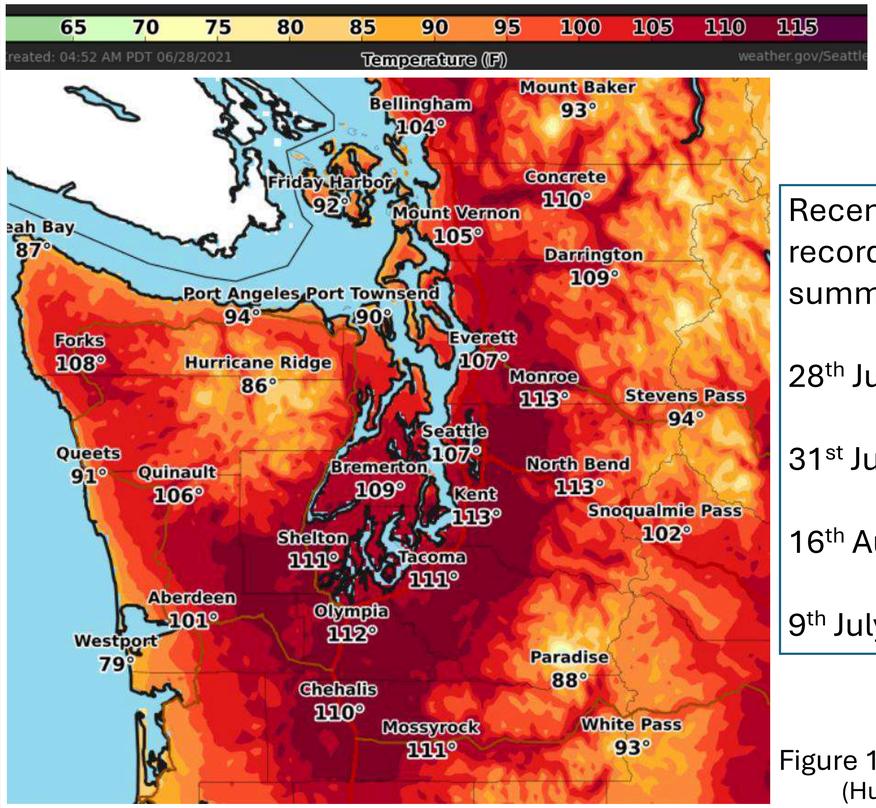



Figure 14: Flood risk map 1
(Mahr, 2019)




Figure 15: Flood risk map 2
(Schultz, 2021)

## 1.4.2 EXTREME HEAT




Recent temperature recording of Seattle in summer

28th June 2021 - 42 deg. C

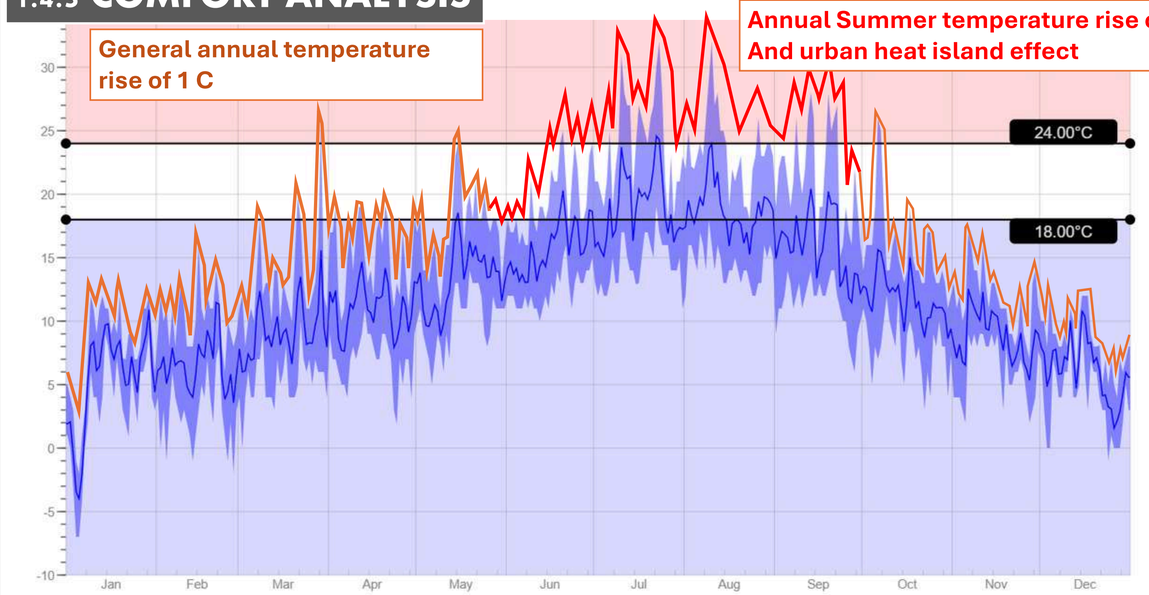
31st July 2022 – 35 deg. C

16th August 2023 – 35 deg. C

9th July 2024 – 36 deg. C

Figure 16: Flood risk map 2
(Hunziker, 2022)

## 1.4.3 COMFORT ANALYSIS




| BELOW COMFORT | | | WITHIN COMFORT | | ABOVE COMFORT | | |
|---|---|---|---|---|---|---|---|
| Hrs | Deg.Hrs | Percent | Hrs | Percent | Hrs | Deg.Hrs | Percent |
| 7312 | 58187.2 | 83.5% | 1276 | 14.6% | 172 | 495.2 | 2.0% |

Figure 17 : Comfort Analysis and temperature rise (Marsh, 2025 with modifications by author)

## 1.5.0 SITE : SEATTLE UNIVERSITY CAMPUS

The proposed project will be situated at the Central campus of Seattle University close to the school of business.

The proposed project will supplement the limited number of students residence at the University.

P Car Park

Bus stop

Campus boundary

Central campus boundary

Existing Students' accommodation on central campus

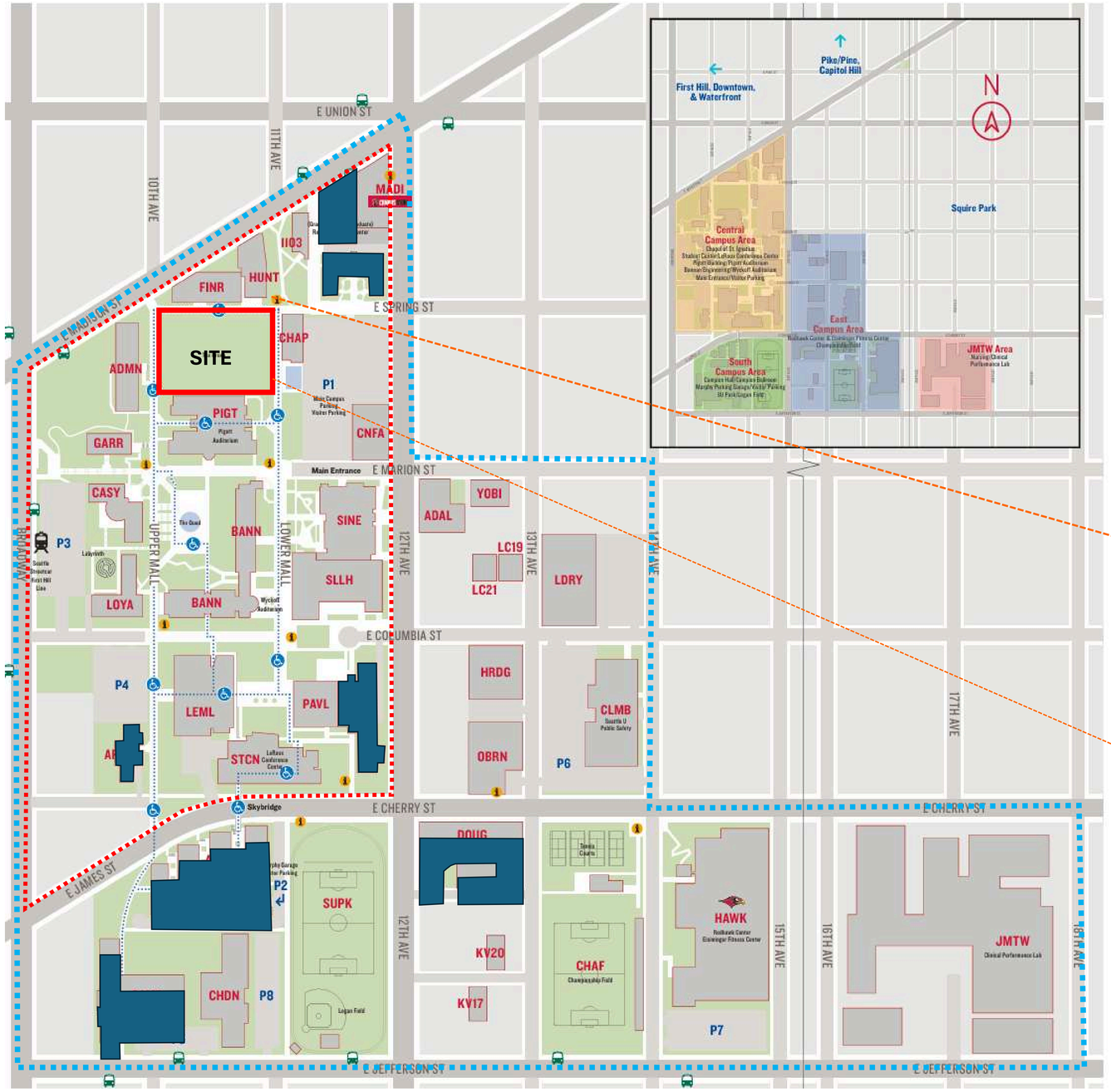



Figure 18 : Site Plan with different elevations (Seattle University, 2025)

## 1.5.1 STUDENT ACCOMMODATION CHALLENGE

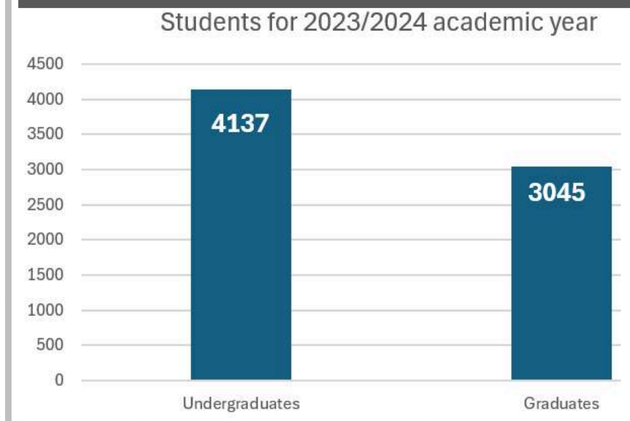



Figure 19 : Undergraduate versus graduate students' numbers

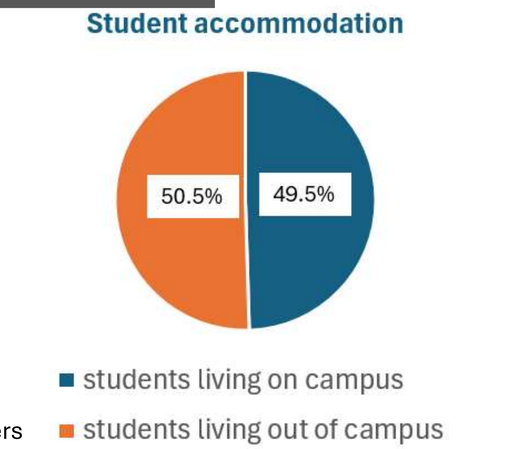



Figure 20 : residential versus non-residential Campus accommodation (Seattle University, 2025)

- Rising rent prices in Seattle
- Student population growth
- Limited On-Campus Housing Availability

**Housing disparity: Graduate students and upperclassmen often face more difficulty in securing on-campus housing, as priority is typically given to first-year students**

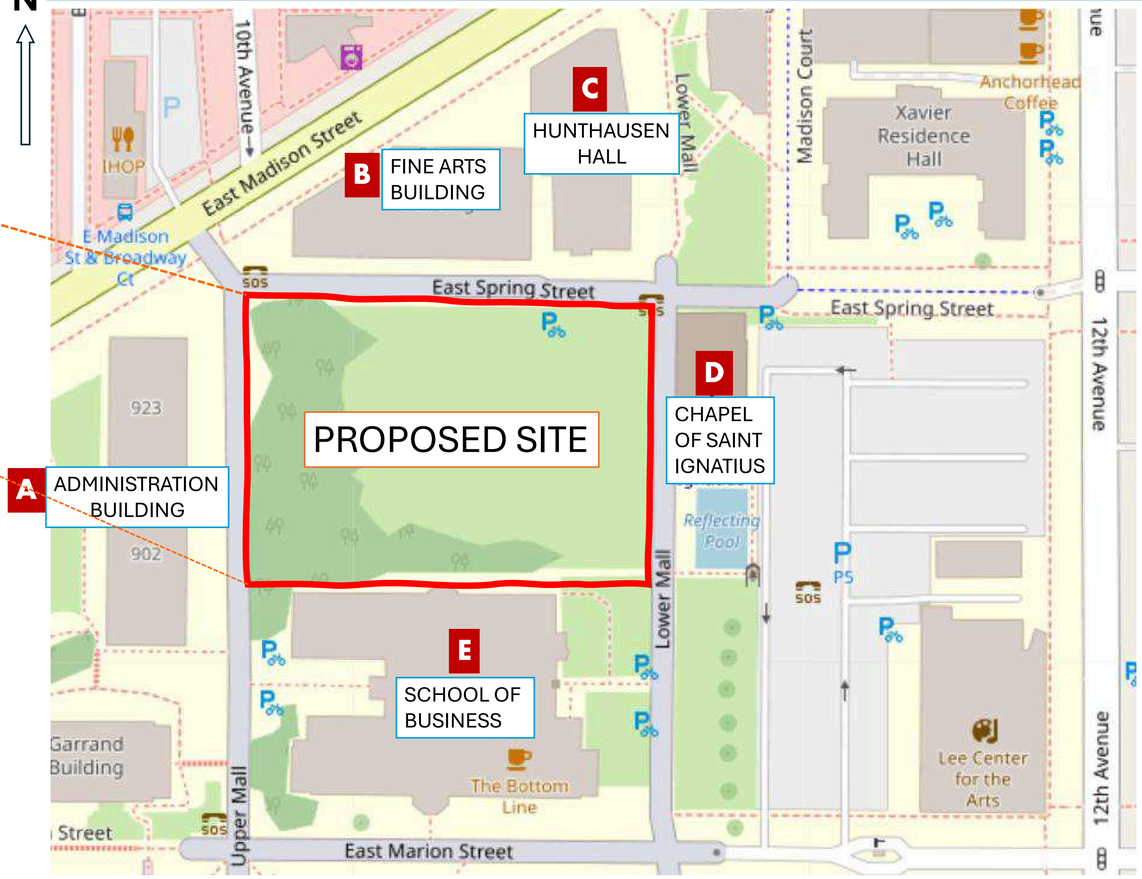



Figure 21 : Site Plan of proposed site (Google, 2025)

## 1.5.2 SITE TOPOGRAPHY AND VEGETATION

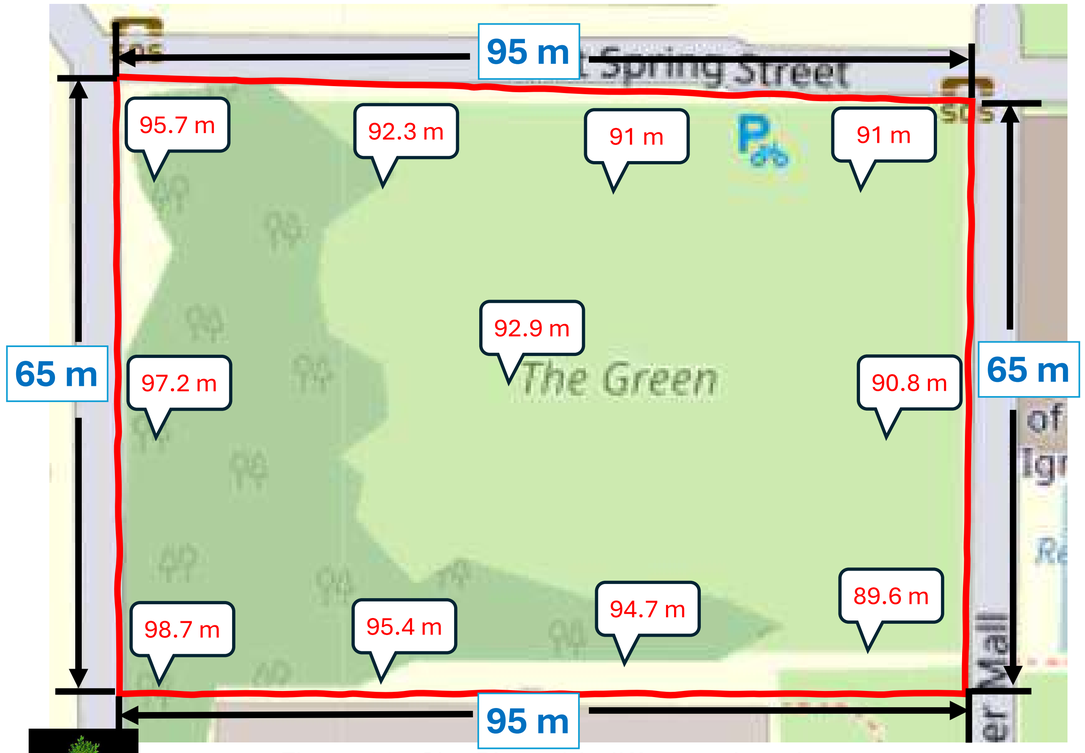



Figure 22 : Site Plan with different elevations (Google, 2025)




Figure 24 : Site Section (Author, 2025)

## 1.5.3 SITE ARIEL VIEW

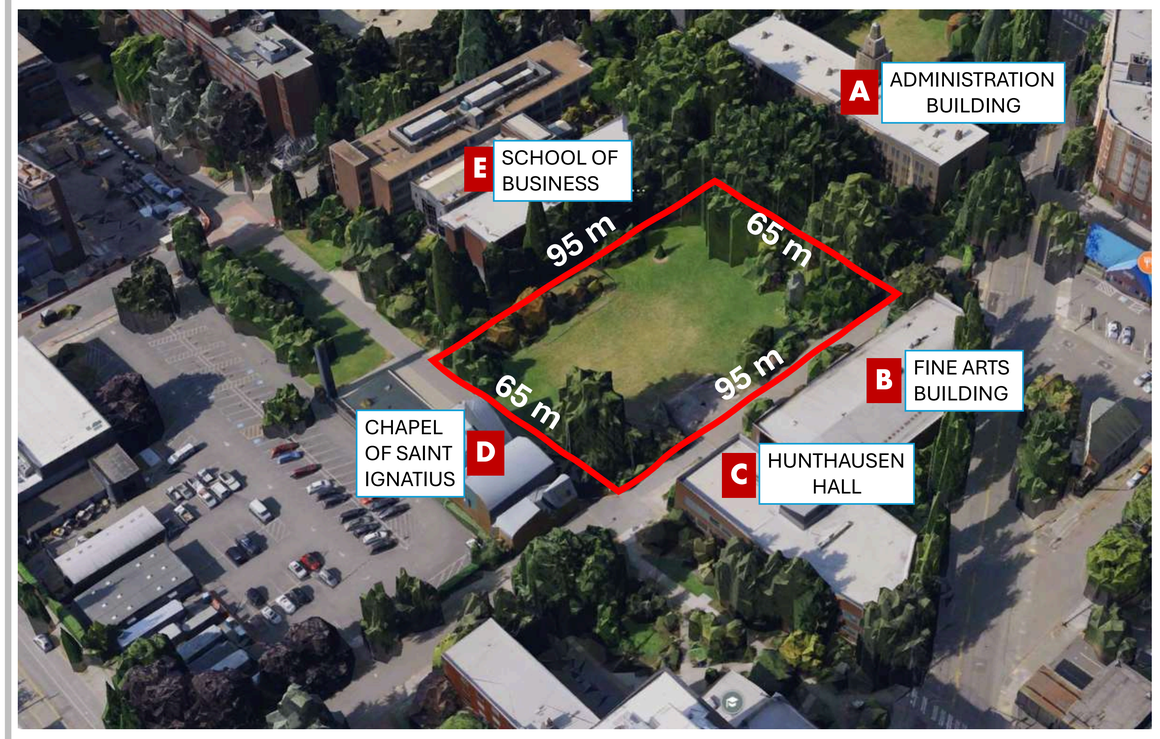



Figure 23: Northeast view of proposed site (Google, 2025)

**EVERGREEN TREES ON SITE**

- ❑ Douglas fir
- ❑ Western Red Cedar

**DECIDUOUS TREES ON SITE**

- ❑ Red maple
- ❑ Bigleaf maple

## 1.5.4 SITE IMAGES



Figure 25 : Grass cover on site (Google, 2025)

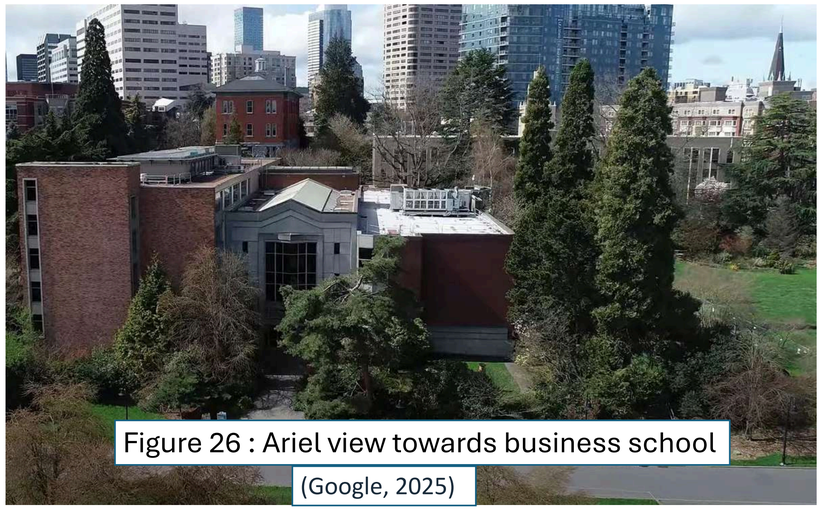


Figure 26 : Ariel view towards business school (Google, 2025)



Figure 27 : View from site towards chapel (Google, 2025)

## 1.5.5 SITE ANALYSIS : SUN PATH AND SHADOW STUDIES

**SUN PATH – 21st JUNE : 9AM**

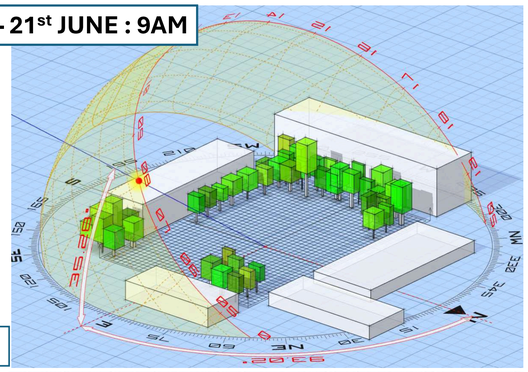


Figure 28 :

**4PM**

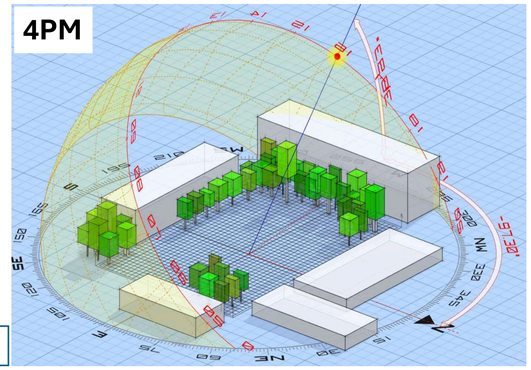


Figure 29 :

**SUN PATH – 21st DECEMBER : 9AM**

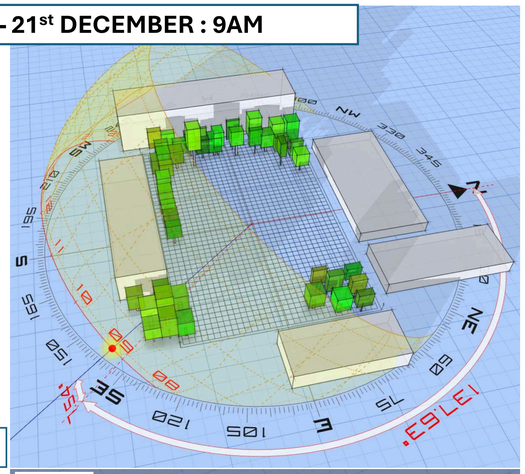


Figure 30 :

**4PM**

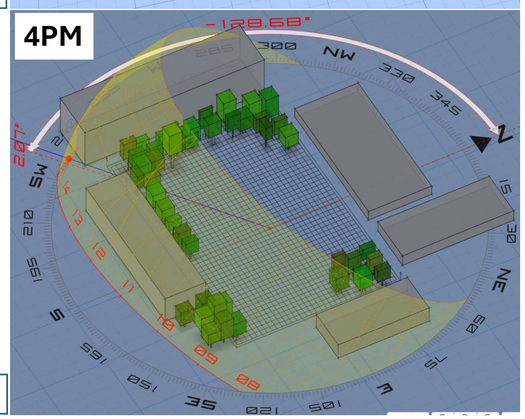


Figure 31 :

**21ST JUNE : 9AM**




Figure 32 : Shadow studies at 9am

**12 NOON**

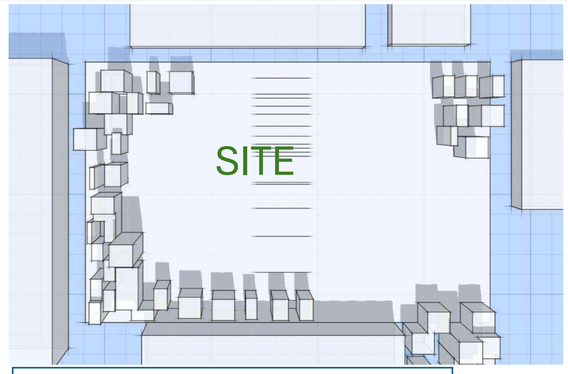



Figure 33 : Shadow studies at 12 noon

**4PM**

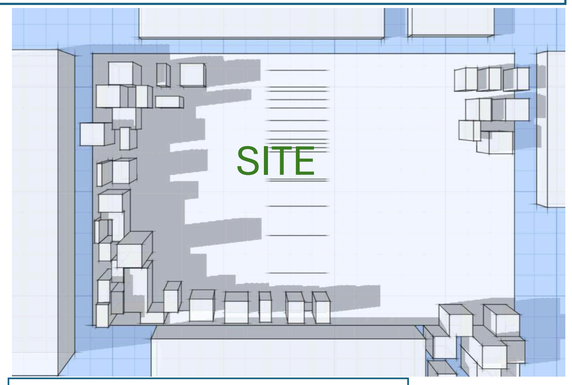



Figure 34 : Shadow studies at 4pm

**22ND SEPTEMBER : 9AM**




Figure 35: Shadow studies at 9am

**12 NOON**




Figure 36 : Shadow studies at 12 noon

**4PM**




Figure 37 : Shadow studies at 4pm

**21ST DECEMBER: 9AM**




Figure 38 : Shadow studies at 9am

**12 NOON**




Figure 39 : Shadow studies at 12noon

**4PM**

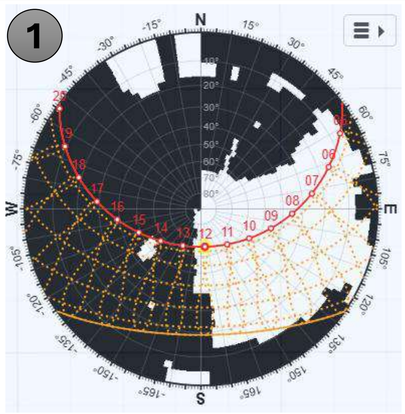


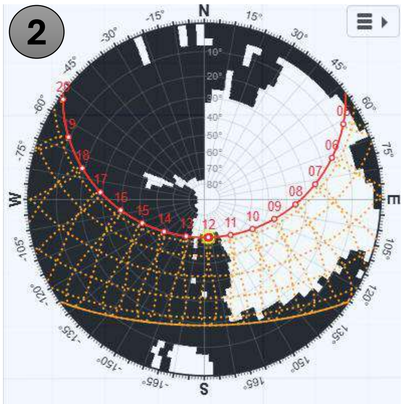

Figure 40 : Shadow studies at 4pm

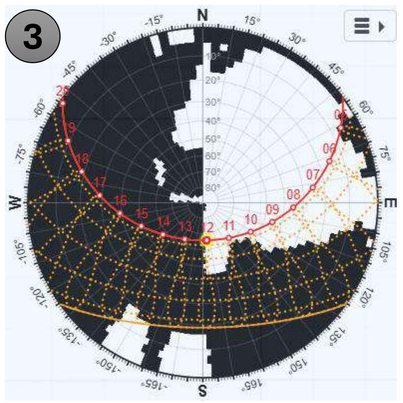
The above shadow studies of the site informs the longest and shortest daylight times. The duration of daylight is longest in summer at 16 hours and winter at 8 hours 25 minutes.
The site is partly shaded by adjacent buildings from Northwest and extended shading from the south and west due to the existing buildings and trees.

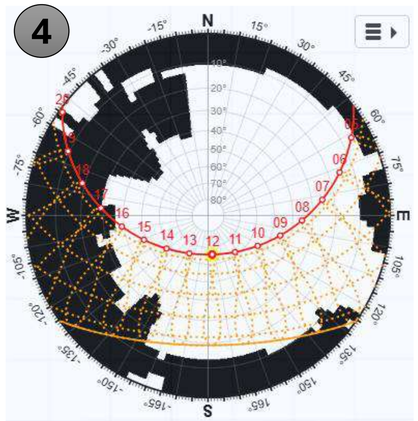
All figures on page: (Marsh, 2025)

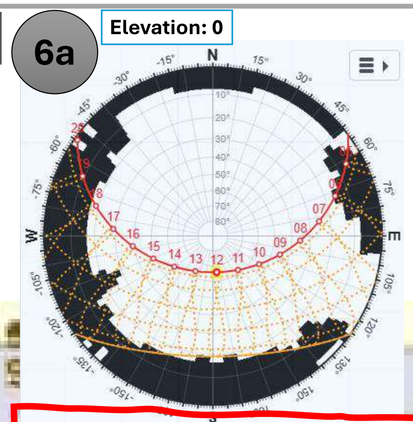
## 1.5.6 SITE ANALYSIS:SHADING MASK

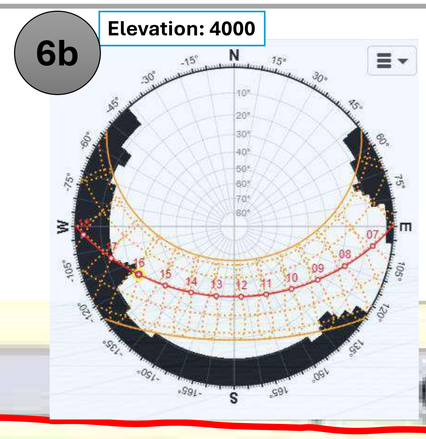


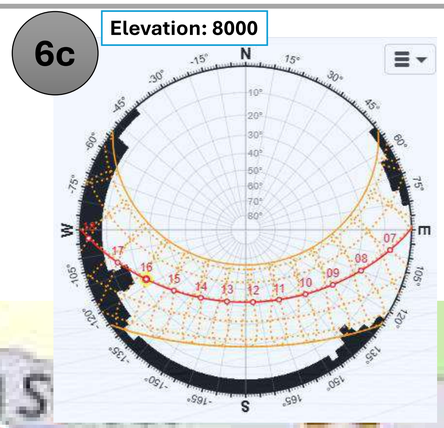



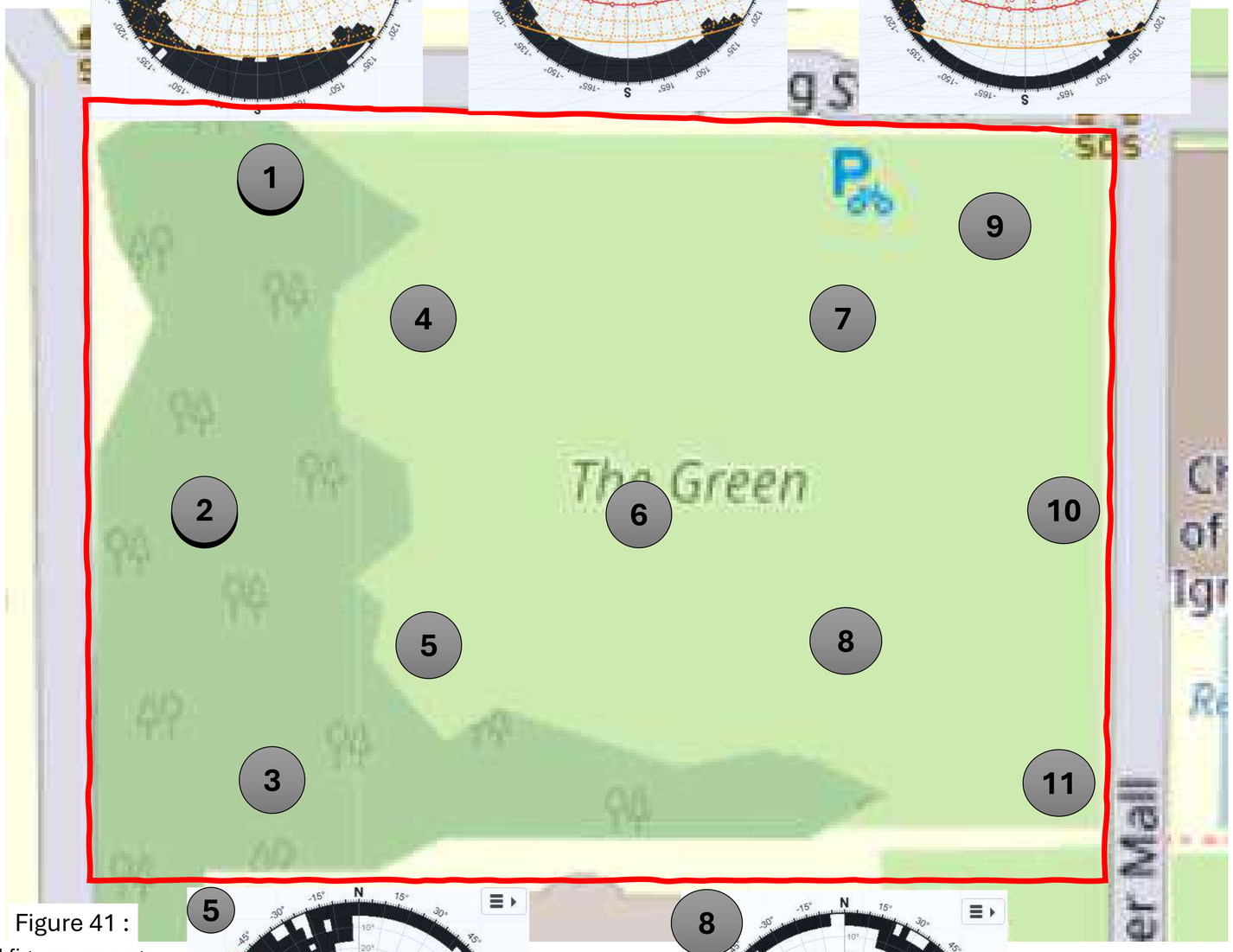
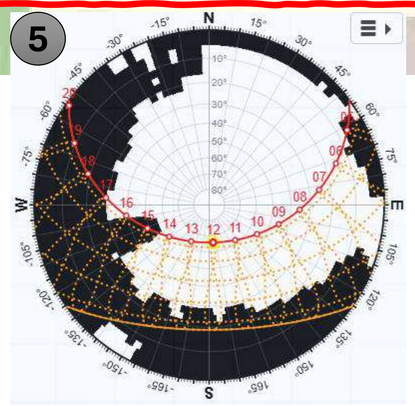
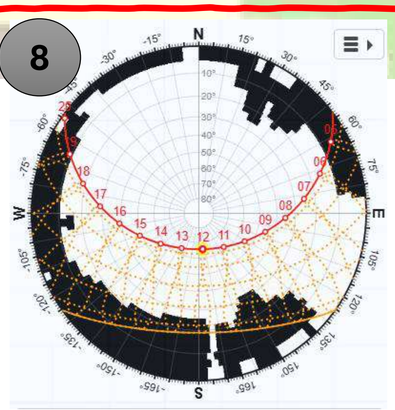
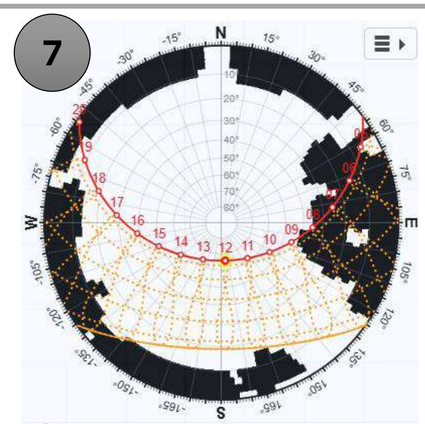
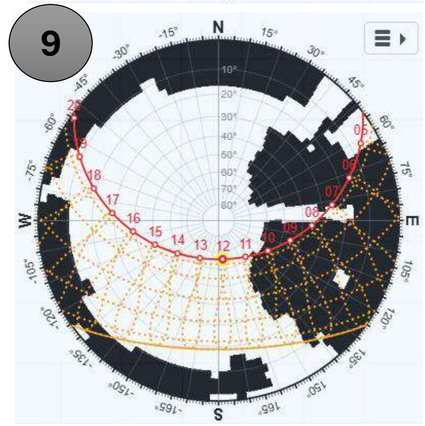



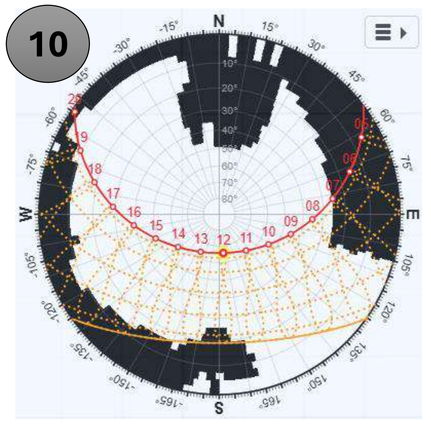

Figure 41 :
(All figures on page: Marsh, 2025)

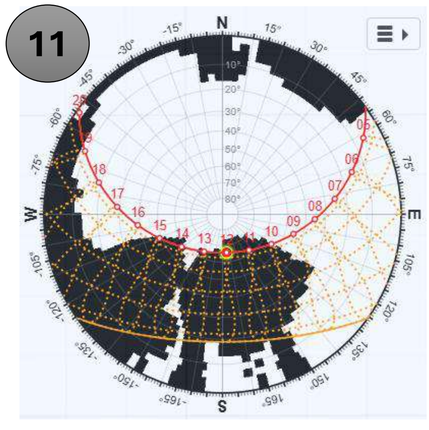


## 1.5.7 SITE ANALYSIS – AGGREGATE SOLAR ANALYSIS : DIRECT + DIFFUSE

### ANNUAL AGGREGATE SOLAR ANALYSIS

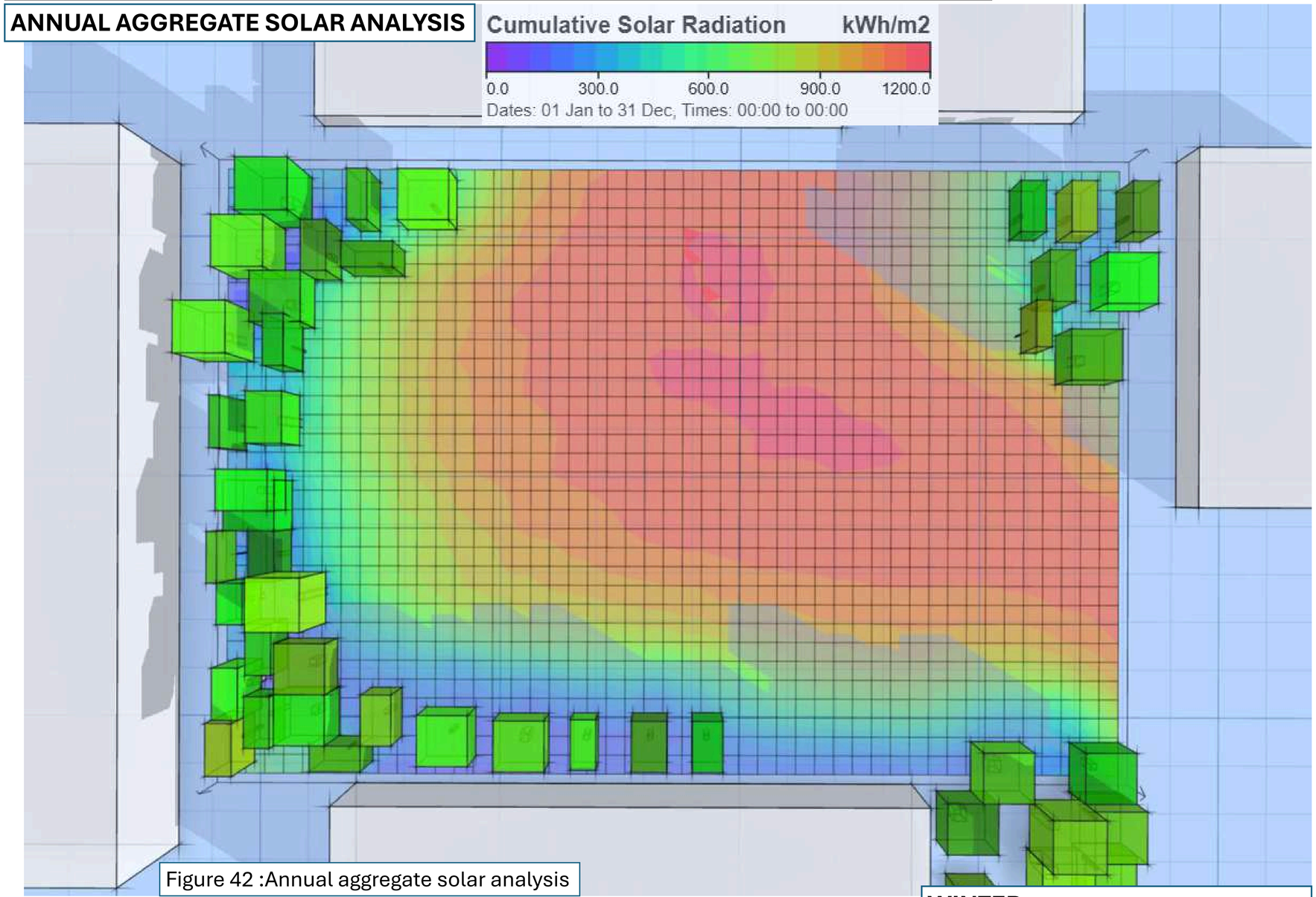



Figure 42 :Annual aggregate solar analysis

All figures on page (Marsh, 2025)

Highest solar radiation in summer : 500kWh/m2
Autumn : 200kWh/m2
Spring : 250kWh/m2
Highest in winter : 150kWh/m2

Solar radiation is very low for 6 months and moderate levels recorded in 3 months.
The highest recorded in summer is double the radiation recorded in spring.

### SUMMER

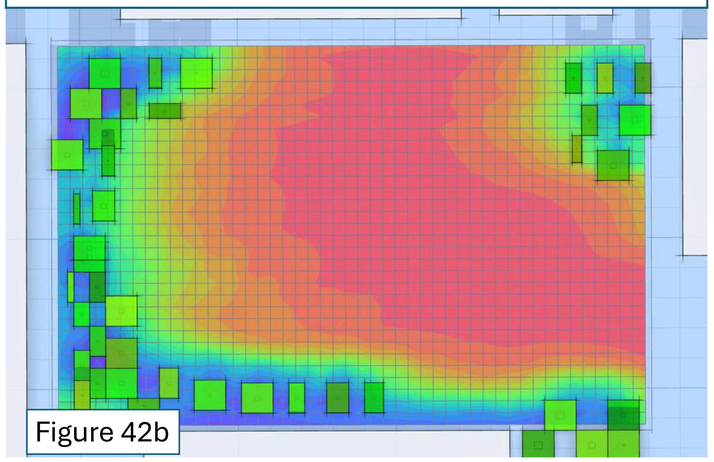


Figure 42b

### AUTUMN

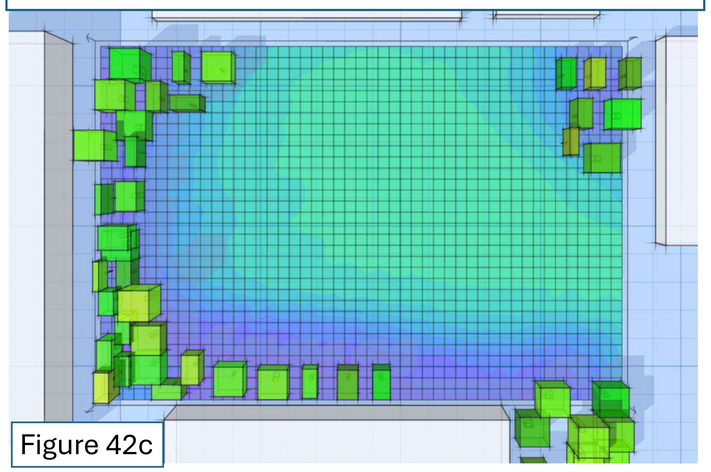


Figure 42c

### SPRING

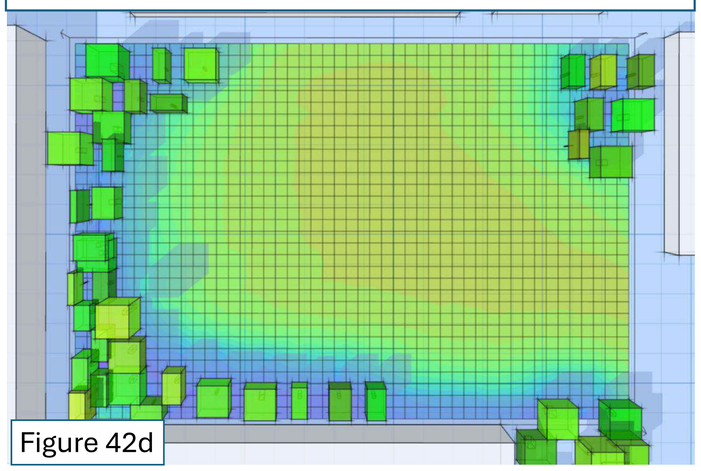


Figure 42d

### WINTER

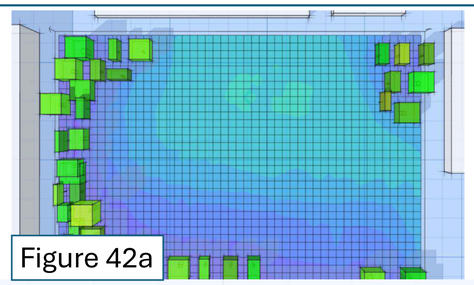
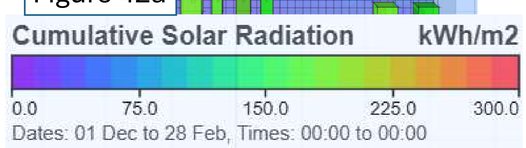
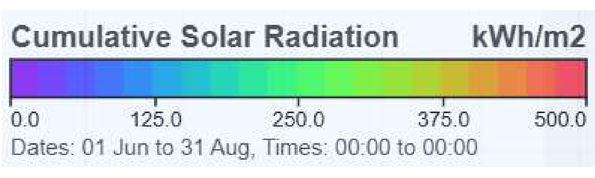


Figure 42a



## 1.5.8 AGGREGATE SOLAR ANALYSIS : PROPOSED BLOCK

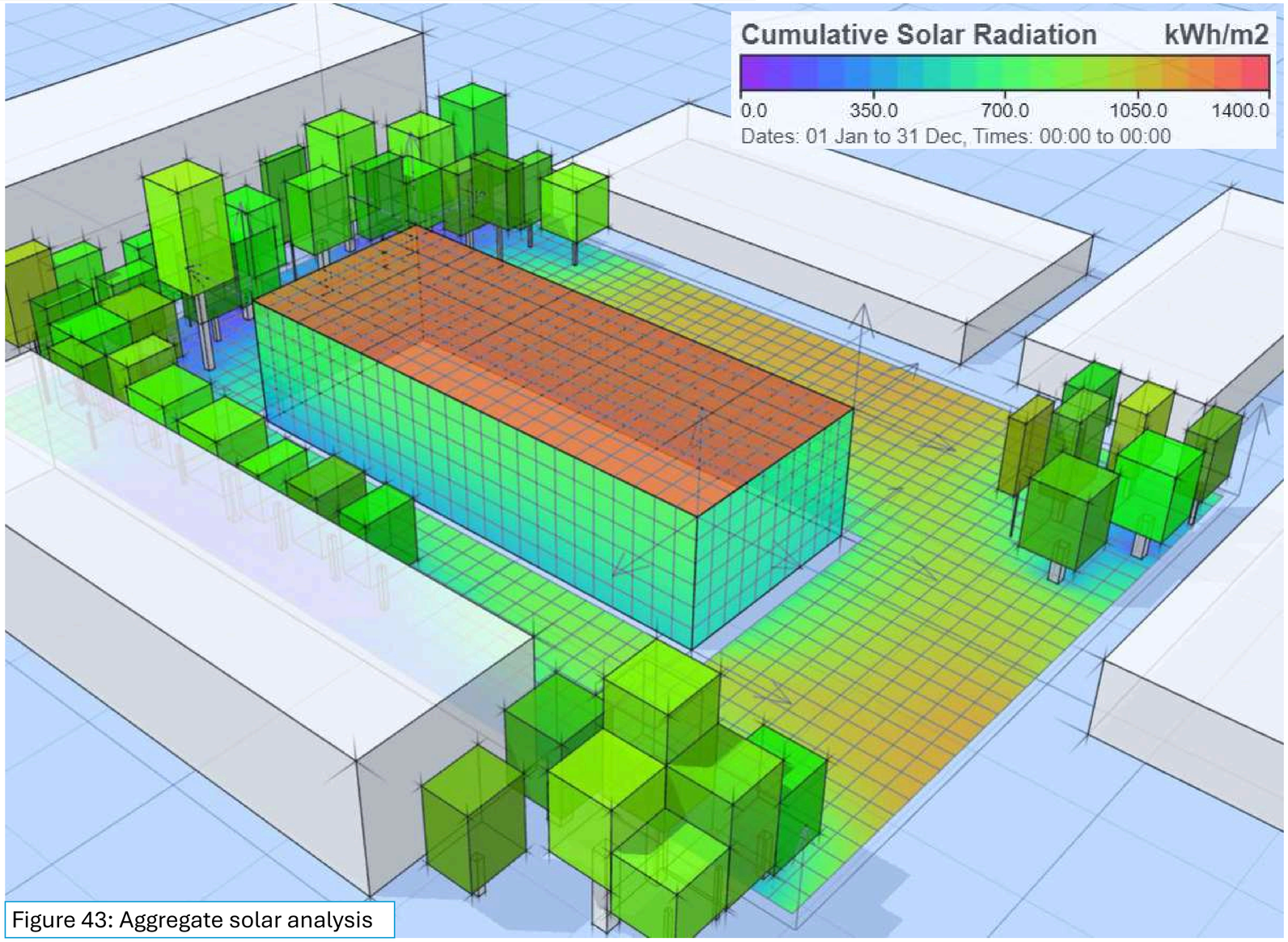



Figure 43: Aggregate solar analysis

All figures on page (Marsh, 2025)

- Minimal solar radiation from the West due to shading from vegetation and building

**Recommendations for proposed project**

- Shading for south facing windows from first floor upwards
- Create outdoor spaces at shaded areas without high solar radiation
- Studio rooms to face south for passive solar heat gain
- Solar radiation on roof is high which is ideal for solar photovoltaic array and solar water heater

**SUMMER**

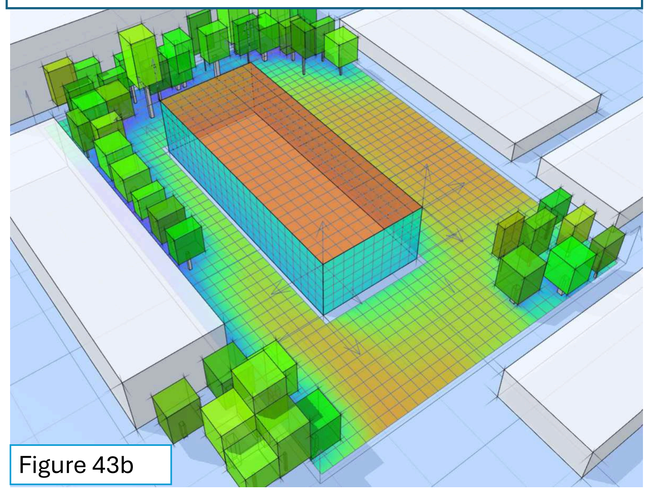


Figure 43b

**AUTUMN**

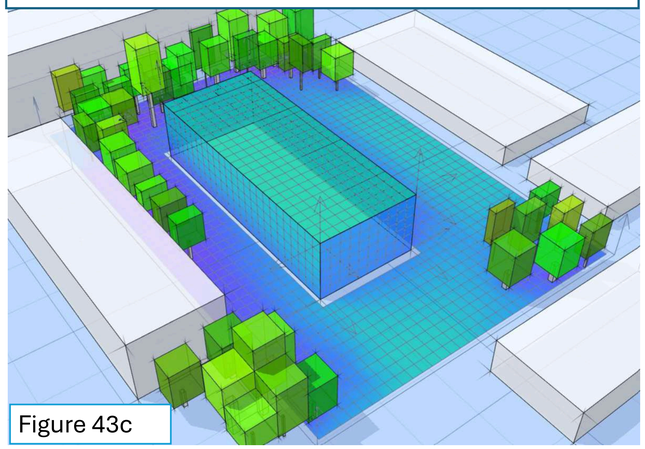


Figure 43c

**SPRING**

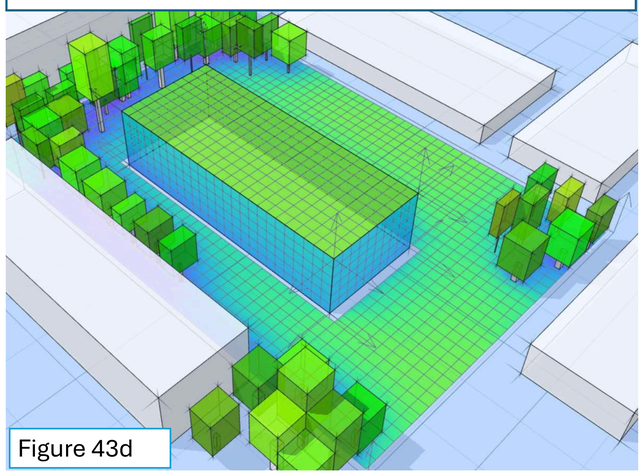


Figure 43d

**WINTER**

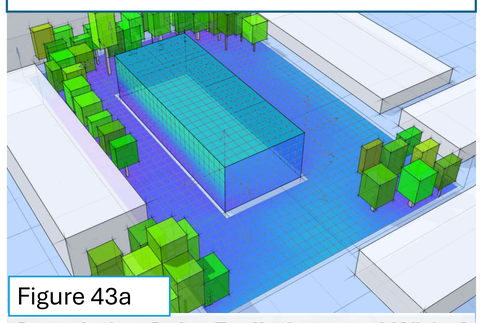
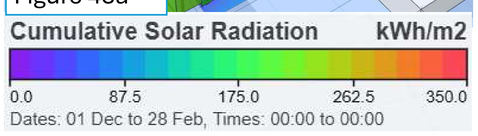
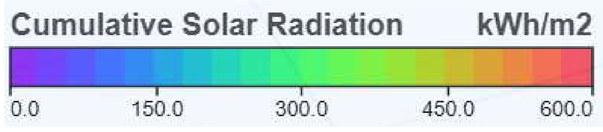


Figure 43a



## 1.5.9 SITE ANALYSIS-WIND PATTERN ON SITE

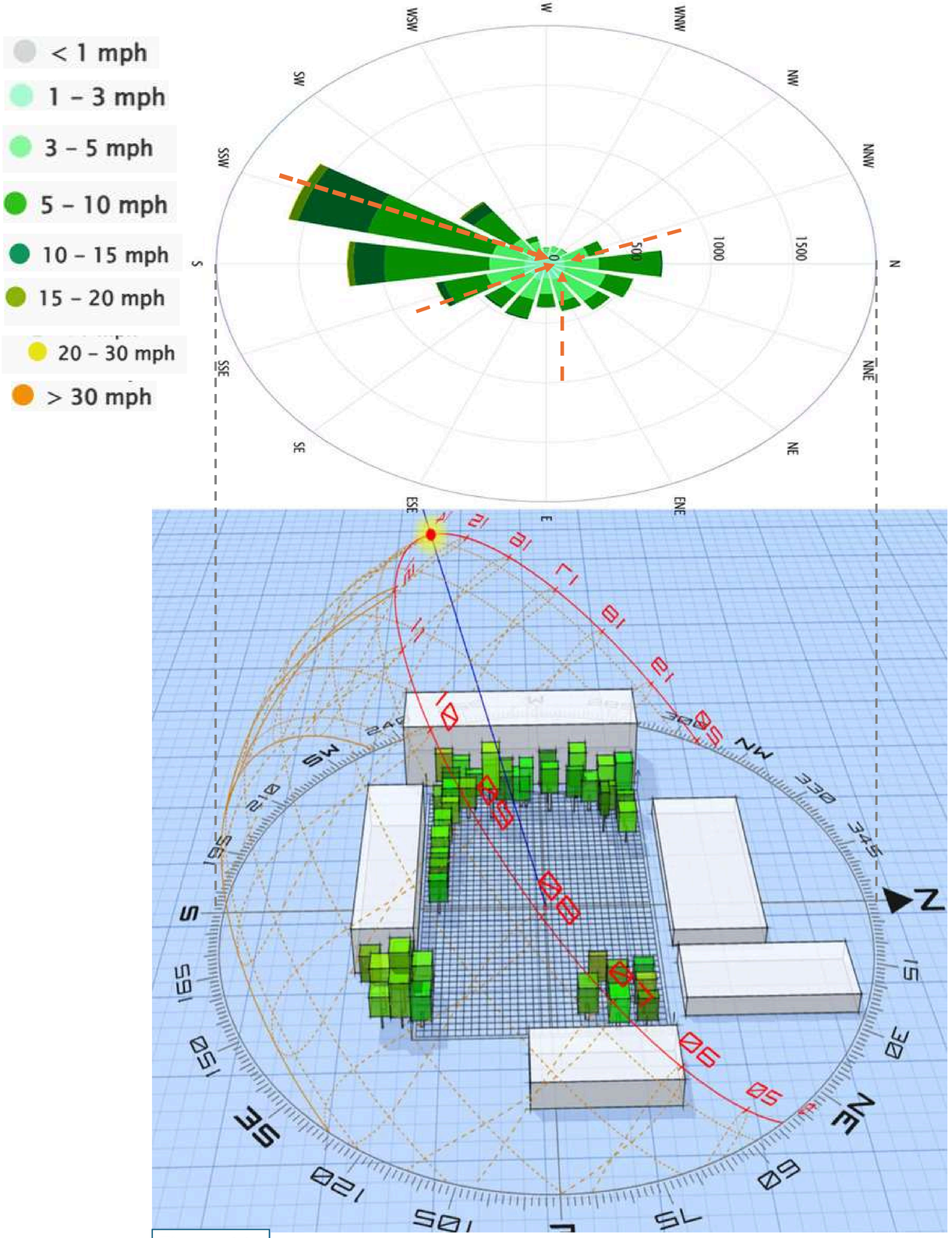



Figure 44 : (Marsh, 2025 and USDA, 2025 ) with modifications by author

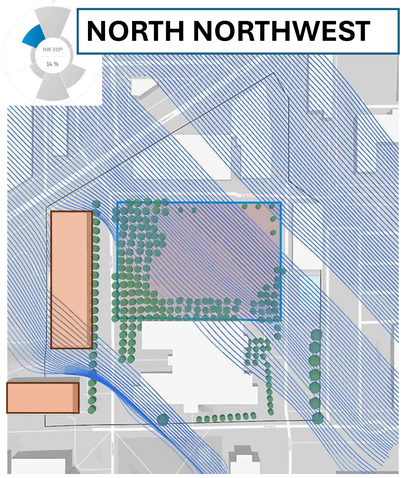



Figure 45: Wind direction from North Northwest

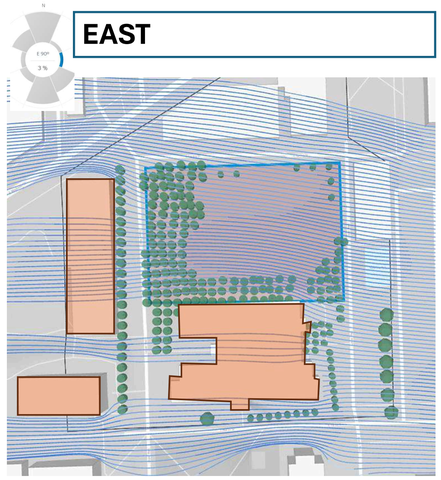



Figure 46: Wind direction from East

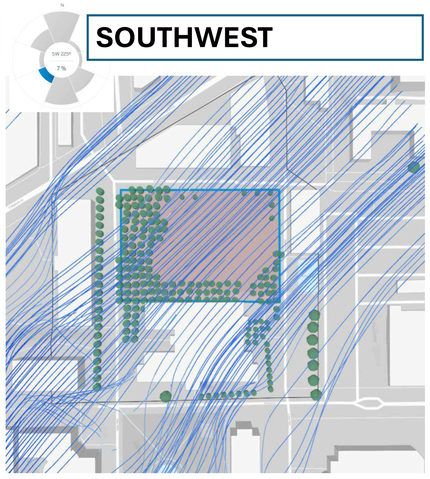



Figure 47: Wind direction from Southwest

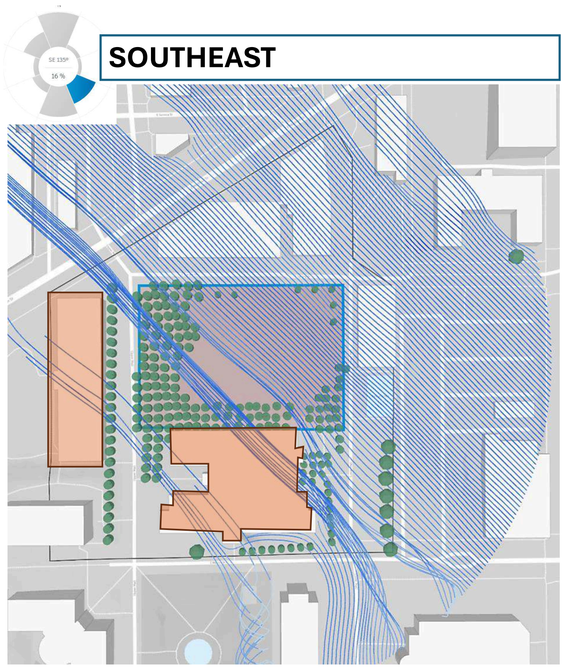



Figure 48: Wind direction from Southeast

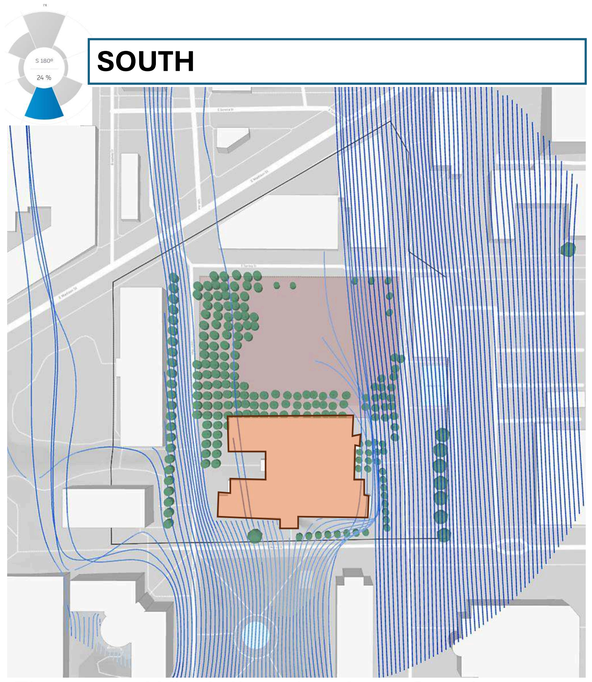



Figure 49: Wind direction from South

Figures 45 to 49: (Autodesk Forma, 2025)

Obstruction to wind flow

**Prevailing winds from Northwest (6-8m/s) East(4-6 m/s) and Southeast offer the best wind directions to site.**
**Minimal south winds to site due to obstruction.**

## 1.5.10 SITE ANALYSIS : STREET NOISE MAPPING

- < 50 db (quite rural background noise, library)
- 50-55 db (office average background noise)
- 55-60 db (conversational speech)
- 60-65 db (piano practice)
- 65-70 db (noisy restaurant)
- 70 > db (vacuum cleaner)

**According to the noise mapping of the site, all three streets have acceptable sound levels of 59 decibels during the day and 52 decibels at night. This indicates the site will have any negative sound impact.**

| STREET NAME | SOUND LEVEL (decibels) | | |
|---|---|---|---|
| East spring street | Avg. 58 dB | Day. 59 dB | Nig. 52 dB |
| Upper mall street | Avg. 58 dB | Day. 59 dB | Nig. 53 dB |
| Lower mall street | Avg. 58 dB | Day. 59 dB | Nig. 53 dB |

(Open Street Map, 2025)

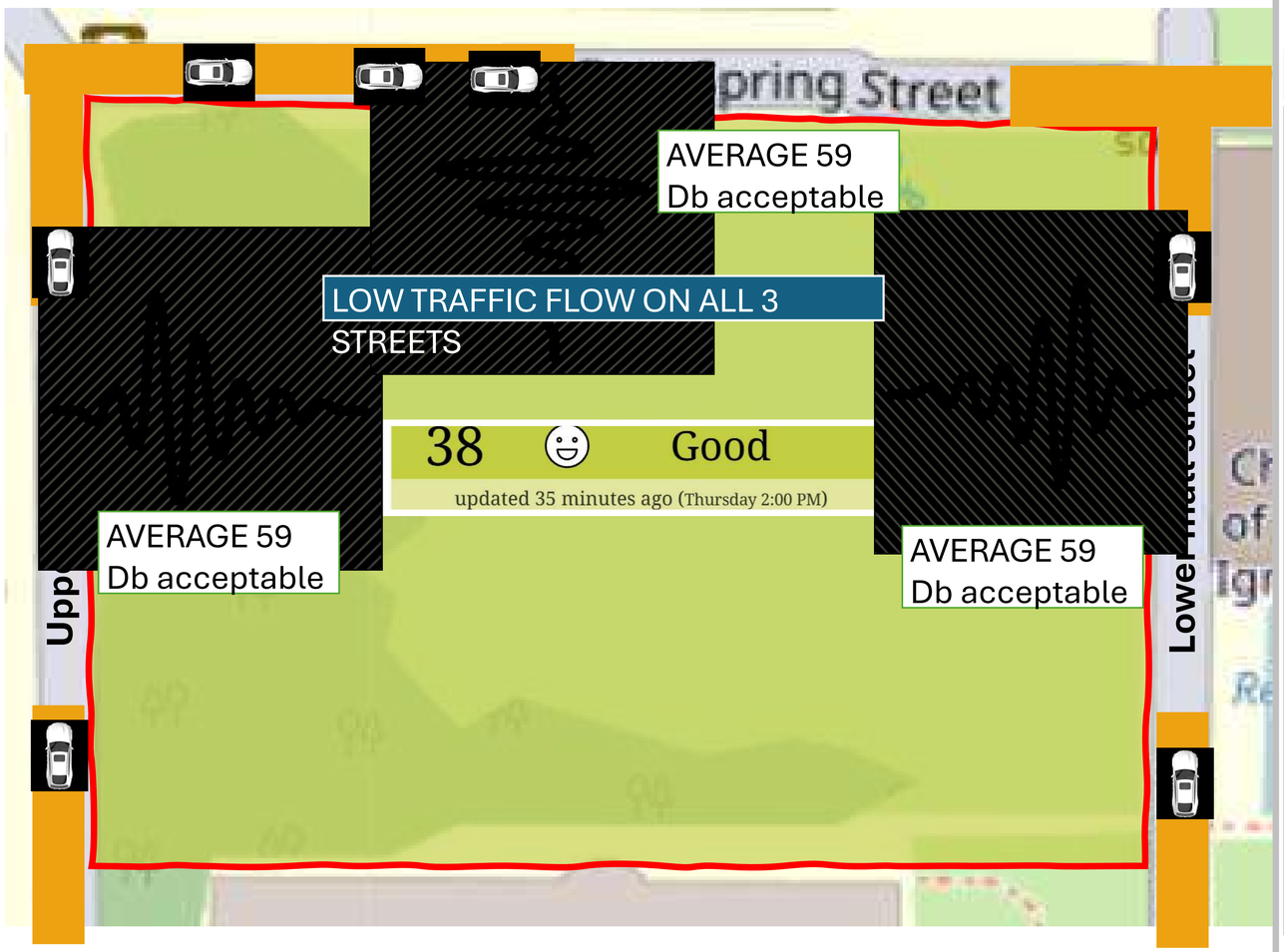
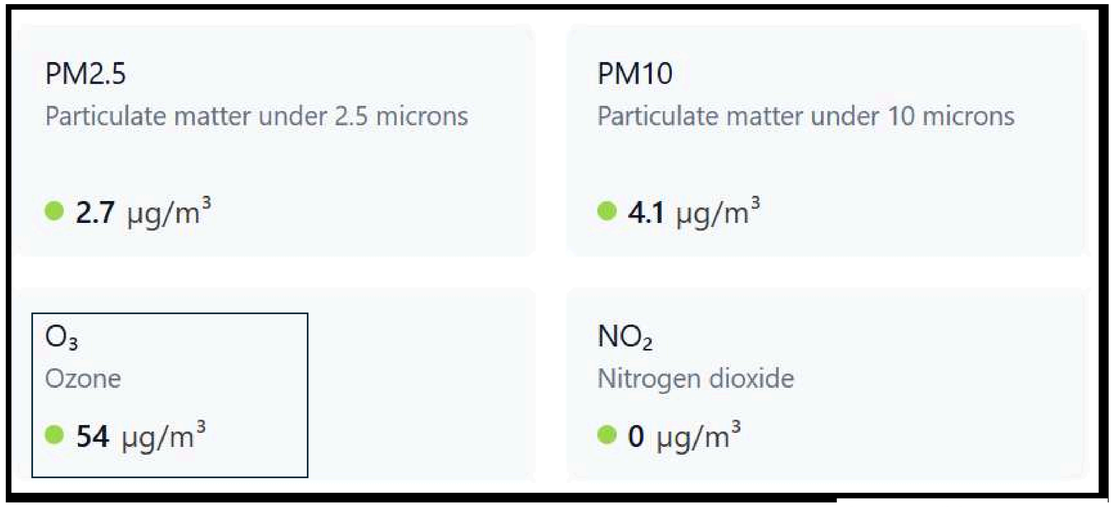



Figure 50 (Google, 2025) ; (IQAir, 2025) ; (Open Street Map, 2025) with modifications by author

## 1.5.11 AIR QUALITY FORECAST

The air quality index at the site is 38 which is generally considered acceptable as there are no high records of pollutants that will impact the health of people.




Source: (IQAir, 2025)

**Table 1 : Air Quality definition scale by US-EPA (2016 Standard)**

| AQI | Air Pollution Level | Health Implications | Cautionary Statement (for PM2.5) |
|---|---|---|---|
| 0 - 50 | Good | Air quality is considered satisfactory, and air pollution poses little or no risk | None |
| 51 -100 | Moderate | Air quality is acceptable; however, for some pollutants there may be a moderate health concern for a very small number of people who are unusually sensitive to air pollution. | Active children and adults, and people with respiratory disease, such as asthma, should limit prolonged outdoor exertion. |
| 101-150 | Unhealthy for Sensitive Groups | Members of sensitive groups may experience health effects. The general public is not likely to be affected. | Active children and adults, and people with respiratory disease, such as asthma, should limit prolonged outdoor exertion. |
| 151-200 | Unhealthy | Everyone may begin to experience health effects; members of sensitive groups may experience more serious health effects | Active children and adults, and people with respiratory disease, such as asthma, should avoid prolonged outdoor exertion; everyone else, especially children, should limit prolonged outdoor exertion |
| 201-300 | Very Unhealthy | Health warnings of emergency conditions. The entire population is more likely to be affected. | Active children and adults, and people with respiratory disease, such as asthma, should avoid all outdoor exertion; everyone else, especially children, should limit outdoor exertion. |
| 300+ | Hazardous | Health alert: everyone may experience more serious health effects | Everyone should avoid all outdoor exertion |

Source: (IQAir, 2025)

## 1.5.12 SITE ANALYSIS-CONCLUSION

View sphere from site : Fig. 51 to Fig. 53

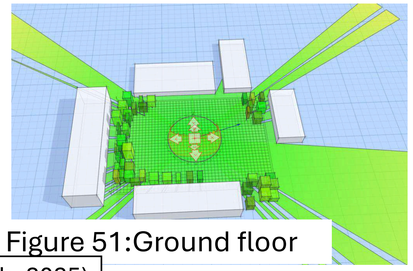


Figure 51:Ground floor (Marsh, 2025)

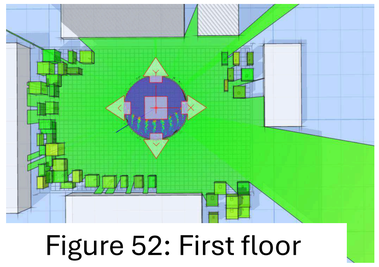


Figure 52: First floor



Figure 53: Second floor

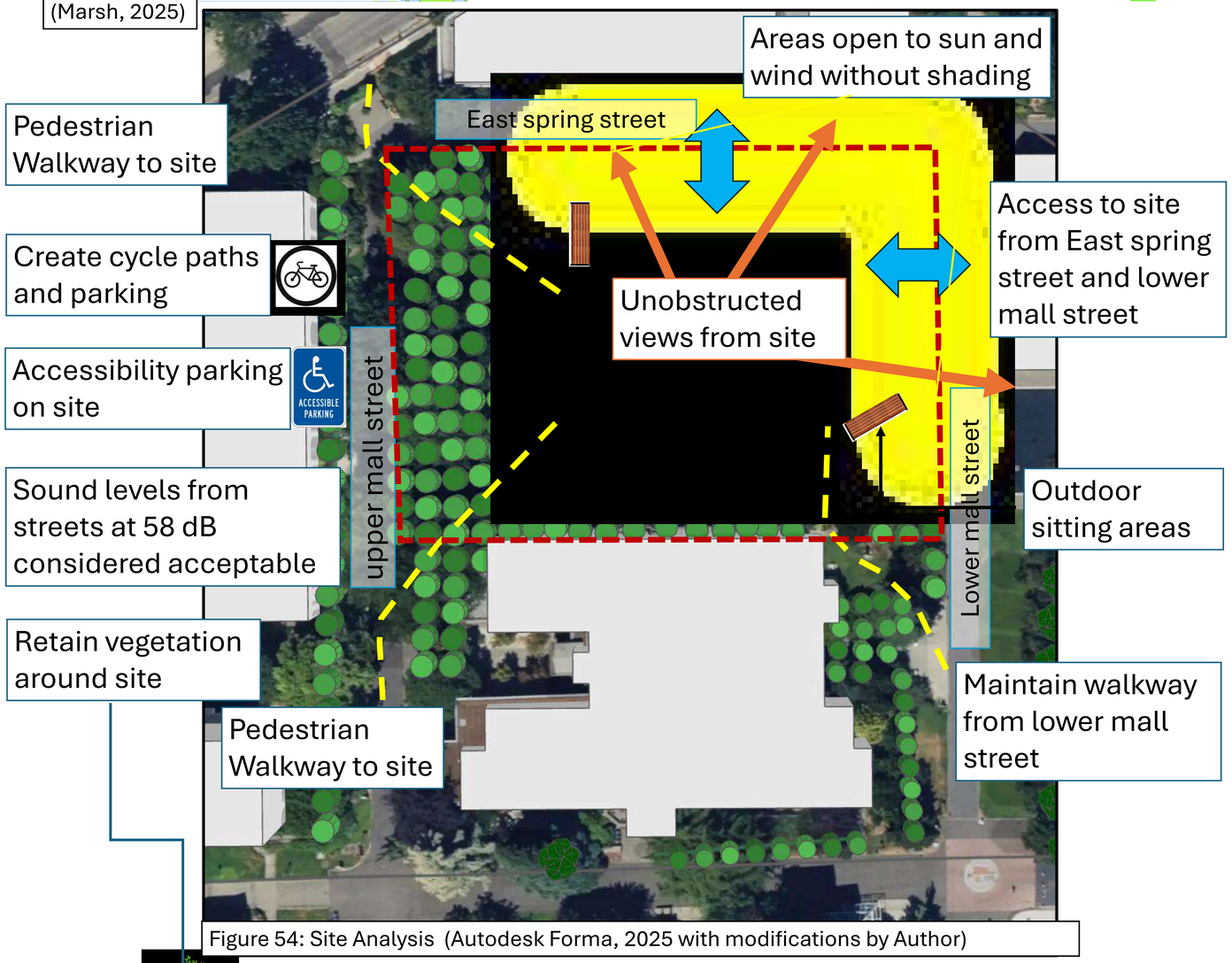



Figure 54: Site Analysis (Autodesk Forma, 2025 with modifications by Author)

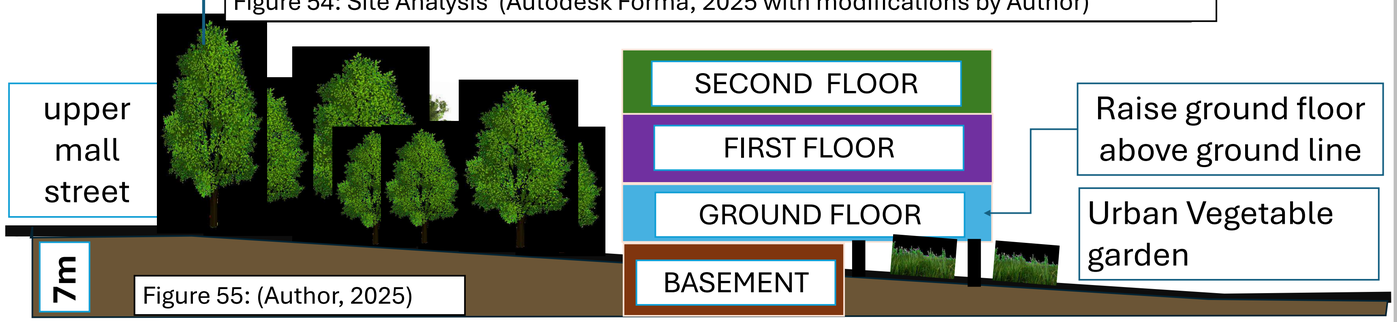



Figure 55: (Author, 2025)

**Building orientation : South facing for passive solar heat gain**

**Shading : first, second and third floor south facing windows**

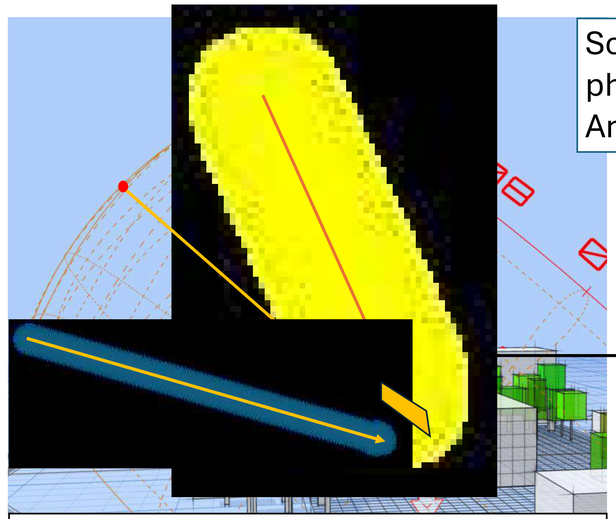



Figure 56:Solar angles and potential solar PV (Marsh, 2025) with modifications by author

Locate entrance on north façade overlooking East spring and lower mall street

Semi-permeable pavers for driveway and walkway

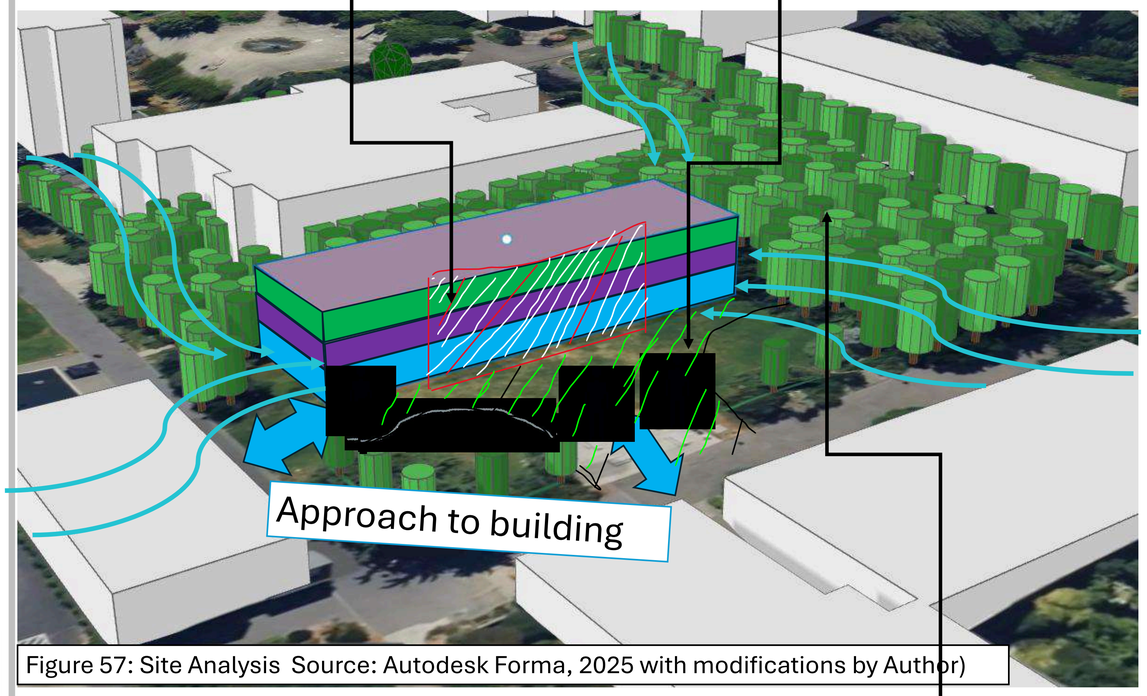



Figure 57: Site Analysis Source: Autodesk Forma, 2025 with modifications by Author)

**Compact form** : To reduce heat loss from the building and efficient use of space

Shading of west and south façade by trees

**Airflow:** From different angles of site towards building for ventilation

# Sustainable Design of Students' Accommodation Block

Sustainable Environmental Design
Afua Ampomah-Amoako



## 2.0 CASE STUDY 1 : THE HOUSE AT CORNELL TECH

| | |
|---|---|
| LOCATION | Cornell University, New York |
| PROJECT TYPE | Residential – Student Housing |
| YEAR OF COMPLETION | 2017 |
| CERTIFICATIONS | Passive House<br>LEED Platinum |
| NO. OF FLOORS | 26 |
| NO. OF UNITS | 352 |

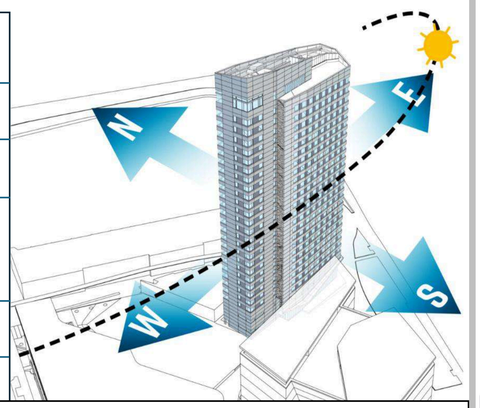



Figure 58: Building orientation(Handle Architects,2017)

### Passive Design Strategies

**Building orientation**: East-west axis for passive solar heat gain.
**Form**: Compact to minimize heat losses
**Insulation**: thick insulated wall around the full height of the building
**Envelope :** Highly insulated prefabricated mega panels with R-value of 19 with windows fixed and sealed
**Window to wall ratio:** 23%

**Overall Source Energy**: 38.1 kBtu/ft/yr
**Heating Energy**: max 4.75 kBtu/ft/yr
**Cooling Energy**: Max 5.39 kBtu/ft/yr
**Air changes per hour** : 0.6 ACH
**Exhaust and supply ventilation** : with energy recovery

### Active Systems

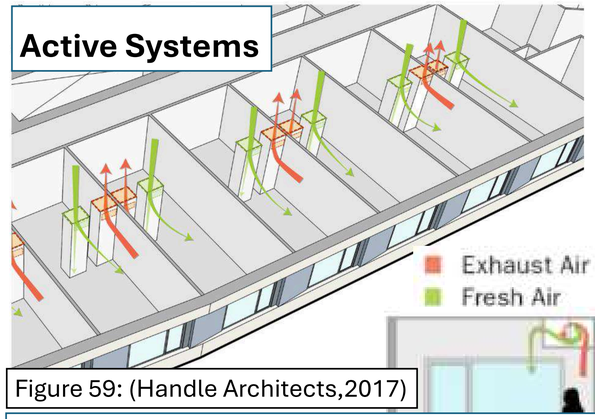



Figure 59: (Handle Architects,2017)

vertical supply and exhaust shafts provide a pathway for air to travel to one of two energy recovery ventilators that sit on the building's rooftop

**Ventilation:** central air distribution system with ERV on the rooftop

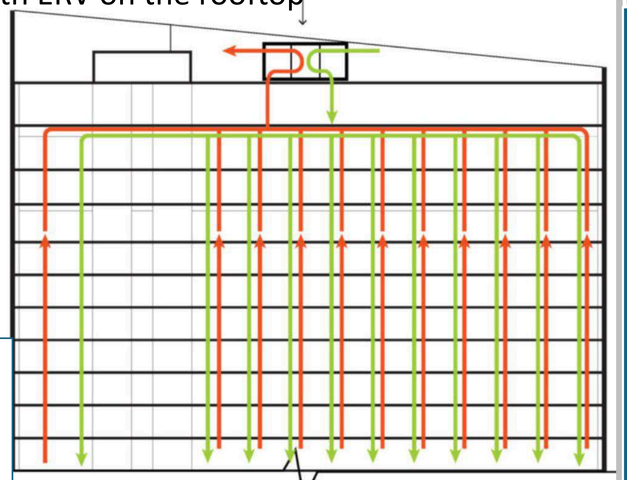


Figure 60: Energy Recovery System (Handle Architects,2017)

**Heating and cooling system**: Low-energy variable refrigerant flow (VRF) system
To extract heat from outside air to provide heat in cool months, or extract heat from inside air in warm months via an outdoor condenser.

## 2.1 CASE STUDY 2 : POMONA COLLEGE STUDENT HOUSING

| | |
|---|---|
| LOCATION | Claremont, California |
| PROJECT TYPE | Residential – Student Housing |
| YEAR OF COMPLETION | 2011 |
| CERTIFICATIONS | LEED Platinum |
| NO. OF FLOORS | 2 / 3 |
| NO. OF UNITS | 150 |
| TOTAL AREA | 81 sf |

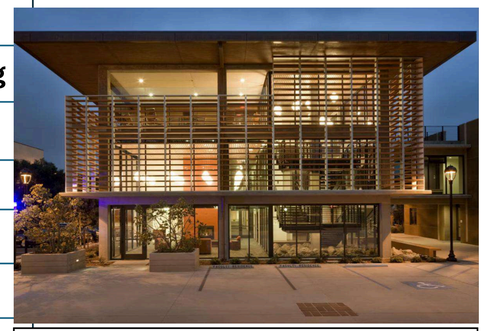


Figure 61: exterior view (Wyatt (2014) )

Solar PV system offsets over 130,000 kWh (14%) of annual electricity consumption

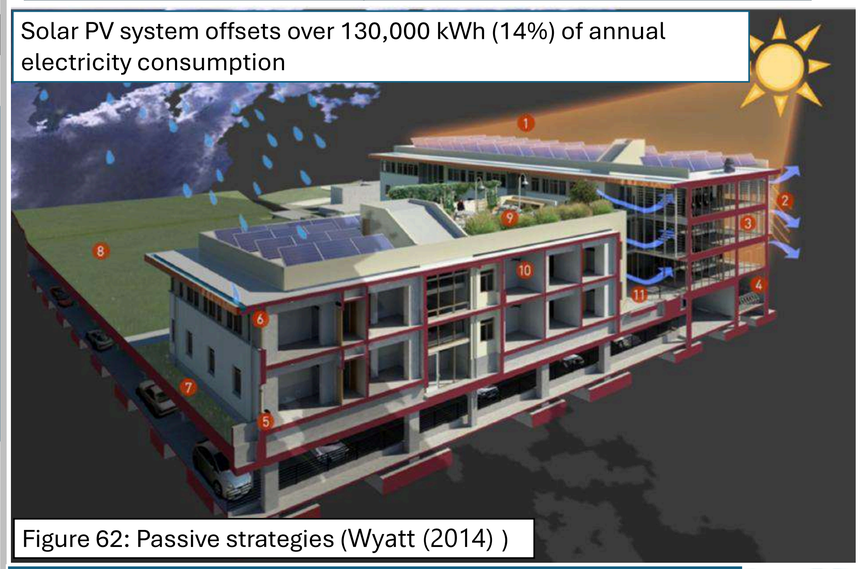


Figure 62: Passive strategies (Wyatt (2014) )

### Performance Results

Energy Use Intensity (EUI) - 53.47 kBTU/sqft/yr

Actual annual energy consumption reduced by 22% compared to California Building Code.

Performance 49% better than ASHRAE 90.1-2004 Standard

**Daylighting**
•Over 95% of occupied spaces have efficient daylight
**Solar Control and Shading**
•8-feet-deep roof overhangs
•Exterior sunscreens consisting of 2 by 8-inches deep horizontal terracotta planks
**Thermal Mass**
•11-inches thick precast concrete panels with 2 wythes (3-inch exterior and 5-inch interior) and 3-inches thick R-19 Polystyrene insulation
**Materials**
- 24% of materials were regionally selected within 500 miles of the project site

•72% of all wood products are FSC-certified
•18% recycled building materials

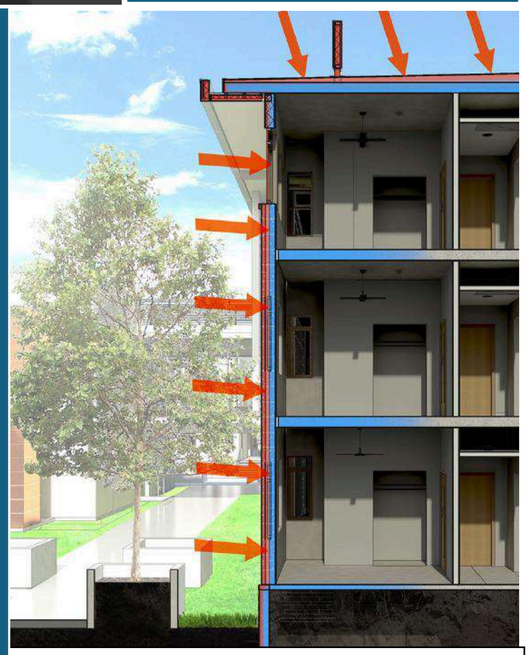


Figure 63: Thermal mass (Wyatt (2014) )

## 3.0 DETAILED BRIEF WITH PERFORMANCE TARGETS




**TARGET : GRADUATE STUDENTS**



42

**STAFF**



8

**Ground floor - Communal Spaces**
Reception
Manager's office
Socialising/play area
Mailboxes zone
Bicycle storage
Circulation: Lift and stair

**1ST, 2ND & 3RD FLOOR - Studios**
Sleeping area
Study
Kitchenette
Toilet and shower
Laundry + utility store
Sun Space

**Basement – Services**

## 3.1 DETAILS OF SPACES AND ACTIVITY SCHEDULE

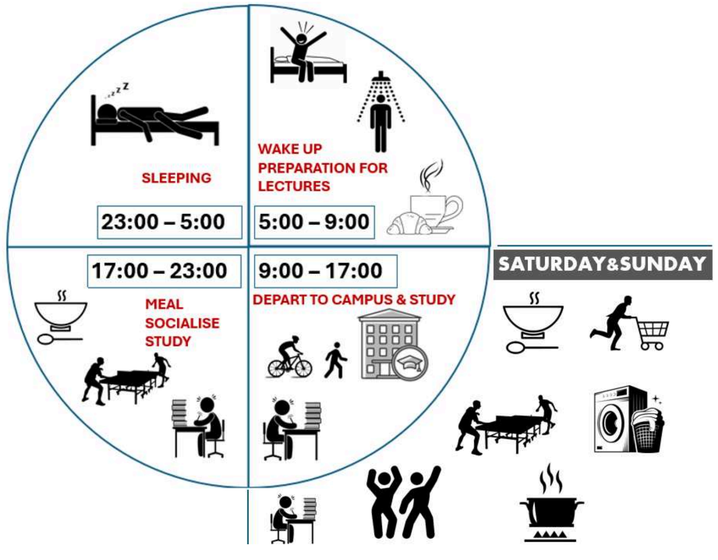



Figure 64: Activity Schedule
Source: (Google 2025 with modifications by Author)

## 3.2 TABLE 2: PERFORMANCE TARGETS

| FLOOR | SPACE | AREA (m2) | NV THERMAL TARGETS | HVAC THERMAL TARGETS | DAYLIGHT TARGETS | | CO2 TARGETS | HUMIDITY |
|---|---|---|---|---|---|---|---|---|
| | | | CIBSE A (°C) | LEED (°C) | LEED | | LEED | LEED |
| TYPICAL FLOOR OF 1,2&3 | STUDIO -sleeping area -study area -Kitchenette | 10.18 (x 14no.) = 142.52 | 18 - 25 | 21-24 | (sDA300,50 %) ≥55% space 300 lux | (ASE1000,250) <10% space 1,000 lux | ≤ 1,000 ppm | 30% to 60% |
| | Toilet and shower | 2.06 (x 14no.) = 28.84 | 18-25 | 21-24 | ≥40% 300 lux | <10% 1,000 lux | ≤ 1,000 ppm | 30% to 60% |
| | Sun Space | 3.43x14 = 48.02 | - | - | | | | |
| | Reading room | 14 | 18 - 25 | 21-24 | ≥55% space 300 lux | <10% space 1,000 lux | ≤ 1,000 ppm | 30% to 60% |
| | Store | 4.68 | 18-25 | 21-24 | ≥40% 300 lux | <10% 1,000 lux | ≤ 1,000 ppm | 30% to 60% |
| | Corridor | 45.24 | - | | | | | |
| **TOTAL/floor** | | **321.31** | | | | | | |
| GROUND FLOOR | Reception | 103.18 | 18 - 25 | 21-24 | ≥55% space 300 lux | <10% space 1,000 lux | ≤ 1,000 ppm | 30% to 60% |
| | Entrance lobby | 14.1 | - | - | ≥40% 300 lux | <10% 1,000 lux | ≤ 1,000 ppm | - |
| | Manager's office | 4.68 | 18 - 25 | 21-24 | ≥55% space 300 lux | <10% space 1,000 lux | ≤ 1,000 ppm | 30% to 60% |
| | Socialising area | 85.58 | 18 - 25 | 21-24 | ≥55% space 300 lux | <10% space 1,000 lux | ≤ 1,000 ppm | 30% to 60% |
| | Stair 1 | 2 | 18-25 | - | ≥40% 300 lux | <10% 1,000 lux | ≤ 1,000 ppm | 30% to 60% |
| | Stair 2 with lobby | 13.93 | 18-25 | - | ≥40% 300 lux | <10% 1,000 lux | ≤ 1,000 ppm | 30% to 60% |
| | Bicycle store | 37.46 | 18-25 | - | | | ≤ 1,000 ppm | - |
| **G. TOTAL** | | **260.93** | | | | | | |
| BASEMENT | services | 200 | - | | | | | |

Sources: LEED version 4.1
ASHRAE 90.1
CIBSE GUIDE A

### TOTAL AREA OF BUILDING

TYPICAL FLOOR PLAN AREA = 321.31 msq. X 3 = 963.93 msq.
GROUND FLOOR = 260.93 msq.

**TOTAL AREA OF BUILDING = 1,224.86 msq.**

### ENERGY USE INTENSITY TARGET

1.Seattle City Council: EUI target of average buildings = 90 kWh/$m^2$/yr

2.LEED version 4.1(refer ASHRAE 90.1) : 60-80 kWh/$m^2$/yr

### TOTAL ENERGY USE TARGET

Annual energy use = EUI x floor area

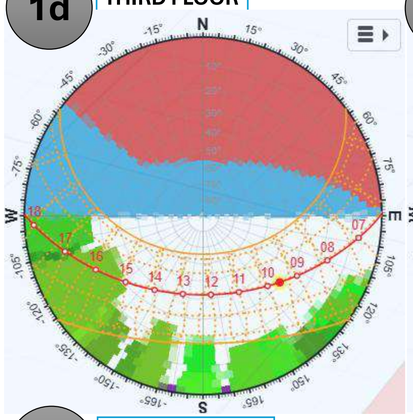
If EUI = 60 kWh/$m^2$/yr

60 kWh/$m^2$/yr x 1224.86 $m^2$ = **73,469.6 kWh**

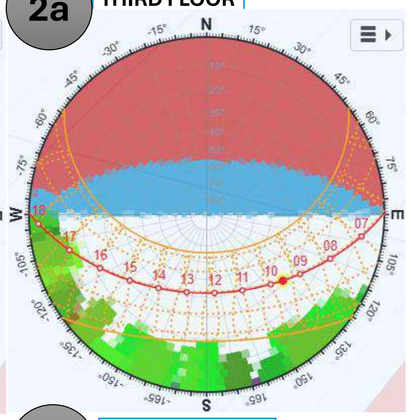
# Sustainable Design of Students' Accommodation Block



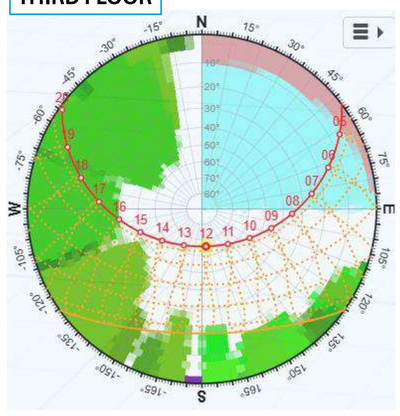
1d THIRD FLOOR



2a THIRD FLOOR



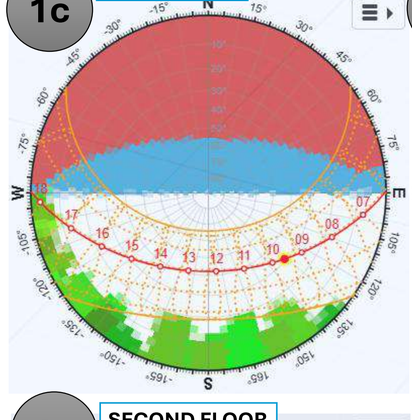
THIRD FLOOR



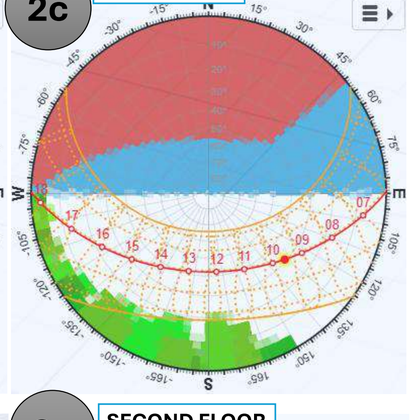
## 4.0 SHADING MASK

- Shading required for first, second and third floors in summer months except the west which receives shading from the trees from 14:00.
- All floors receive winter sun available for passive solar heating.

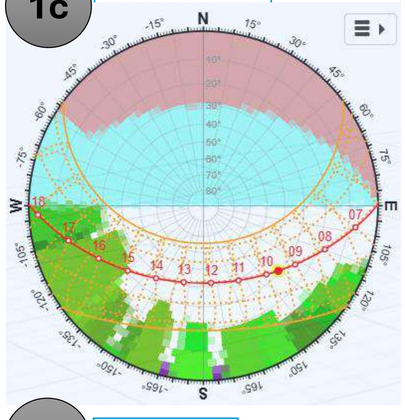
1c THIRD FLOOR



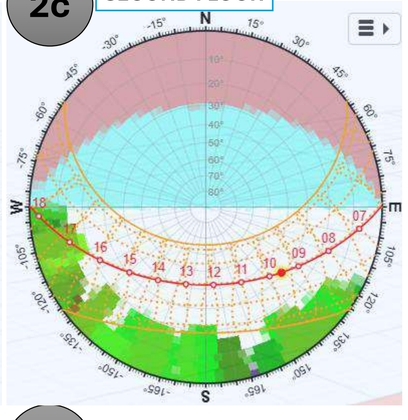
2c THIRD FLOOR

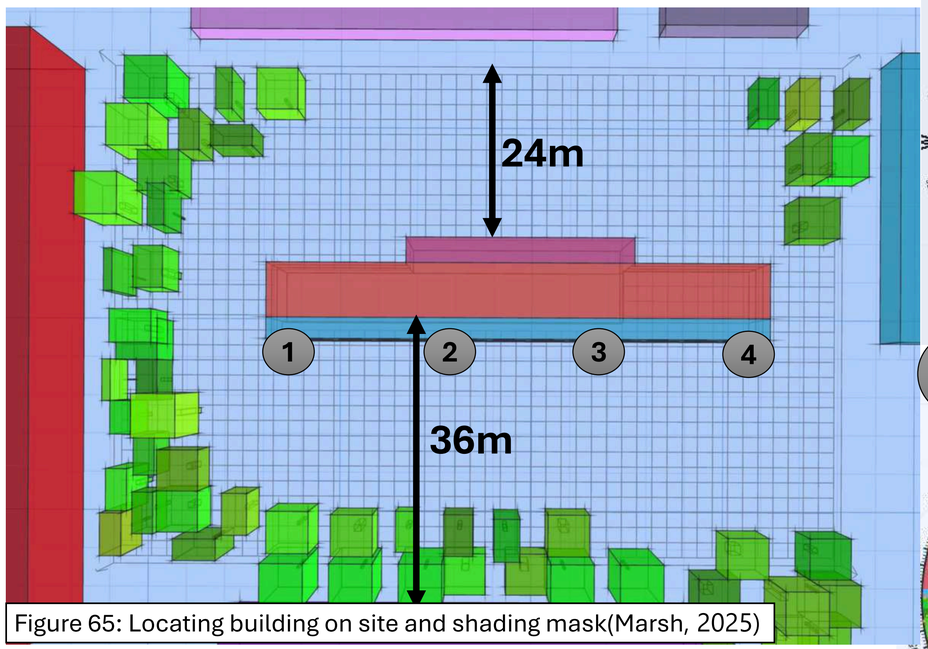


1c SECOND FLOOR

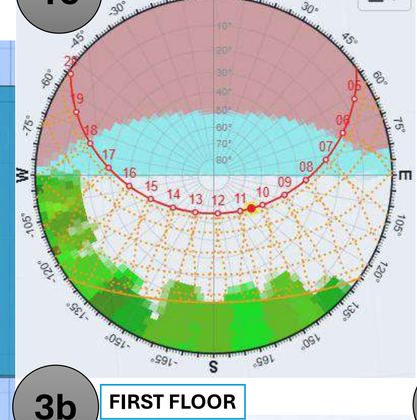


2c SECOND FLOOR

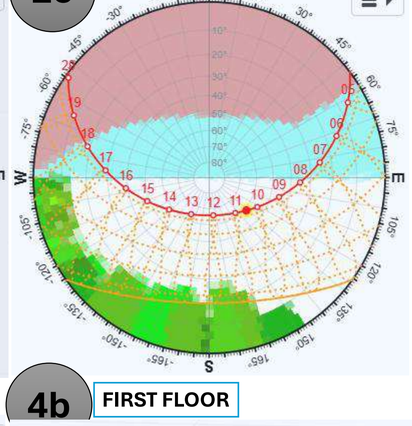




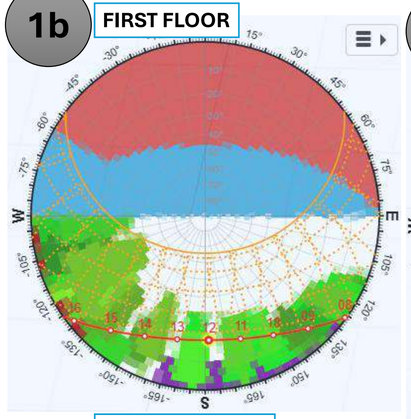

1c SECOND FLOOR



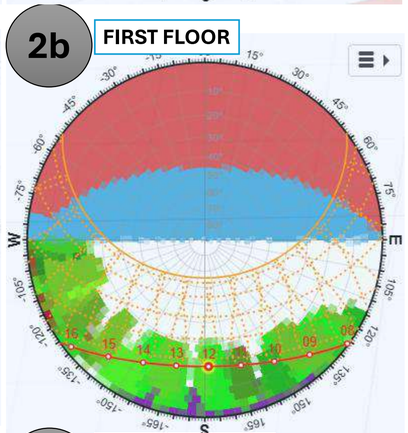
2c SECOND FLOOR



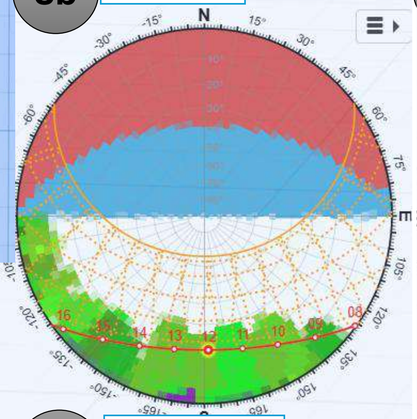
1b FIRST FLOOR



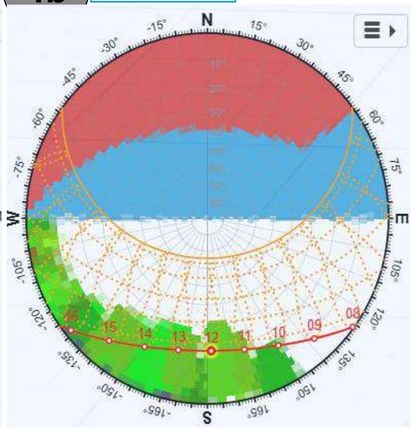
2b FIRST FLOOR



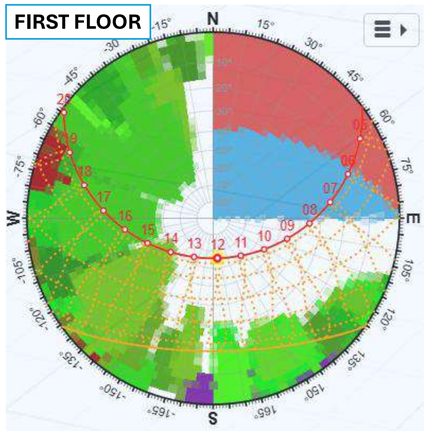
3b FIRST FLOOR



4b FIRST FLOOR



FIRST FLOOR



SECOND FLOOR

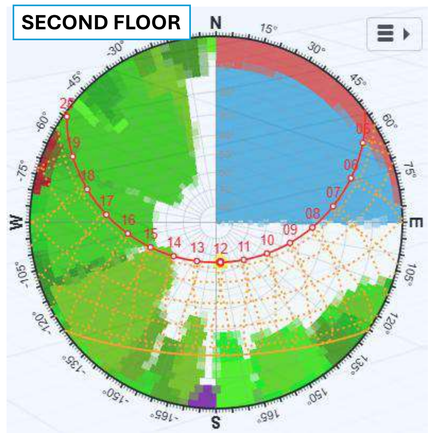
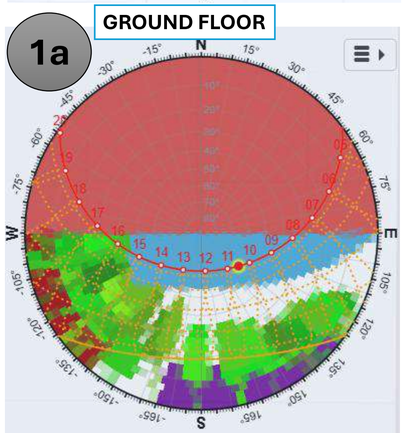
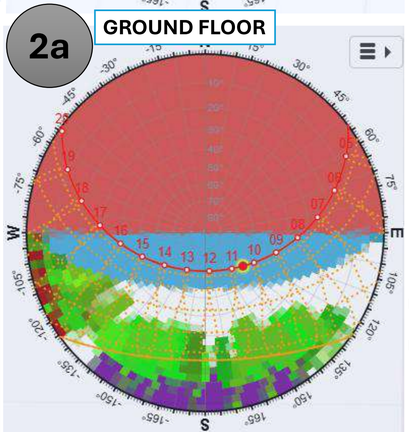


Figure 65: Locating building on site and shading mask(Marsh, 2025)

1a GROUND FLOOR

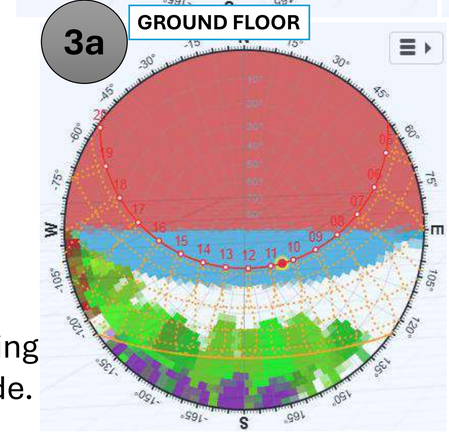


2a GROUND FLOOR



- Ground floor shaded by 1st floor cantilever in summer.
- Ground floor - winter sun available for passive solar heating except from 8-9am when south obstruction shades façade.

3a GROUND FLOOR



- Trees on site are a mixture of evergreen and deciduous trees.
- In winter, the sun will come through to the building when the trees shed their leaves.

## 4.1 SITE OPPORTUNITIES AND RESTRICTIONS

**Simple and compact form**: fewer thermal bridges and to minimize surface area of building to reduce heat loss.

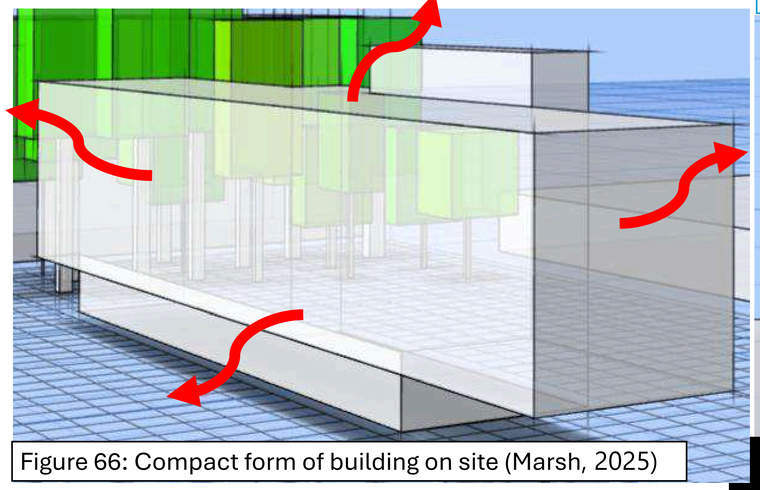
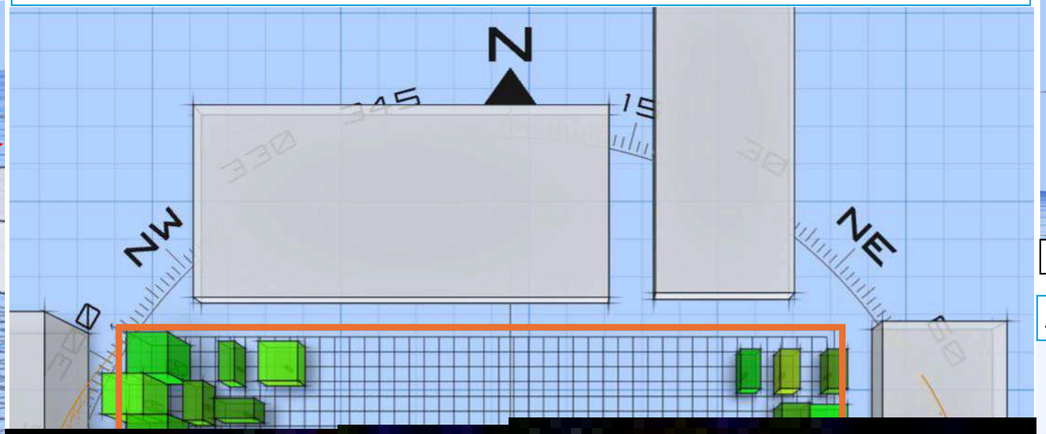


Figure 66: Compact form of building on site (Marsh, 2025)

**BUILDING FORM :** Linear building to complement the shape of site.

Site gives an opportunity for the longest side of the building to face south for passive solar heating in winter.




**MAJOR CONSTRAINT**: Building located at the south obstructs winter sun if building is close.

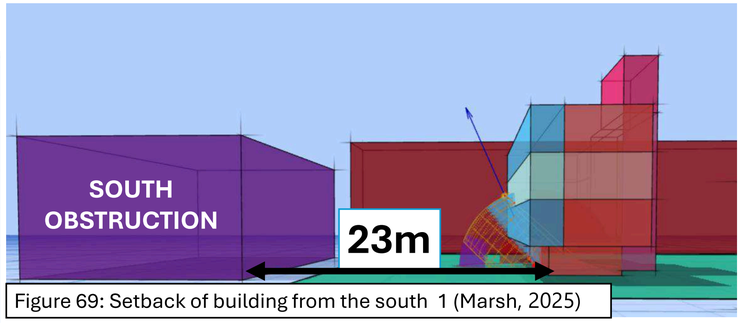
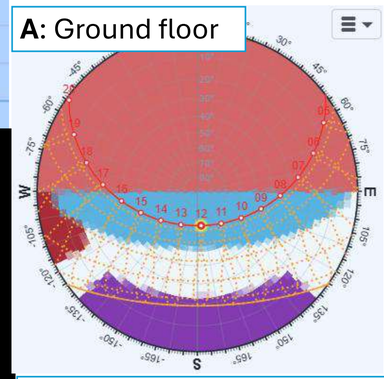
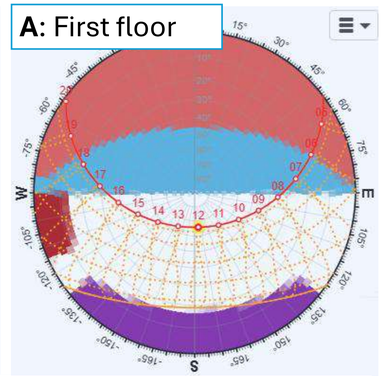



Figure 69: Setback of building from the south 1 (Marsh, 2025)




**South**

Obstruction prevents winter sun on the ground floor for all hours.

First floor receives winter sun from11am-13:00.

**Recommendation**:

**Move building 36m away from south obstruction till ground floor receives winter sun.**

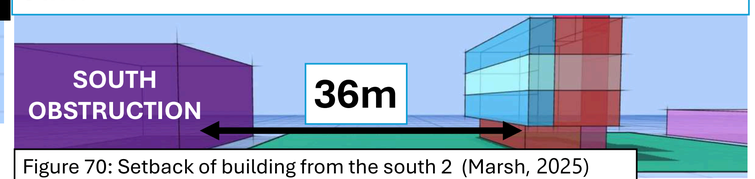



Figure 70: Setback of building from the south 2 (Marsh, 2025)

**West**

Trees around the site completely shades the west in Summer from 13.30 on ground floor so no shading required for that floor. The need for shading is for second and third floor from 14:00 and 14:30 in the afternoon.

Trees around the site completely shading the west will aid passive radiative cooling at night.

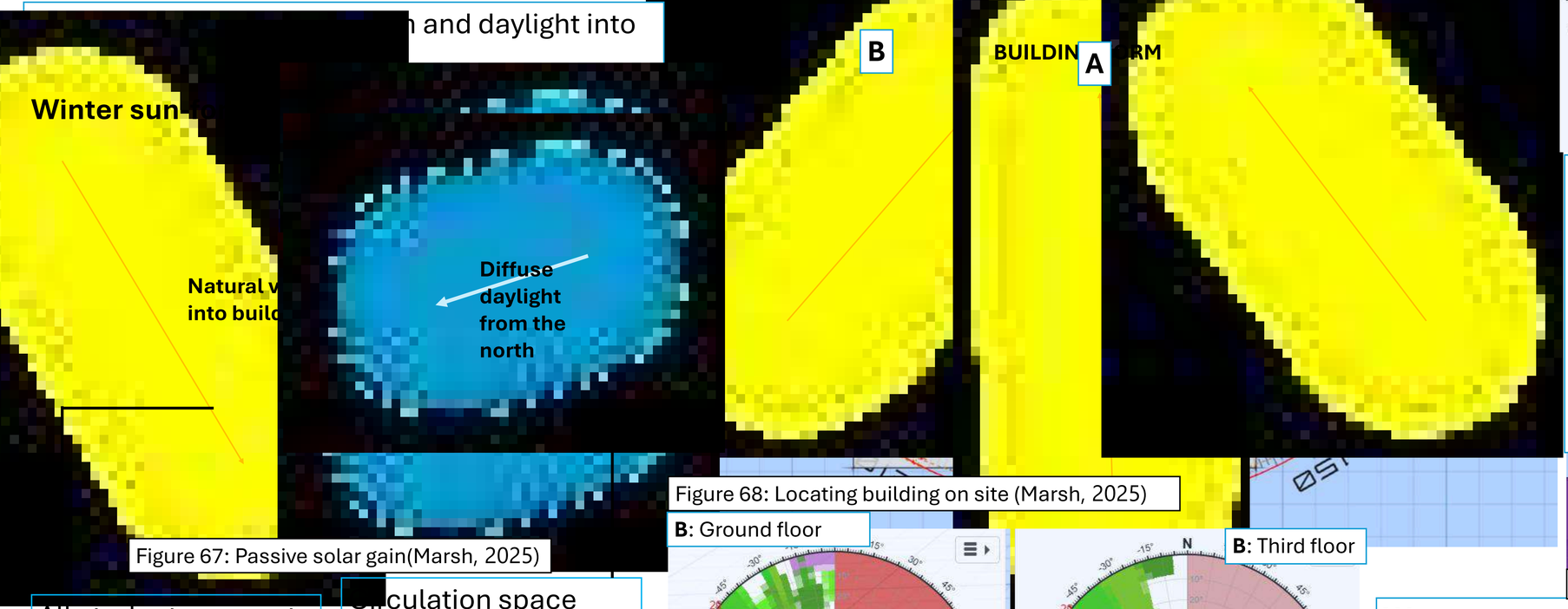
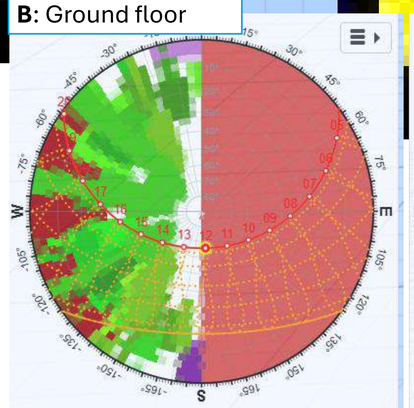


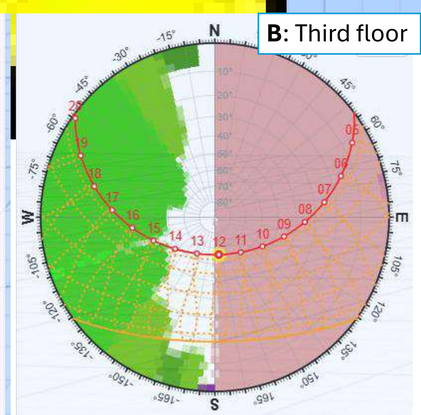

Figure 67: Passive solar gain(Marsh, 2025)

Figure 68: Locating building on site (Marsh, 2025)

**B**: Ground floor



**B**: Third floor



All students rooms to face South

- Sleeping and living area

Circulation space And less occupied spaces to north

- Entrance
- Laundry
- Stair and lift
- corridor
- Store

## 5.1 PASSIVE HEATING AND COOLING STRATEGIES

### 5.1.1 SUN SPACE + THERMAL MASS

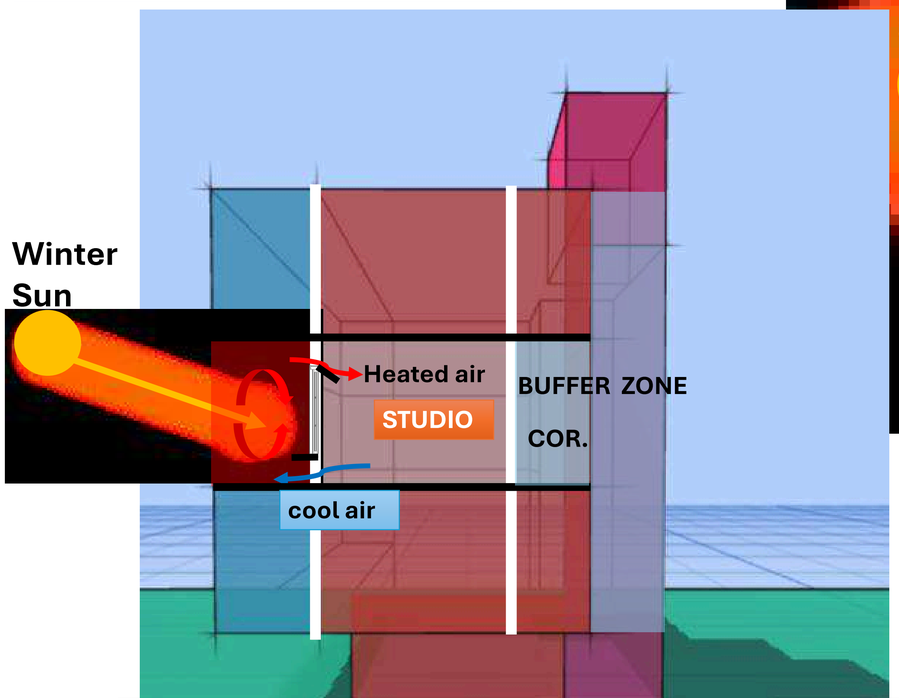



Figure 71: Sunspace and thermal mass (Marsh, 2025 with modifications by author)

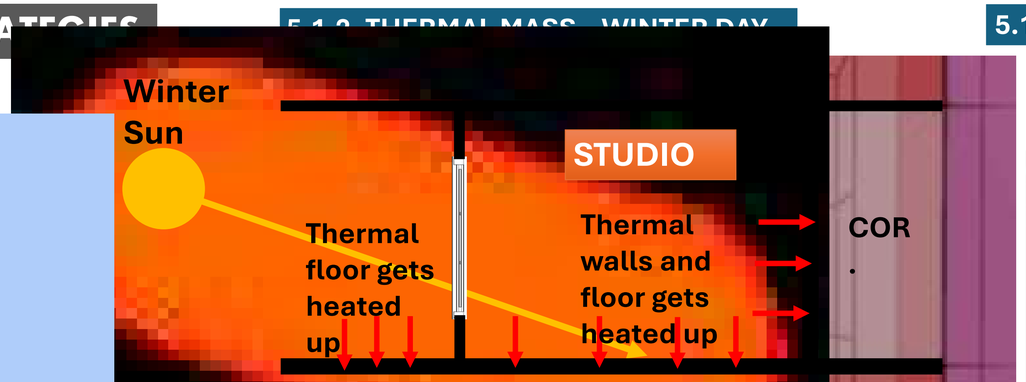



Figure 72: Thermal mass – winter day (Marsh, 2025 with modifications by author)

### 5.1.3. THERMAL MASS – WINTER NIGHT

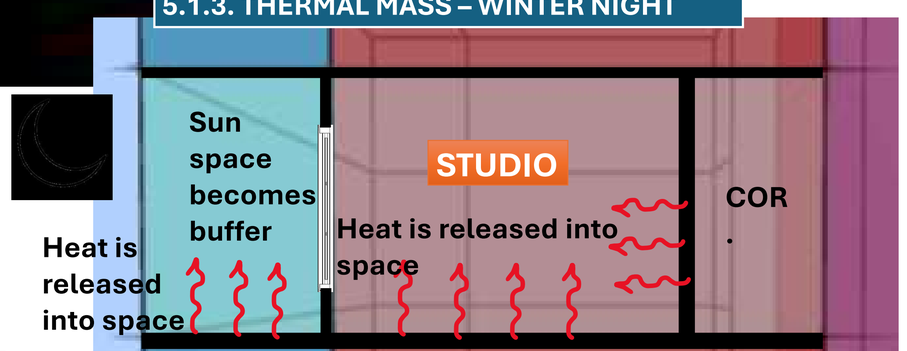



Figure 73: Thermal mass – winter night (Marsh, 2025 with modifications by author)

### 5.1.4.SUN SPACE – SUMMER VENTILATION AND COOLING

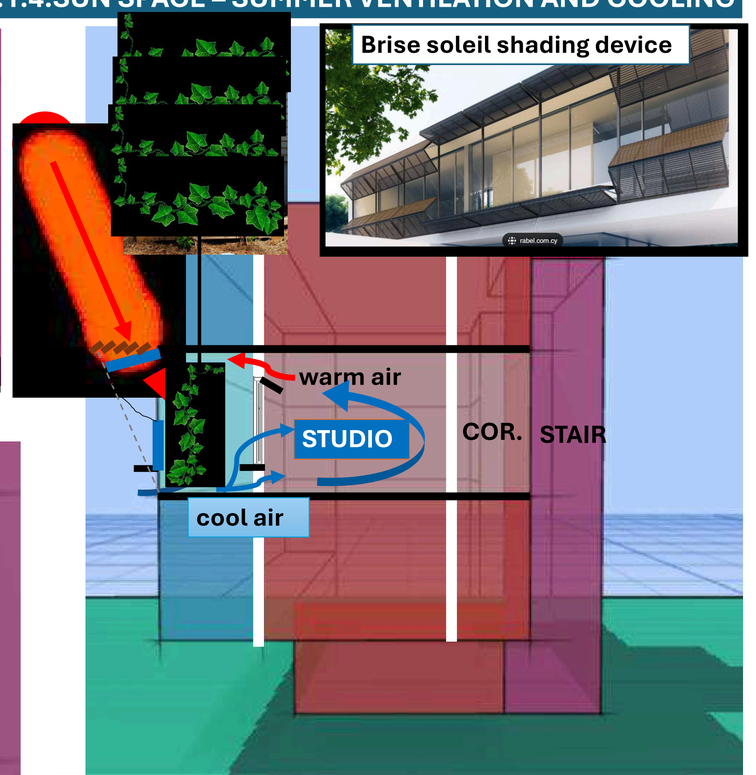



Figure 74: Summer cooling (Marsh, 2025 with modifications by author)

### 5.1.5 CORRIDOR VENTILATION

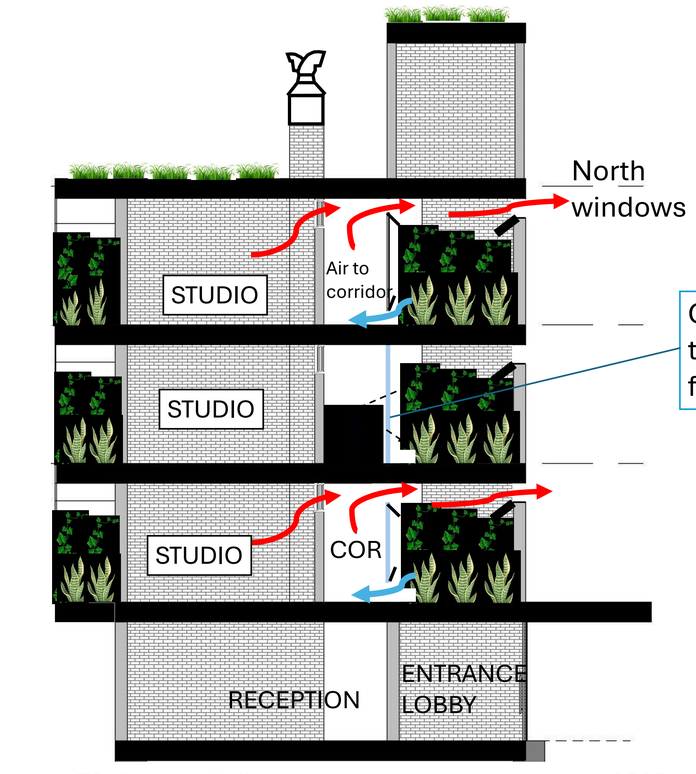
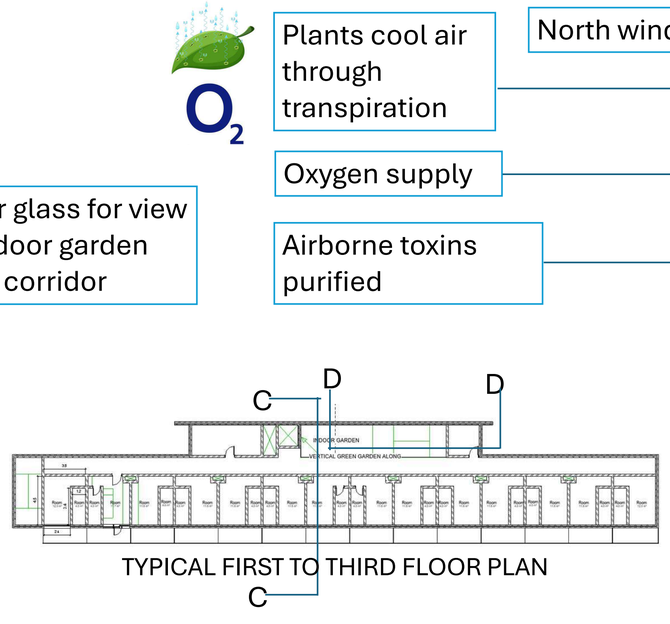



Figure 75: Section C-C : Ventilation of corridor (Author, 2025)




### 5.1.6.STAIR AND INDOOR GARDEN

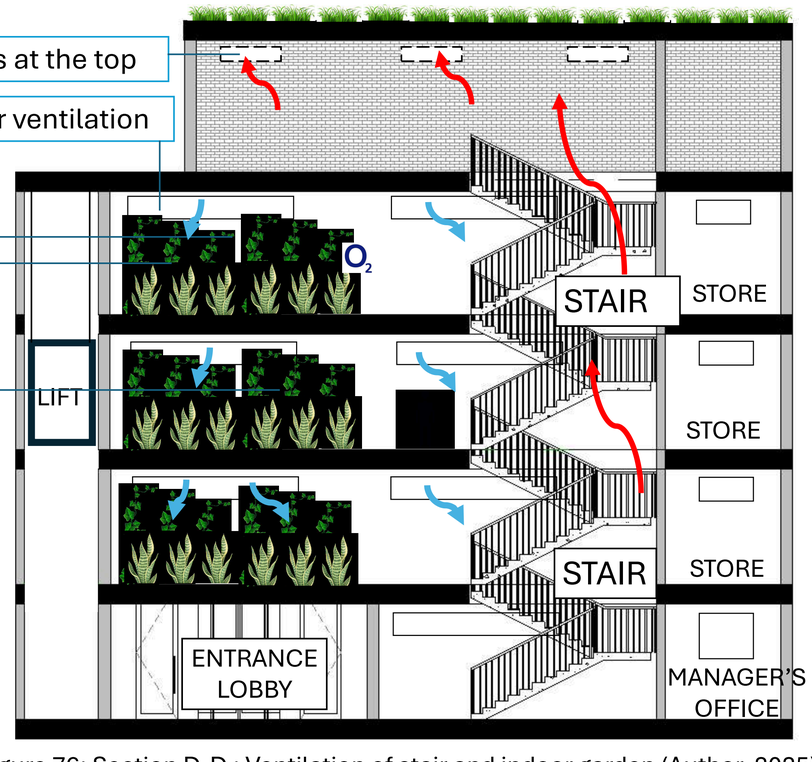



Figure 76: Section D-D : Ventilation of stair and indoor garden (Author, 2025)

## 5.2 PASSIVE COOLING STRATEGIES

### 5.2.1. CROSS VENTILATION




Typical first to third floor plan

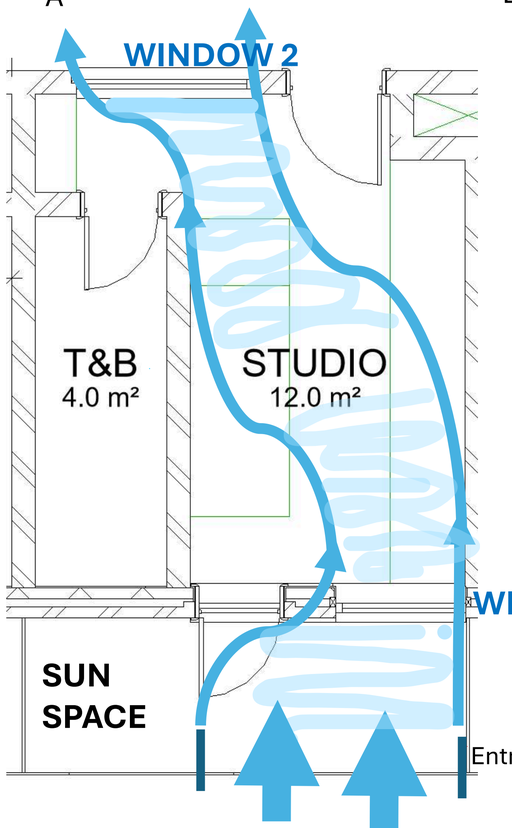



Figure 77: Typical Floor Plan Of Studio (Author, 2025)

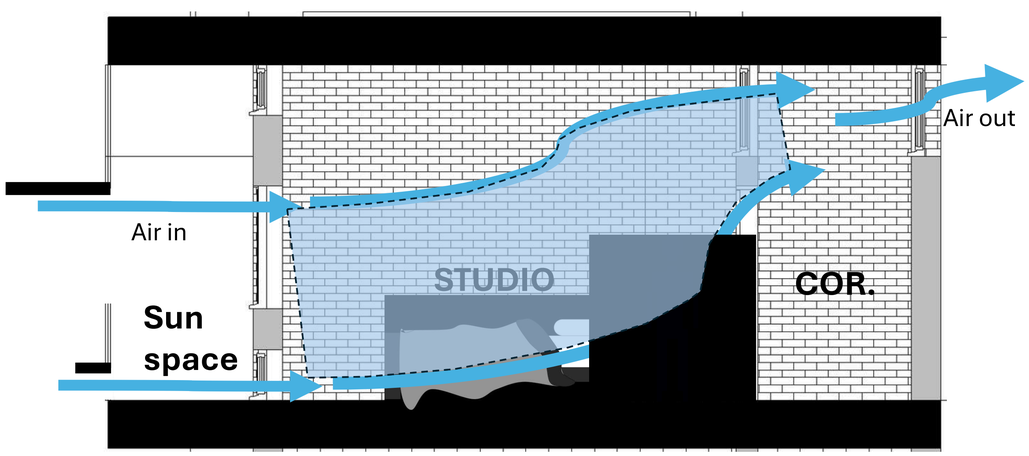



Figure 78: Section A-A – This is a typical windy day (Author, 2025)

### 5.2.2 EVAPORATIVE COOLING IN SUMMER

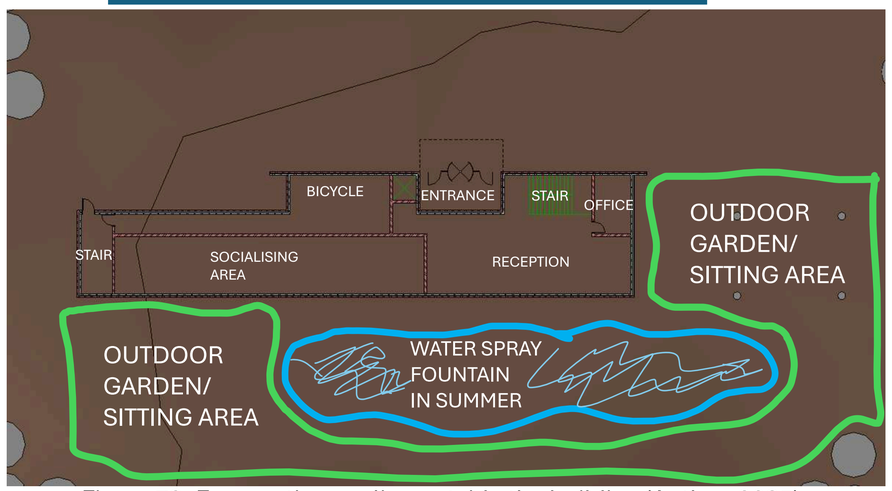



Figure 79: Evaporative cooling outside the building (Author, 2025)

Evaporative cooling with water spray and vegetation to change microclimate of the site in summer to be cooler.

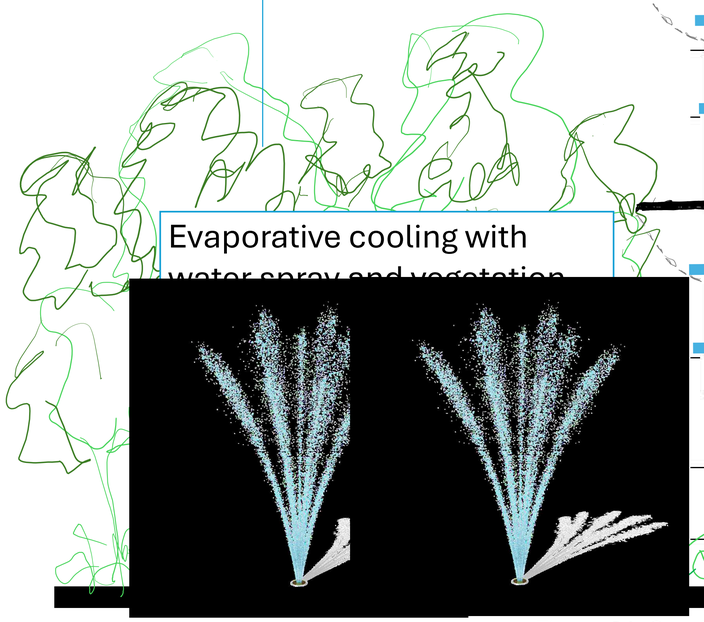


Plants cool air through transpiration

$O_2$

Oxygen supply

Airborne toxins purified




### 5.2.3. WIND DRIVEN VENTILATION – WIND COWLS



Wind Cowls (Inhabitat, 2008)

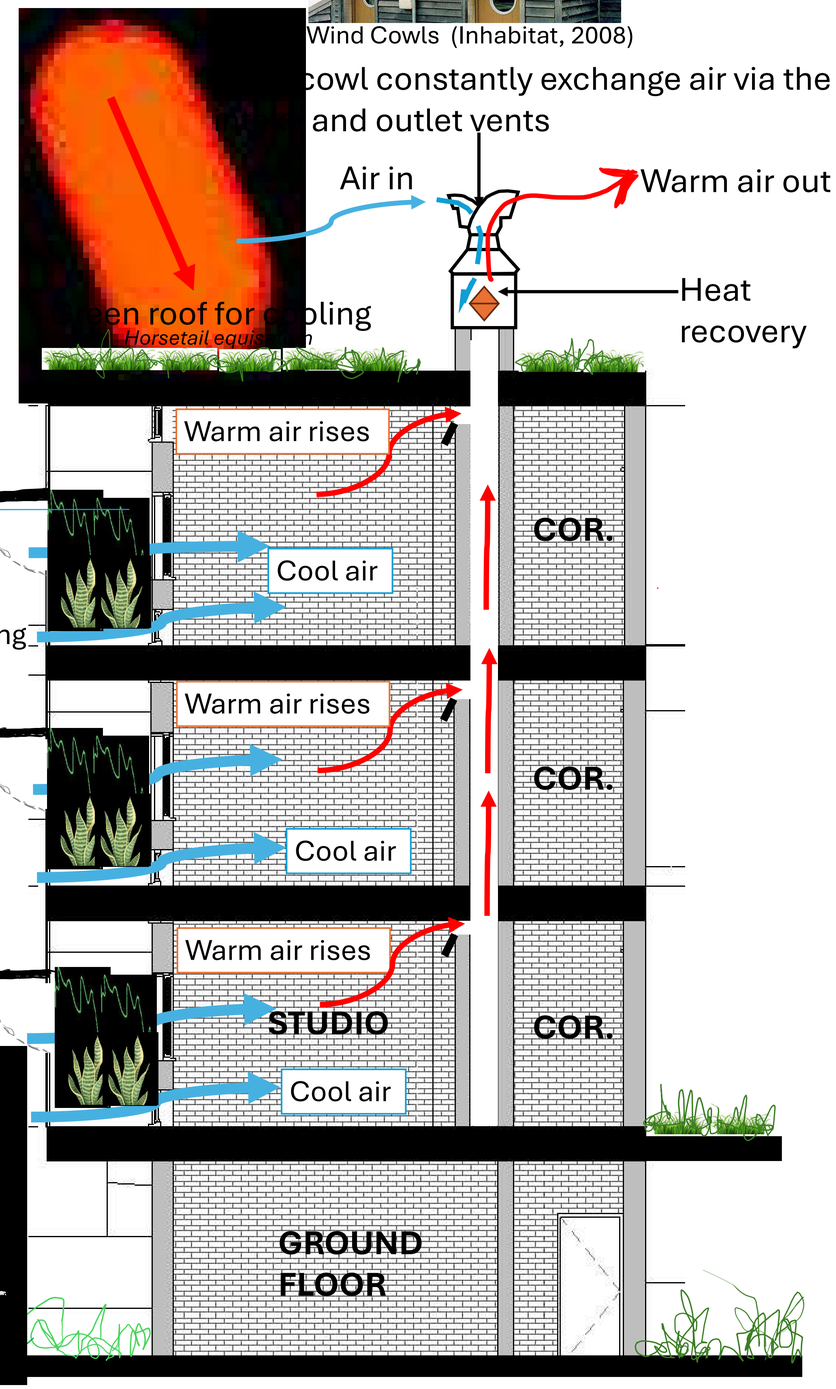



Figure 81: Section B-B – This is a typical still day (Author, 2025)

## 6.0 FLOOR PLANS

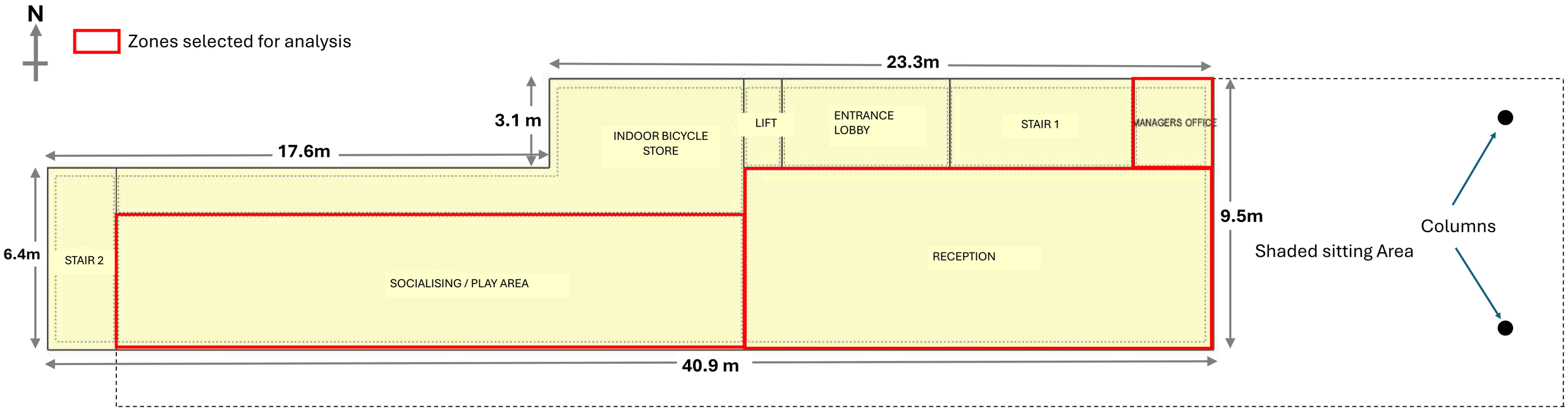



Figure 82: GROUND FLOOR PLAN (Author, 2025)

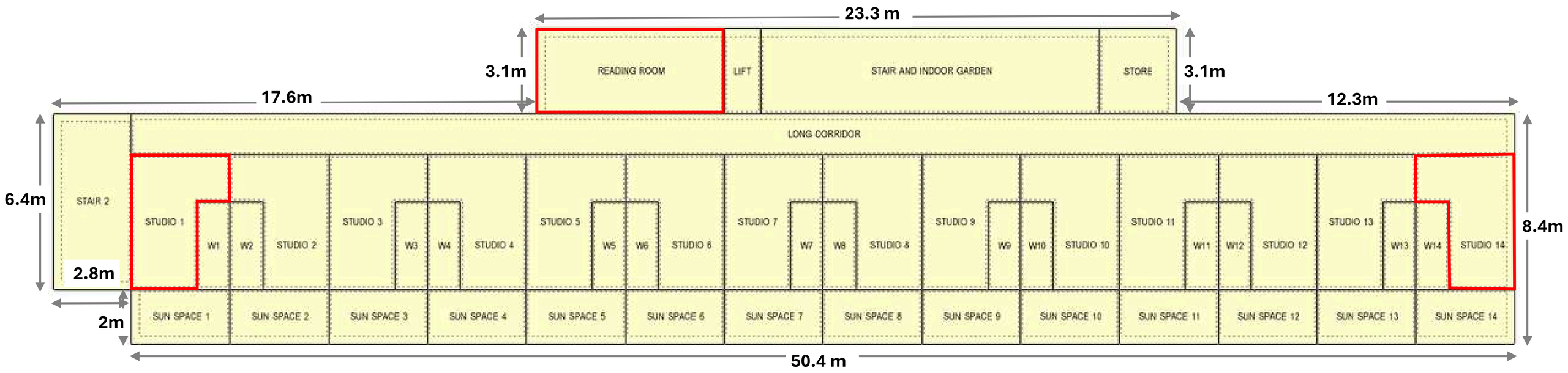



Figure 83: TYPICAL FIRST TO THIRD FLOOR PLAN (Author, 2025)

## 6.1 FINAL MODEL OF STUDENTS ACCOMMODATION BLOCK

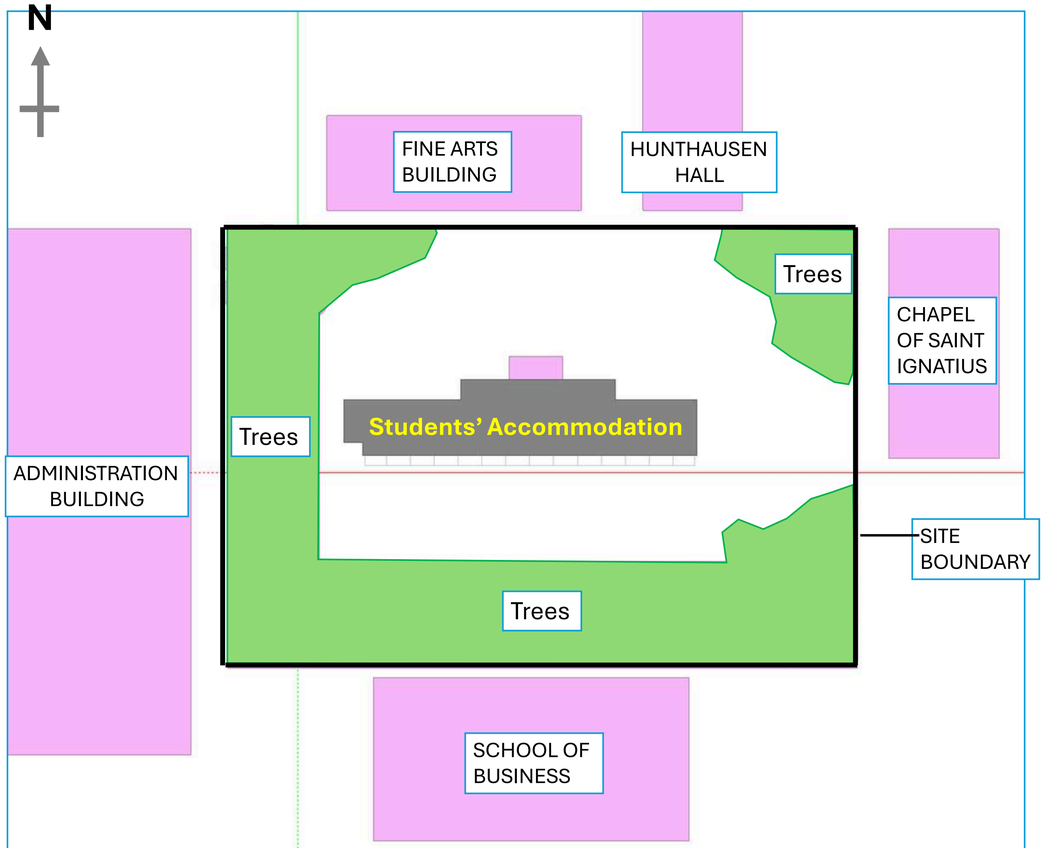



Figure 84: Site Layout of Students accommodation block (Author, 2025)

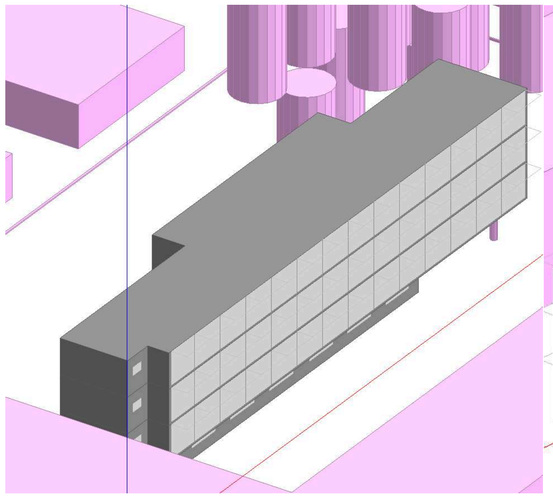


Figure 85: South-West View (Author, 2025)

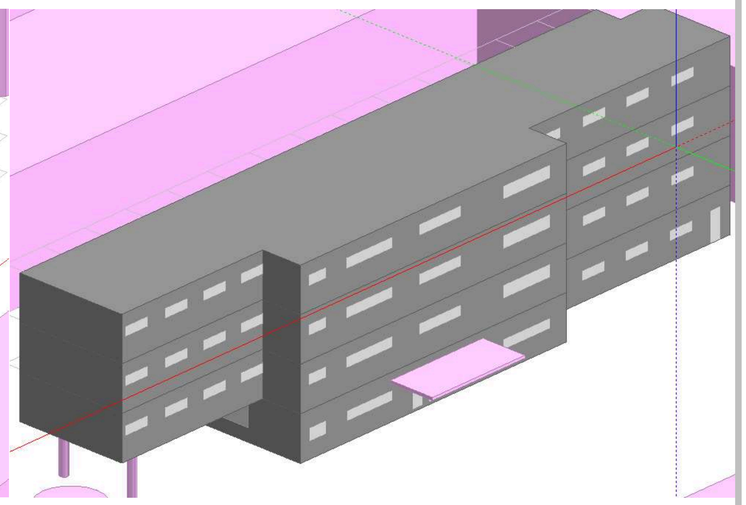


Figure 86: North-East View (Author, 2025)

## 6.2 THREE-DIMENSIONAL IMPRESSIONS OF BUILDING

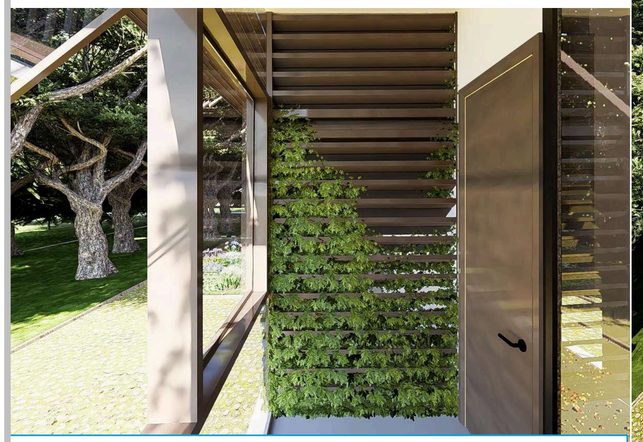


Figure 87 : View from sun space towards southern side (Author, 2025)

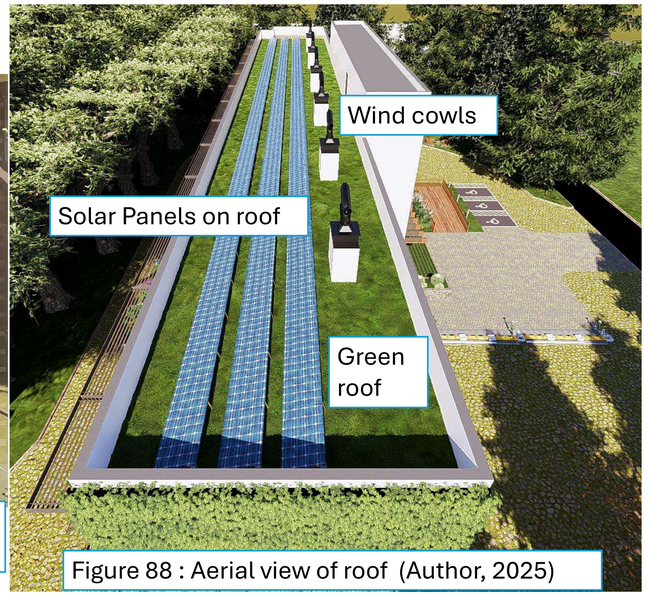



Figure 88 : Aerial view of roof (Author, 2025)




Figure 89 : South-East view of building (Author, 2025)



Figure 90 : North-West view of building (Author, 2025)

## 6.3 INTERIOR DETAILS OF STUDIO APARTMENT

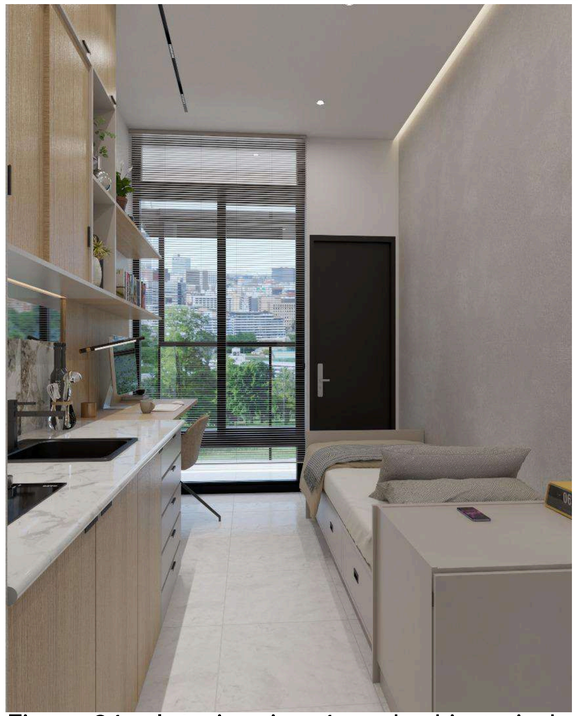


Figure 91a: Interior view 1overlooking window and door to sun space (Author, 2025)

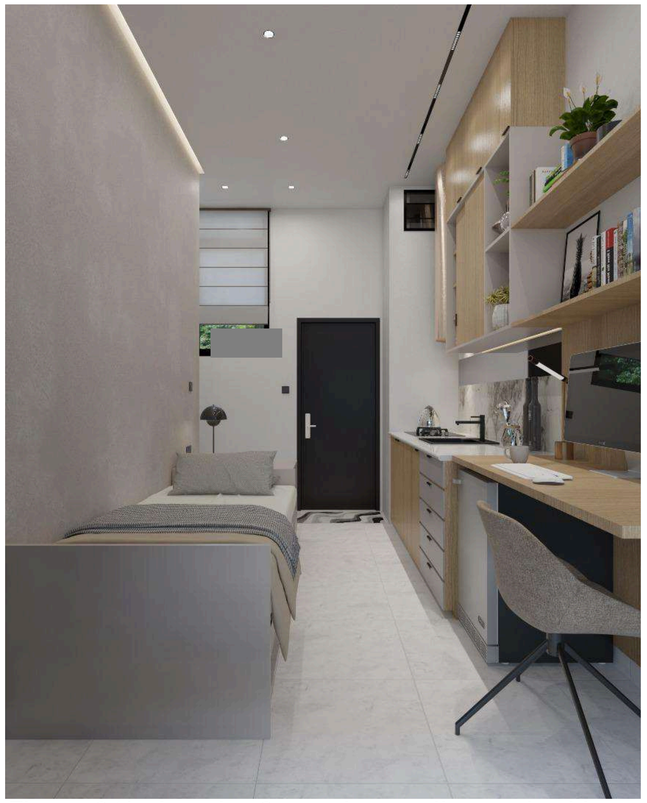


Figure 91b: Interior 2 – entire studio overlooking Sleeping and study area (Author, 2025)

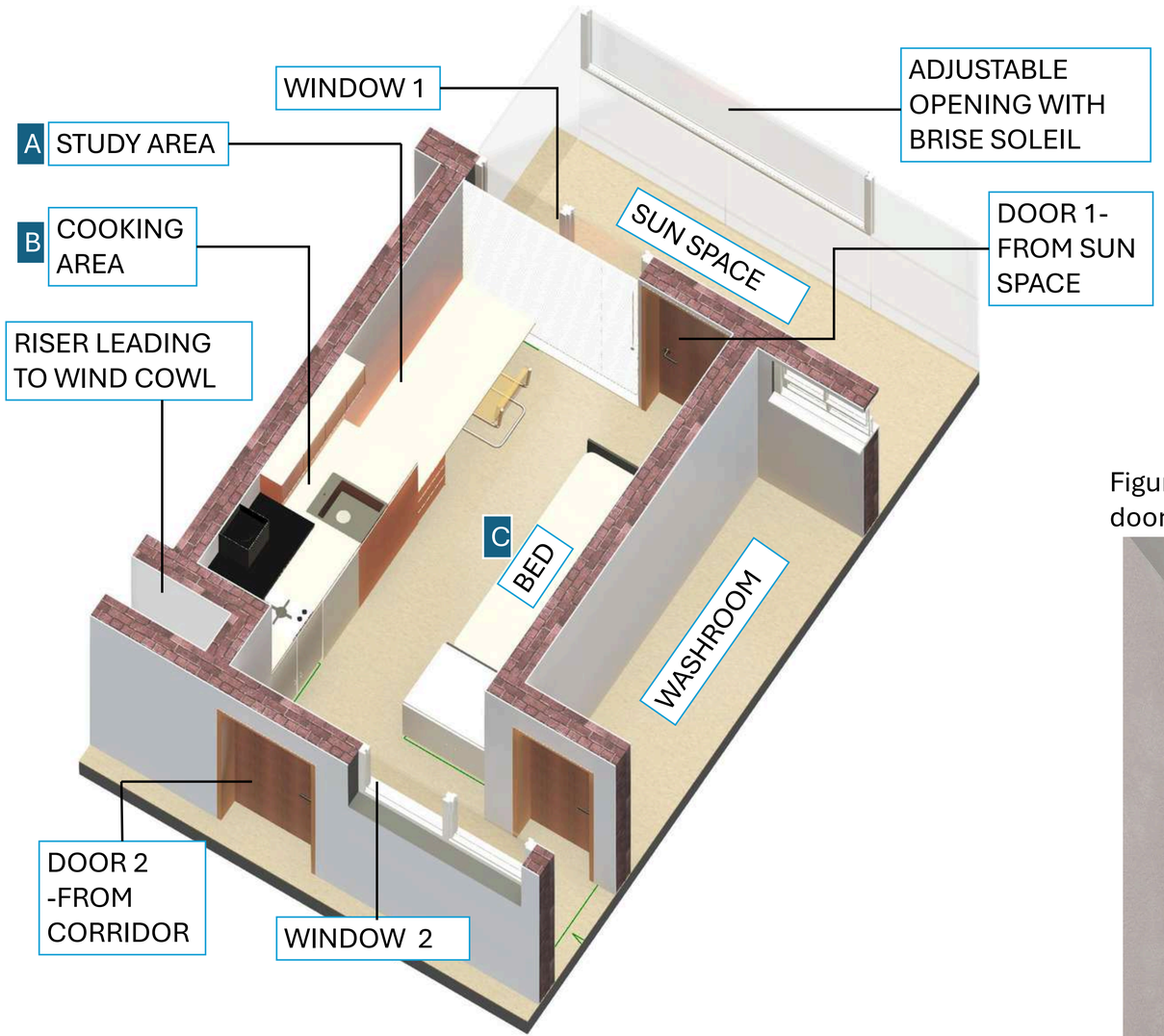



Figure 91: Axonometric view of a Typical Studio Apartment (Author, 2025)

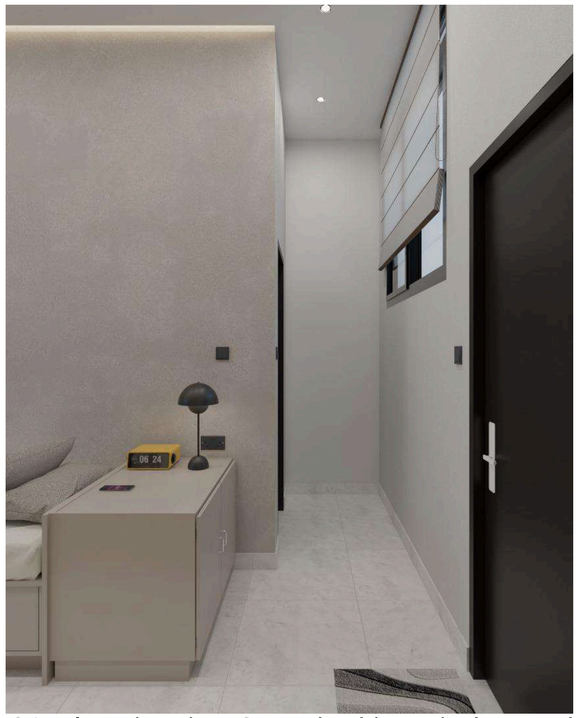


Figure 91c: Interior view 3 overlooking window and door leading to corridor (Author, 2025)

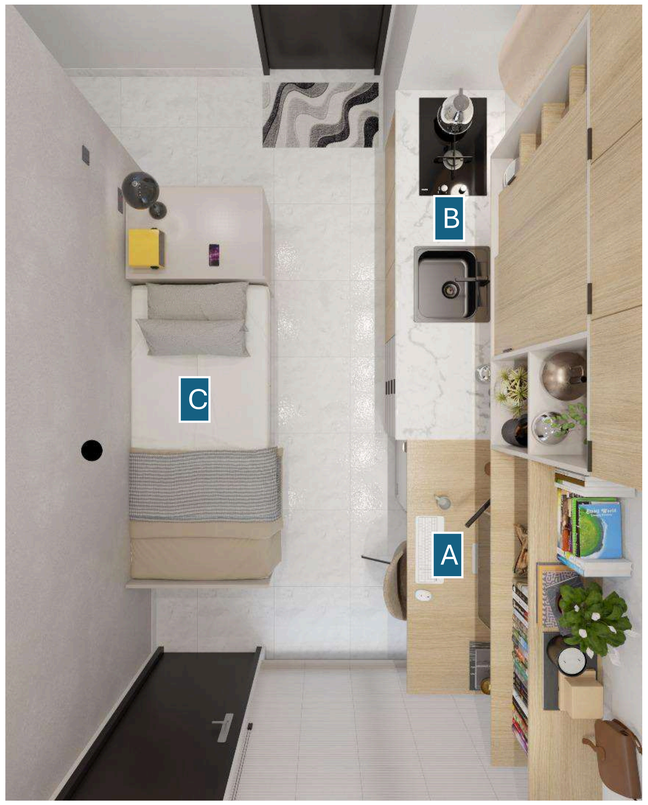
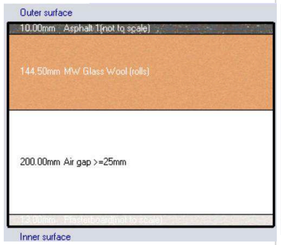



Figure 91d: Aerial view of Studio (Author, 2025)

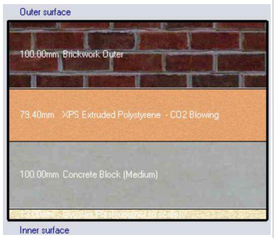
## 7.1.1 BASE CASE – ITERATION 1




**Roof (367.5mm)**

10mm asphalt
144.5mm glass wool
200mm air gap
13mm plasterboard

**U-value(0.25)**




**External wall (292mm)**

100mm brickwork
79.4mm XPS extruded polystyrene insulation
100mm concrete block (medium)
13mm gypsum plastering

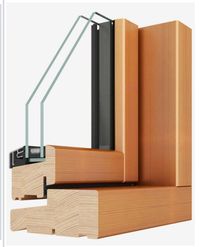
**U-value(0.351)**




**Ground floor slab (300mm)**

100mm UREA fomaldehyde foam
100mm cast concrete
70mm floor screed
30mm timber flooring

**U-value(0.314)**



**Window:Double Clear** 3mm/13mmAir
Frame:painted wood

G-value: 0.764
VT: 0.812

**U-value(2.7)**

Table 3 : WINDOW SIZES (NO SHADING)

| ZONE | WINDOW SIZE | NUMBER | LOCATION |
|---|---|---|---|
| READING ROOM | 1m x 0.7m | 2 | North |
| STUDIO 1 | 1m x 0.9m | 1 | South |
| STUDIO 14 | 1m x 0.9m | 1 | South |
| RECEPTION | (South)1.5mx1m ; 1.5x1(East) | (South)4 ; (East)1 | South & East |
| MANAGER'S OFFICE | 1m x 0.7m | 1 | North |
| SOCIALISING AREA | 1.5m | 5 | South |

**RESULT:** BASE CASE PASSES TM AND ASE BUT EXTREMELY LOW SDA VALUES

ACHIEVE DAYLIGHTING TARGET BY INCREASING WINDOW SIZES AND CHANGING GLASS TYPE

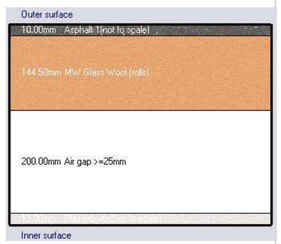
BASE CASE PASS CIBSE TM59

| Zone | Criterion A (%) | Criterion B (hr) | Pass/Fail |
|---|---|---|---|
| MANAGERSOFFICE | 0.00 | N/A | Pass |
| RECEPTION | 0.28 | N/A | Pass |
| SOCIALISINGAREA | 0.00 | N/A | Pass |
| READINGROOM | 0.00 | 0.00 | Pass |
| STUDIO1 | 0.00 | 0.00 | Pass |
| STUDIO14 | 0.00 | 0.00 | Pass |

BASE CASE PASS ASE BUT NOT SDA

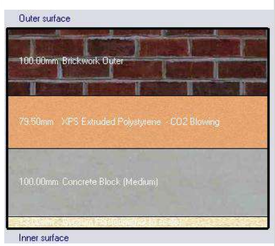
| Zone | Floor area (m2) | sDA area in range (m2) | sDA % in Range% | ASE area in range (m2) | ASE % in range |
|---|---|---|---|---|---|
| READING ROOM | 17.7 | 0.00 | 0.00 | 17.66 | 100.0 |
| STUDIO 1 | 12.5 | 0.00 | 0.00 | 12.55 | 100.0 |
| STUDIO 14 | 11.5 | 0.11 | 0.95 | 11.51 | 100.0 |
| RECEPTION | 113.7 | 56.42 | 49.63 | 113.49 | 99.8 |
| MANAGERS OFFICE | 6.6 | 0.00 | 0.00 | 6.65 | 100.0 |
| SOCIALISING AREA | 95.8 | 8.76 | 9.14 | 95.74 | 99.9 |
| | **257.9** | **65.28** | **25.32** | **257.59** | **99.9** |

## 7.1.2 ITERATION 2




**Roof (367.5mm)**
10mm asphalt
144.5mm MW glass wool
200mm air gap
13mm plasterboard

**U-value(0.25)**



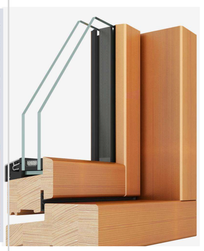

**External wall (292mm)**
100mm brickwork
79.5mm XPS insulation
100mm concrete block
13mm gypsum plastering

**U-value(0.350)**




**Ground floor slab (300mm)**
30mm timber flooring
70mm floor screed
100mm cast concrete
132.7mm UREA fomaldehyde
Foam

**U-value(0.25)**



**Window:Double Clear**
3mm/13mmAir
Frame:painted wood
G-value: 0.691
VT: 0.744

**U-value(1.96)**

Table 4: WINDOW SIZES ( 1m SHADING)

| ZONE | WINDOW SIZE | NUMBER | LOCATION |
|---|---|---|---|
| READING ROOM | 1mx0.9 | 4 | North |
| STUDIO 1 | 1.13mx3.1m | 1 | South |
| STUDIO 14 | 1.13mx3.1m | 1 | South |
| RECEPTION | (S)1.95mx1.78m ; 1.9mx1.65m(E) | (S)8 ; (E)2 | South & East |
| MANAGER'S OFFICE | 1.7m x 0.9m | 1 | North |
| SOCIALISING AREA | 1.9mx2.1m | 9 | South |

**RESULT:** OVERHEATING OF SPACES AFTER INCREASING WINDOW SIZES. IMPROVED SDA BUT ASE SHOULD BE IMPROVED

ADDRESS OVERHEATING AND IMPROVE ASE. INCREASE U-VALUE OF BUILDING FABRIC.
ADDRESS OVERHEATING, INCREASE WINDOW SHADING AND CHANGE GLASS TYPE TO IMPROVE ASE.

ITERATION 2 – TWO SPACES FAIL CIBSE TM59

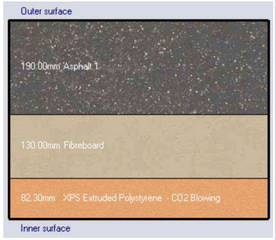
| Zone | Criterion A (%) | Criterion B (hr) | Pass/Fail |
|---|---|---|---|
| MANAGERSOFFICE | 0.00 | N/A | Pass |
| RECEPTION | 5.68 | N/A | Fail |
| SOCIALISINGAREA | 6.01 | N/A | Fail |
| READINGROOM | 0.00 | 4.00 | Pass |
| STUDIO1 | 0.57 | 112.50 | Fail |
| STUDIO14 | 0.00 | 6.00 | Pass |

ITERAION 2 PASS BOTH ASE AND SDA

| Zone | Floor area (m2) | sDA area in range (m2) | sDA % in Range% | ASE area in range (m2) | ASE % in range |
|---|---|---|---|---|---|
| READING ROOM | 17.7 | 14.37 | 81.38 | 17.66 | 100.0 |
| STUDIO 1 | 12.5 | 9.10 | 72.50 | 11.08 | 88.3 |
| STUDIO 14 | 11.5 | 7.68 | 66.67 | 10.09 | 87.6 |
| RECEPTION | 113.7 | 113.21 | 99.59 | 101.29 | 89.1 |
| MANAGERS OFFICE | 6.6 | 4.77 | 71.83 | 6.65 | 100.0 |
| SOCIALISING AREA | 95.8 | 95.17 | 99.30 | 40.73 | 42.5 |
| | **257.9** | **244.30** | **94.73** | **187.51** | **72.7** |

Source of all images and tables : DesignBuilder version 7 and EnergyPlus

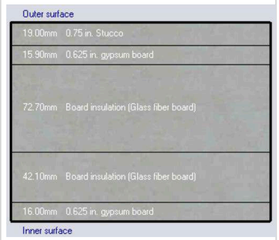
## 7.1.3 ITERATION 3




**Roof (402mm)**

190mm asphalt
130mm fibreboard
82.3mm XPS

**U-value(0.20)**




**External wall (166mm)**

19mm Stucco
16mm Gypsumboard
72.7mm glass fibre insulation
42.1mm glass fibre insulation
16mm gypsumboard

**U-value(0.28)**




**Ground floor slab (358mm)**

50mm floor screed
182.4mm EPS insul.
100mm concrete
25mm brick slips
Clay underfloor

**U-value(0.20)**



**Window: LoE Clr**
3mm double – 13mm Air

G-value: 0.65
VT: 0.76

**U-value(1.78)**

Table 5: WINDOW SIZES ( 1.5m SHADING)

| ZONE | WINDOW SIZE | NUMBER | LOCATION |
|---|---|---|---|
| READING ROOM | 1mx0.9 | 4 | North |
| STUDIO 1 | 1.13mx3.1m | 1 | South |
| STUDIO 14 | 1.13mx3.1m | 1 | South |
| RECEPTION | (S)1.95mx1.78m ; 1.9mx1.65m(E) | (S)8 ; (E)2 | South & East |
| MANAGER'S OFFICE | 1.7m x 0.9m | 1 | North |
| SOCIALISING AREA | 1.9mx2.1m | 9 | South |

**RESULT:** SDA ACHIEVED BUT ASE IN SOME SPACES HAS TO BE IMPROVED AND OVERHEATING REMEDIED FOR RECEPTION

IMPROVE U-VALUE OF FABRIC AND CHANGE GLASS TYPE

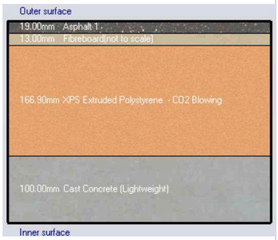
ITERATION 3 – ONE SPACE FAIL CIBSE TM59

| Zone | Criterion A (%) | Criterion B (hr) | Pass/Fail |
|---|---|---|---|
| MANAGERSOFFICE | 0.41 | N/A | Pass |
| RECEPTION | 3.34 | N/A | Fail |
| SOCIALISINGAREA | 2.95 | N/A | Pass |
| READINGROOM | 0.00 | 1.00 | Pass |
| STUDIO1 | 0.00 | 6.50 | Pass |
| STUDIO14 | 0.00 | 2.50 | Pass |

ITERAION 3 PASES BOTH ASE AND SDA

| Zone | Floor area (m2) | sDA area in range (m2) | sDA % in Range% | ASE area in range (m2) | ASE % in range |
|---|---|---|---|---|---|
| READING ROOM | 18.8 | 15.73 | 83.51 | 18.84 | 100.0 |
| STUDIO 1 | 12.5 | 8.05 | 64.17 | 11.08 | 88.3 |
| STUDIO 14 | 12.1 | 6.87 | 56.67 | 10.70 | 88.3 |
| RECEPTION | 117.2 | 112.85 | 96.27 | 99.07 | 84.5 |
| MANAGERS OFFICE | 7.3 | 4.12 | 56.34 | 7.32 | 100.0 |
| SOCIALISING AREA | 98.6 | 95.77 | 97.13 | 36.74 | 37.3 |
| | **266.7** | **243.39** | **91.28** | **183.75** | **68.9** |

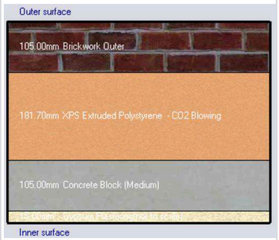
## 7.1.4 ITERATION 4




**Roof (300mm)**

19mm asphalt
13mm fibreboard
167mm XPS
100mm concrete

**U-value(0.18)**




**External wall (407mm)**

105mm brickwork
182mm XPS insulation
105mm concrete
15mm gypsum plaster

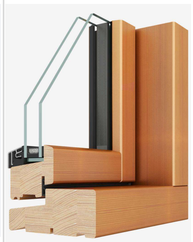
**U-value(0.17)**




**Ground floor slab (440mm)**

30mm timber floor
70mm floor screed
100mm concrete
240mm fomaldehyde foam

**U-value(0.15)**



**Window:LoE Elec Abs bleached 6mm double Argon**

G-value: 0.48
VT: 0.657

**U-value(1.3)**

Table 6: WINDOW SIZES( 1.5m SHADING)

| ZONE | WINDOW SIZE | NUMBER | LOCATION |
|---|---|---|---|
| READING ROOM | 1mx0.9 | 4 | North |
| STUDIO 1 | 1.13mx3.1m | 1 | South |
| STUDIO 14 | 1.13mx3.1m | 1 | South |
| RECEPTION | (S)1.95mx1.78m ; 1.9mx1.65m(E) | (S)8 ; (E)2 | South & East |
| MANAGER'S OFFICE | 1.7m x 0.9m | 1 | North |
| SOCIALISING AREA | 1.9mx2.1m | 9 | South |

RESULT: OVERHEATING SOLVED BUT THE SDA FOR MANAGER'S OFFICE AND ASE FOR SOCIALISING AREA SHOULD BE IMPROVED

IMPROVE U-VALUE OF FABRIC TO IMPROVE CIBSE VALUES FOR RECEPTION, SOCIALISING AREA AND STUDIO 1.
CHANGE WINDOW SIZES FOR MANAGER'S OFFICE AND SOCIALISING AREA.
CHANGE GLASS TYPE TO IMPROVE SDA AND ASE

ITERATION 4 – ALL SPACES PASS CIBSE TM59

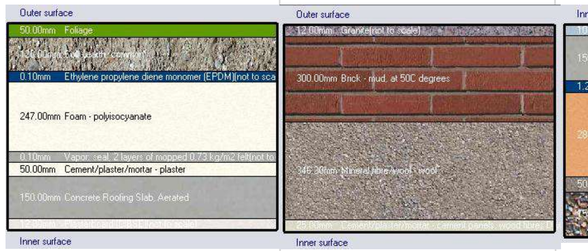
| Zone | Criterion A (%) | Criterion B (hr) | Pass/Fail |
|---|---|---|---|
| MANAGERSOFFICE | 0.00 | N/A | Pass |
| RECEPTION | 2.70 | N/A | Pass |
| SOCIALISINGAREA | 2.62 | N/A | Pass |
| READINGROOM | 0.00 | 0.00 | Pass |
| STUDIO1 | 0.00 | 3.00 | Pass |
| STUDIO14 | 0.00 | 0.00 | Pass |

ITERAION 4 PASES BOTH ASE AND SDA – IMPROVEMENT REQUIRED

| Zone | Floor area (m2) | sDA area in range (m2) | sDA % in Range% | ASE area in range (m2) | ASE % in range |
|---|---|---|---|---|---|
| READING ROOM | 16.6 | 9.95 | 59.88 | 16.62 | 100.0 |
| STUDIO 1 | 12.5 | 6.69 | 53.33 | 11.40 | 90.8 |
| STUDIO 14 | 11.0 | 5.54 | 50.48 | 10.34 | 94.3 |
| RECEPTION | 110.5 | 104.16 | 94.26 | 107.37 | 97.2 |
| MANAGERS OFFICE | 6.1 | 2.98 | 49.09 | 6.07 | 100.0 |
| SOCIALISING AREA | 93.3 | 87.86 | 94.14 | 48.01 | 51.4 |
| | **250.0** | **217.18** | **86.86** | **199.81** | **79.9** |

Source of all images and tables : DesignBuilder version 7 and EnergyPlus

## 7.1.5 ITERATION 5




**Green Roof (629mm)**

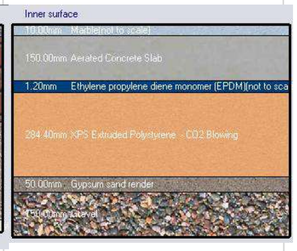
50mm vegetation
120mm soil
Waterproofing
247mm polysiocyanate insulation
Vapour seal
50mm screed
150mm concrete
12mm plasterboard
**U-value(0.1)**




**External wall (683mm)**

glass-fiber-reinforced concrete panels
300mm brick mud
346.3mm mineral fibre wool
12mm white granite

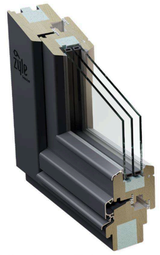
**U-value(0.1)**




**Ground floor slab (540mm)**

10mm marble
150mm aerated concrete
1.2mm damp proof
284.4mm extruded polystrene
50mm blinding
150mm hardcore

**U-value(0.149)**



**Window: Triple lowE clear 3mm/13mm Argon**
**Frame: Wood**

G-value: 0.47
VT: 0.66

**U-value(0.78)**

Table 7: WINDOW SIZES ( 1.5m SHADING)

| ZONE | WINDOW SIZE | NUMBER | LOCATION |
|---|---|---|---|
| READING ROOM | 1mx0.9 | 4 | North |
| STUDIO 1 | 1.13mx3.1m | 1 | South |
| STUDIO 14 | 1.13mx3.1m | 1 | South |
| RECEPTION | (S)1.95mx1.65m ; 1.9mx1.65m(E) | (S)8 ; (E)2 | South & East |
| MANAGER'S OFFICE | 2m x 0.9m | 1 | North |
| SOCIALISING AREA | 1.7mx1.65m | 9 | South |

RESULT: MANAGER'S OFFICE SDA AND SOCIALISING AREA ASE ACHIEVED
RESULT: CIBSE TM 59 VALUES IMPROVED, SDA AND ASE TARGETS ACHIEVED.

ITERATION 5 – ALL SPACES PASS CIBSE TM59

| Zone | Criterion A (%) | Criterion B (hr) | Pass/Fail |
|---|---|---|---|
| MANAGERSOFFICE | 0.00 | N/A | Pass |
| RECEPTION | 0.99 | N/A | Pass |
| SOCIALISINGAREA | 0.10 | N/A | Pass |
| READINGROOM | 0.00 | 0.00 | Pass |
| STUDIO1 | 0.00 | 0.00 | Pass |
| STUDIO14 | 0.00 | 0.00 | Pass |

ITERATION 5 ACHIEVES ASE AND SDA

| Zone | Floor area (m2) | sDA area in range (m2) | sDA % in Range% | ASE area in range (m2) | ASE % in range |
|---|---|---|---|---|---|
| READING ROOM | 14.0 | 7.36 | 52.52 | 14.02 | 100.0 |
| STUDIO 1 | 10.2 | 5.79 | 56.84 | 9.64 | 94.7 |
| STUDIO 14 | 8.7 | 4.40 | 50.62 | 8.49 | 97.5 |
| RECEPTION | 103.2 | 83.46 | 80.88 | 101.19 | 98.1 |
| MANAGERS OFFICE | 4.7 | 2.62 | 56.10 | 4.68 | 100.0 |
| SOCIALISING AREA | 85.6 | 58.38 | 68.22 | 80.94 | 94.6 |
| | **226.3** | **162.02** | **71.58** | **218.95** | **96.7** |

Source of all images and tables : DesignBuilder version 7 and EnergyPlus

## 7.1.6 SUMMARY OF ALL ITERATIONS

The base case passed CIBSE with extremely low window-to-wall ratio but gave poor daylighting results.

The following changes were made to finally achieve all requirements for CIBSE, SDA and ASE.

- **Increasing glazing areas** to maximize passive solar heat gain in cold months
- **Thermal mass** (natural stone flooring, concrete and brick) to absorb, store and release the heat
- **Increasing insulation** with extruded polystyrene(XPS) and making the envelope more airtight to prevent infiltration
- **Windows**: Some of the above-mentioned steps affected daylight targets. Solution was changing the windows' glass type from double 3mm clear glass to triple low-E clear 3mm with13mm Argon, achieving good U-value whiles attaining SDA and ASE daylighting targets.

Combining the above strategies also improved the operative temperatures comparing outside dry bulb temperature, operative temperature of the base case till iteration 5.

## 8.0 TEMPERATURE RANGE COMPARISON OF ITERATIONS

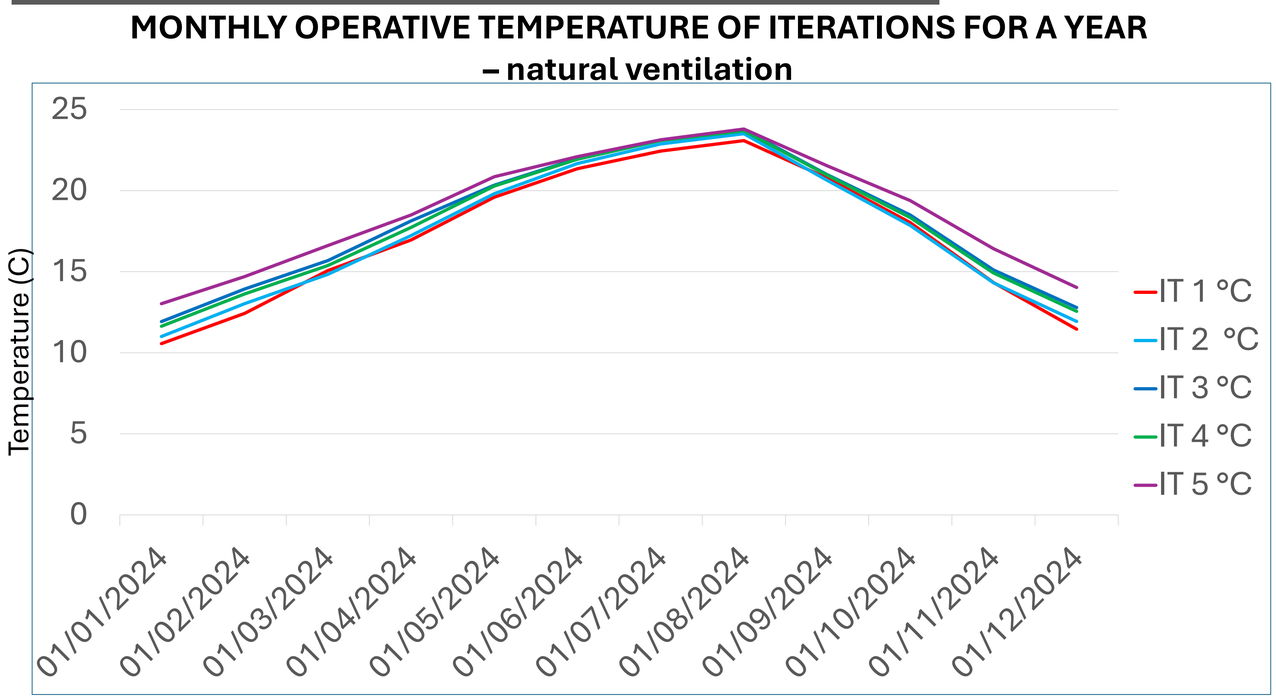



Figure 92: monthly operative temperature of all iterations for one year

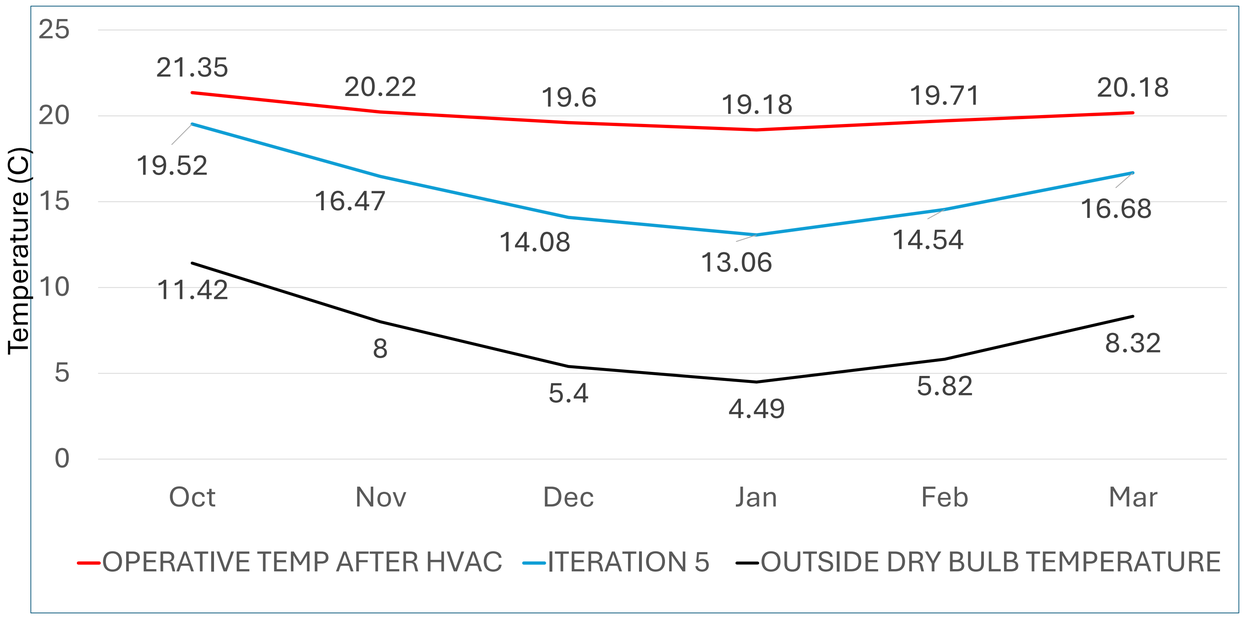



Figure 94: monthly operative temperatures from October to March

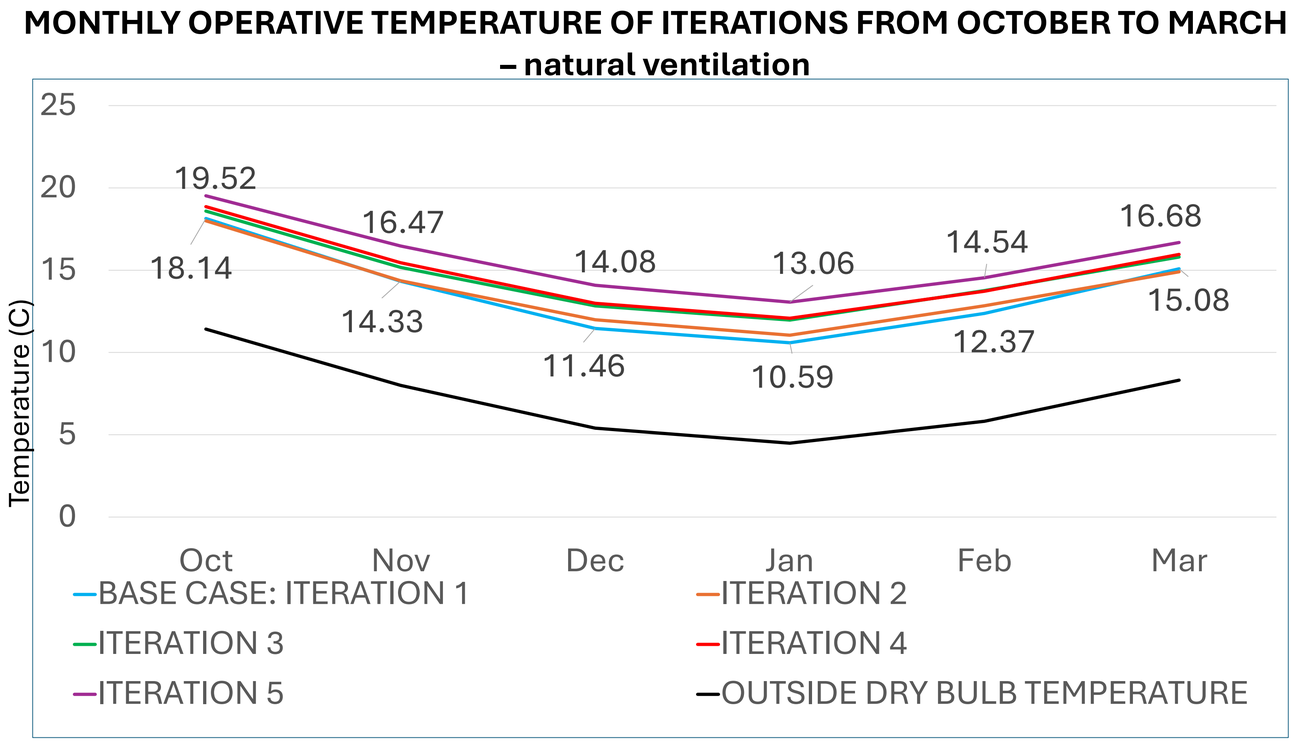


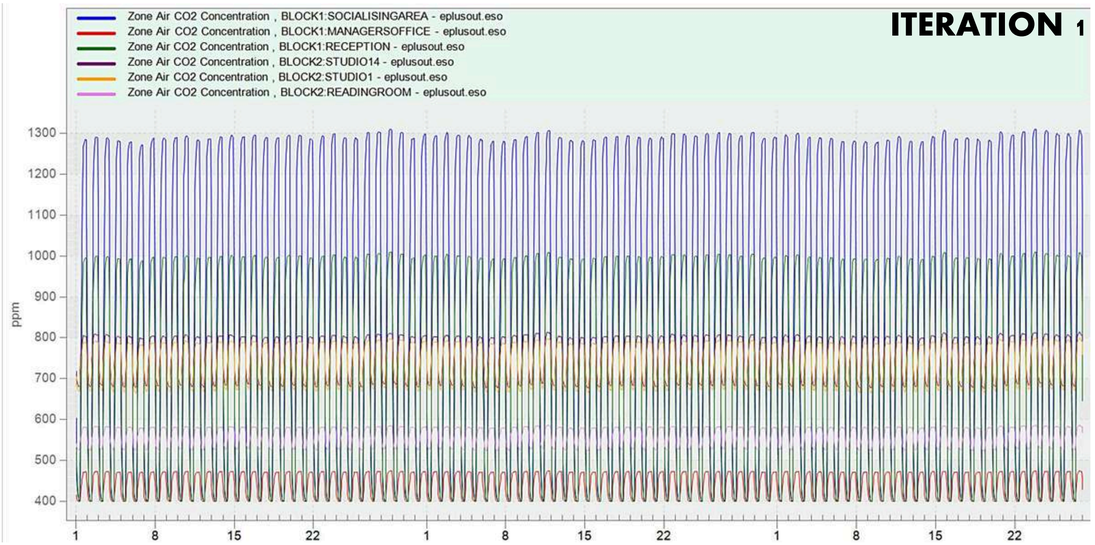

Figure 93: monthly operative temperatures from October to March

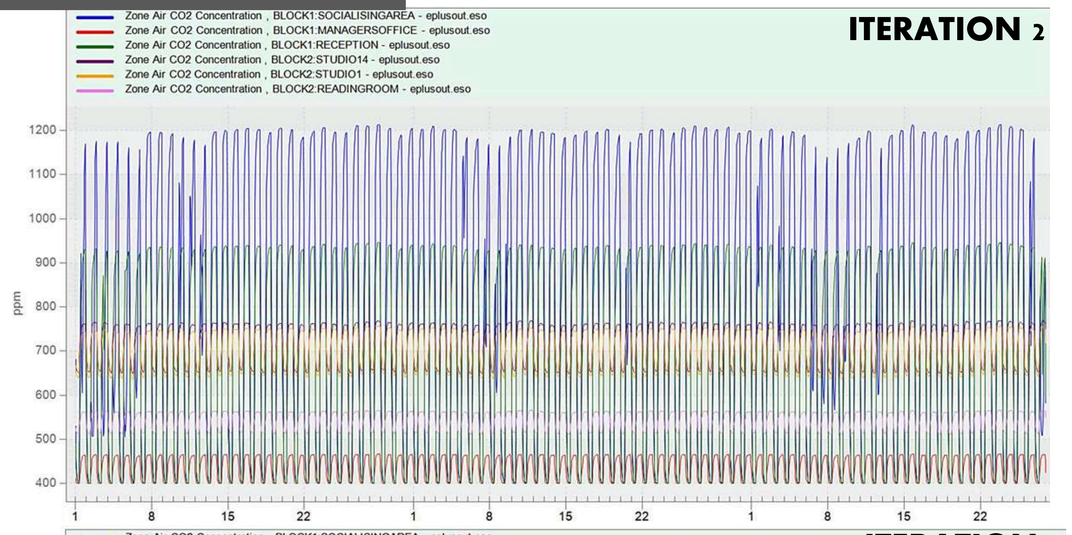
**The operative temperatures of the project increased from iteration 1 (Base Case) to 5 due to passive strategies adopted. Comfort levels are achieved between the months of April and October where operative temperatures are between 18 and 25 degrees. Natural ventilation will be employed during these months to achieve indoor thermal comfort for occupants as seen in Figure 92.**

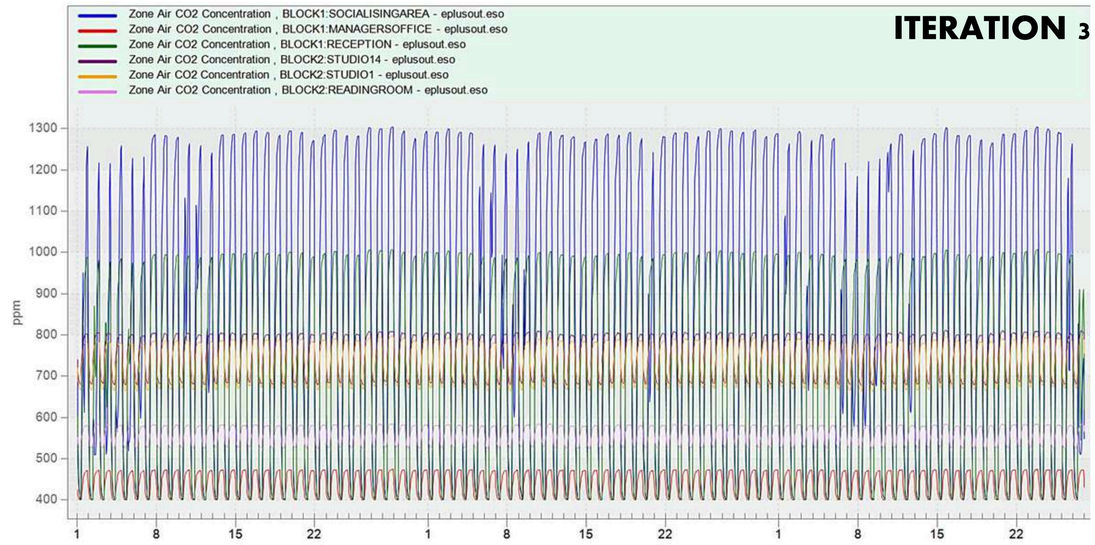
**Figure 93 shows improvements in operative temperatures compared to the base case. The coldest month in January improved from 10.59 degrees(base case) to 13.06 degrees (Iteration 5) which shows a significant increase in operative temperature of 2.47 degrees, to increase the indoor comfort temperature.**

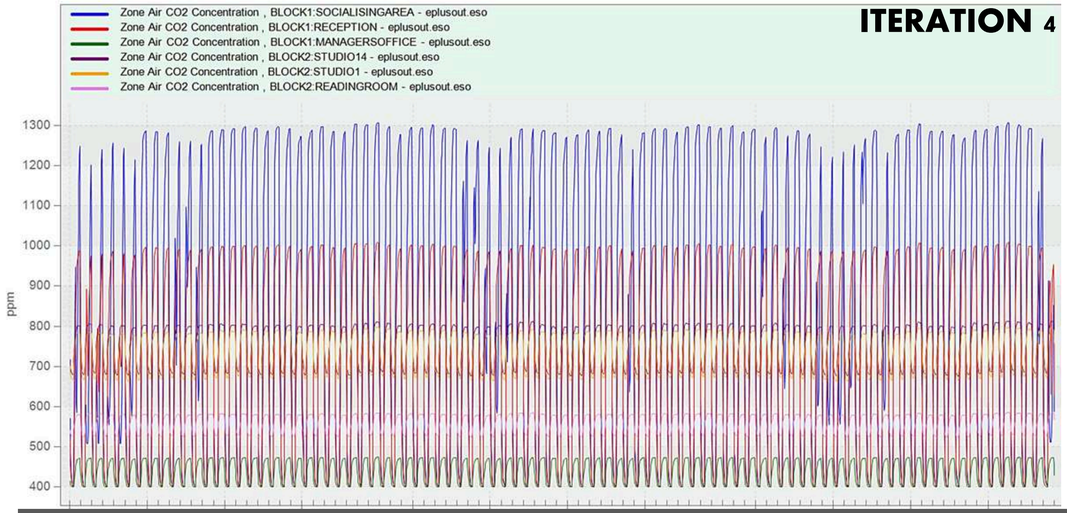
**Figure 94 shows the difference between the outdoor dry bulb temperature, the improvements made for Iteration 5 and the operative temperature drastically improved after a mechanical system was introduced to keep comfort levels within 21 to 24 degrees as the set point temperatures.**

## 8.1 HOURLY $C_{02}$ LEVELS OF ZONES FROM DECEMBER TO FEBRUARY - NATURAL VENTILATION MODE



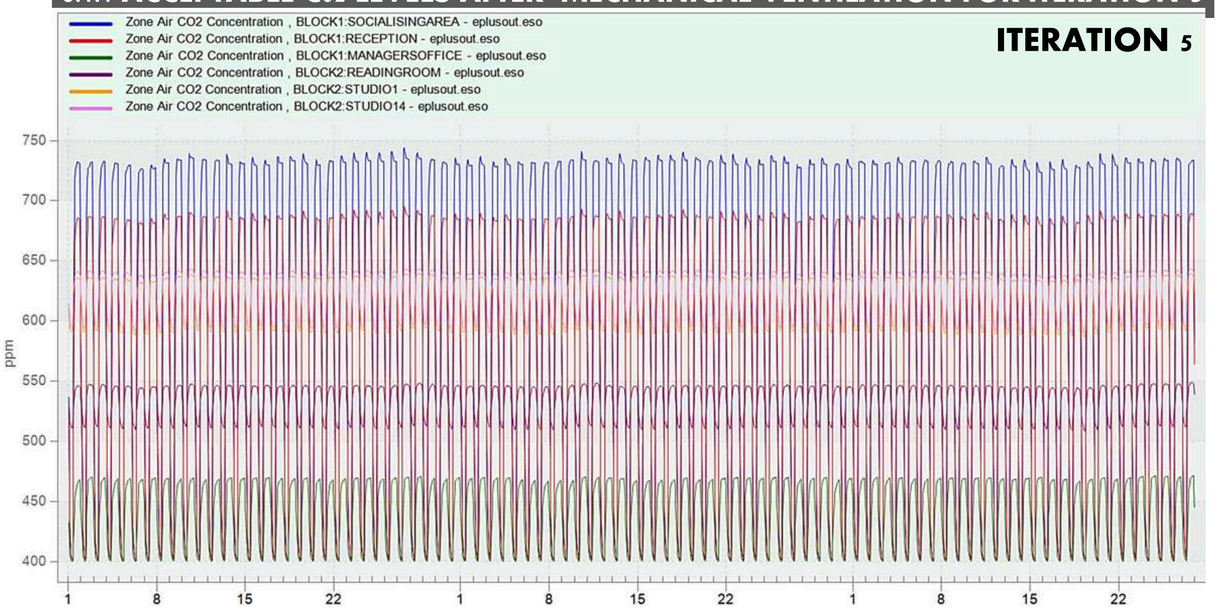







### 8.1.1 ACCEPTABLE $C_{02}$ LEVELS AFTER MECHANICAL VENTILATION FOR ITERATION 5



## 8.2 PERFORMANCE TARGETS VERSUS ACTUAL OUTCOMES

### TOTAL ANNUAL COMFORT HOURS VERSUS DISCOMFORT HOURS FOR BUILDING

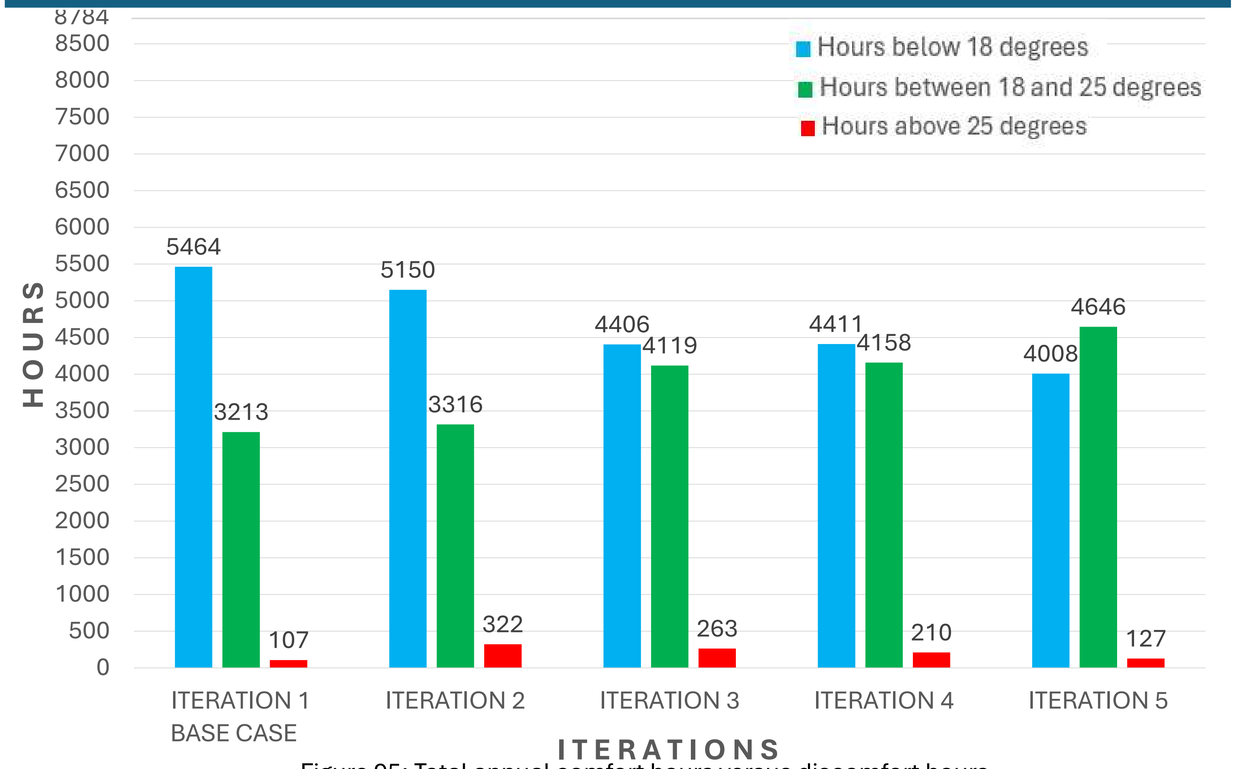



Figure 95: Total annual comfort hours versus discomfort hours

### SDA, ASE AND CIBSE TM 59 TARGETS FOR 6 ZONES

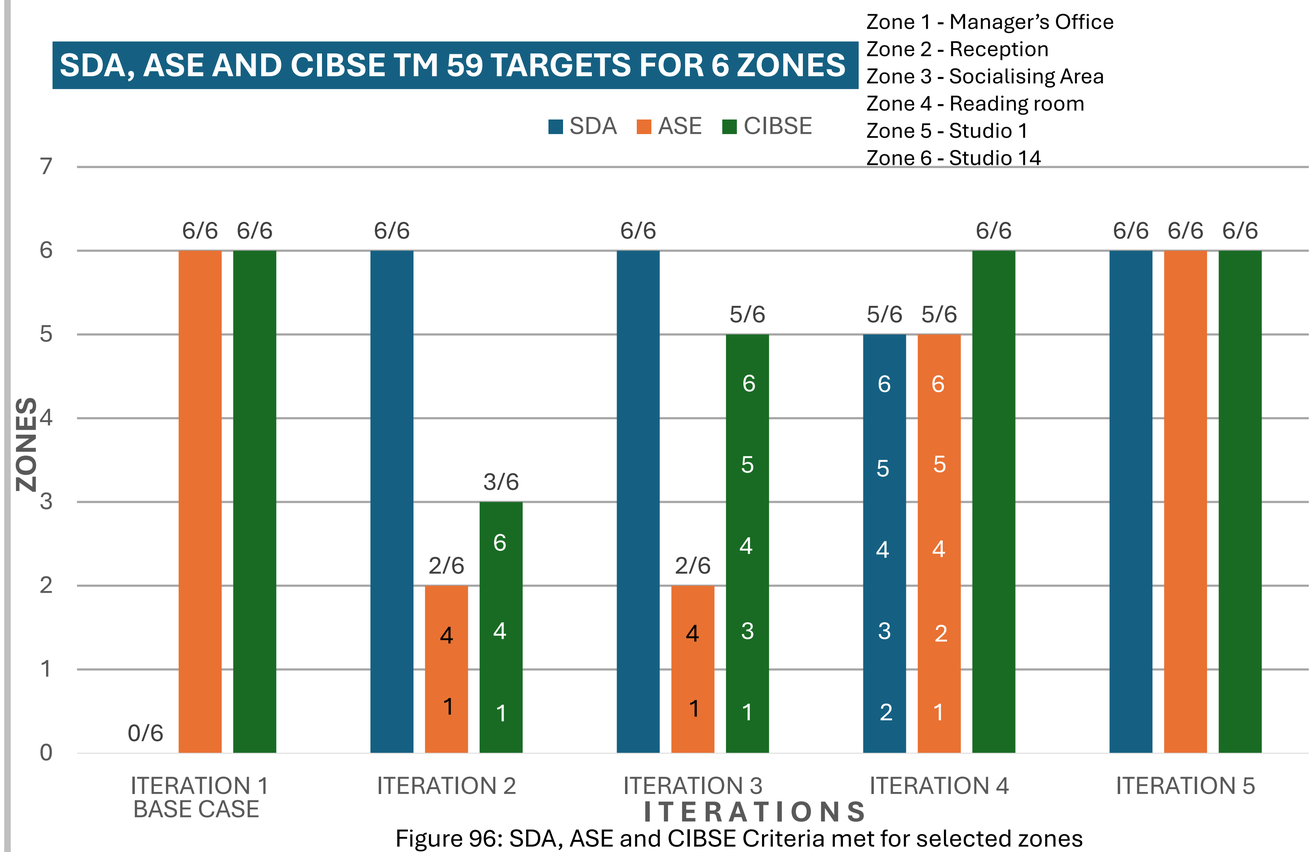



Figure 96: SDA, ASE and CIBSE Criteria met for selected zones

**Table 3: Natural ventilation comfort range based on CIBSE GUIDE A.**

| | Temperature Range | Implication |
|---|---|---|
| **Below Comfort** | T<18 degrees | Too cold. Requires heating to achieve comfort |
| **Within Comfort** | 18 degrees ≥ T ≤ 25 degrees | Ideal range. Natural Ventilation acceptable |
| **Above Comfort** | T > 25 degrees | Too warm. Requires cooling to maintain comfort |

### Conclusion of design optimization from Base Case to Iteration 5

1. **Hours below 18°C decreased** by **26.7%** (from 5,464 hours to 4,008 hours).
2. **Hours between 18–25°C increased** by **44.6%** (from 3,213 hours to 4,646 hours).
3. **Hours above 25°C decreased** by **8.5%** (from 107 hours to 127 hours).

Iteration 5 no more has summer overheating because it passed the CIBSE criteria for overheating. The temperature above 25 degrees is 1.4% of the total hours, which is below the 3% stated by CIBSE TM 59, therefore it is acceptable.
52.8% of total hours in the year is within the comfort range and 45.6% below comfort.
**The design requires a heating system to increase the temperature below comfort to the acceptable comfortable range.**

### INDOOR CO2 LEVELS BELOW 1000ppm FOR 6 ZONES

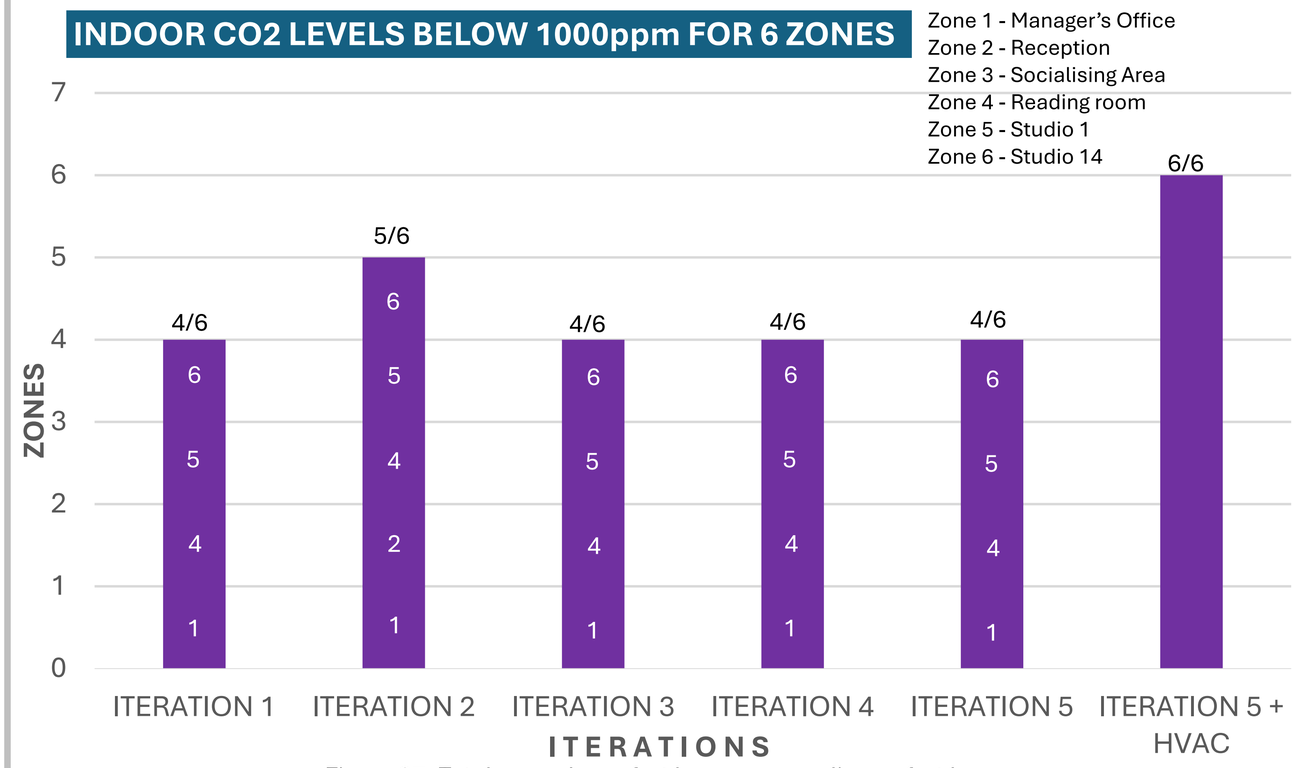



Figure 97: Total annual comfort hours versus discomfort hours

## 8.3 MECHANICAL SYSTEMS

| | **Total energy of building (kWh) if HVAC system heating only** | Energy per total building area (kWh/m2) | **Total energy of building (kWh) if HVAC system heating and cooling** | Energy per total building area (kWh/m2) |
|---|---|---|---|---|
| **1. Variable Refrigerant Flow (VRF)** (air cooled) with heat recovery and Dedicated Outside Air System | 69096.29 | 56.41 | 91550.22 | 74.74 |
| **2. Ground Source Heat Pump (GSHP):** water to water heat pump with under floor heating | 56591.96 | 46.20 | 69610.89 | 56.83 |

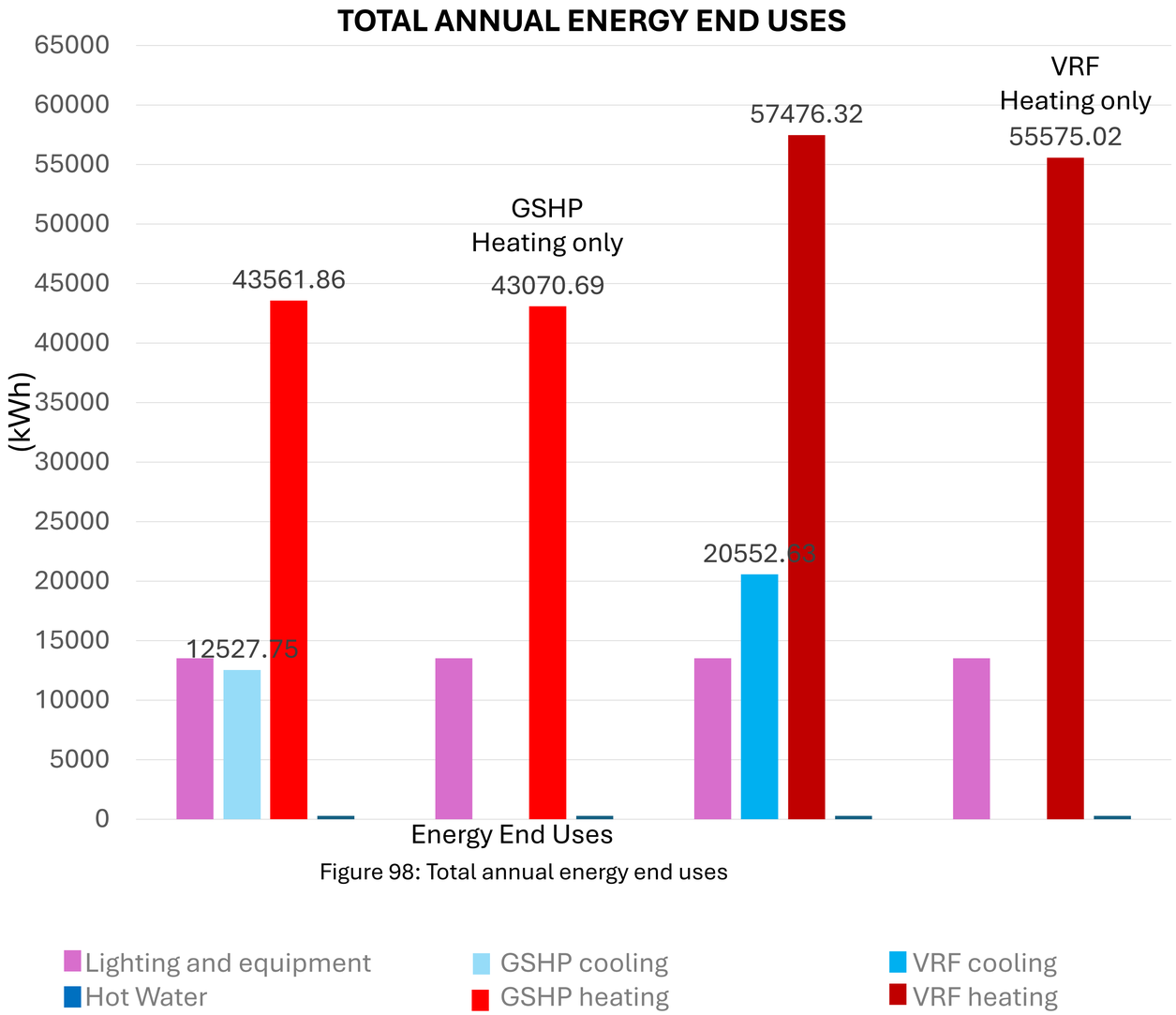



Figure 98: Total annual energy end uses

According to figure 98, In heating only mode, the GSHP uses 22.5% less energy compared to the VRF (ie. 43,070.69kWh versus 55,575.02kWh).

When heating and cooling is combined, the GSHP energy use is 24.2% less than the VRF.

In cooling mode, the GSHP outperforms the VRF because it consumes 64% less energy than the VRF (ie. 12,527.75kWh versus 20,552.63 kWh).

**SUMMARY: The GSHP presents better efficiency and more energy savings and is the recommended mechanical system.**

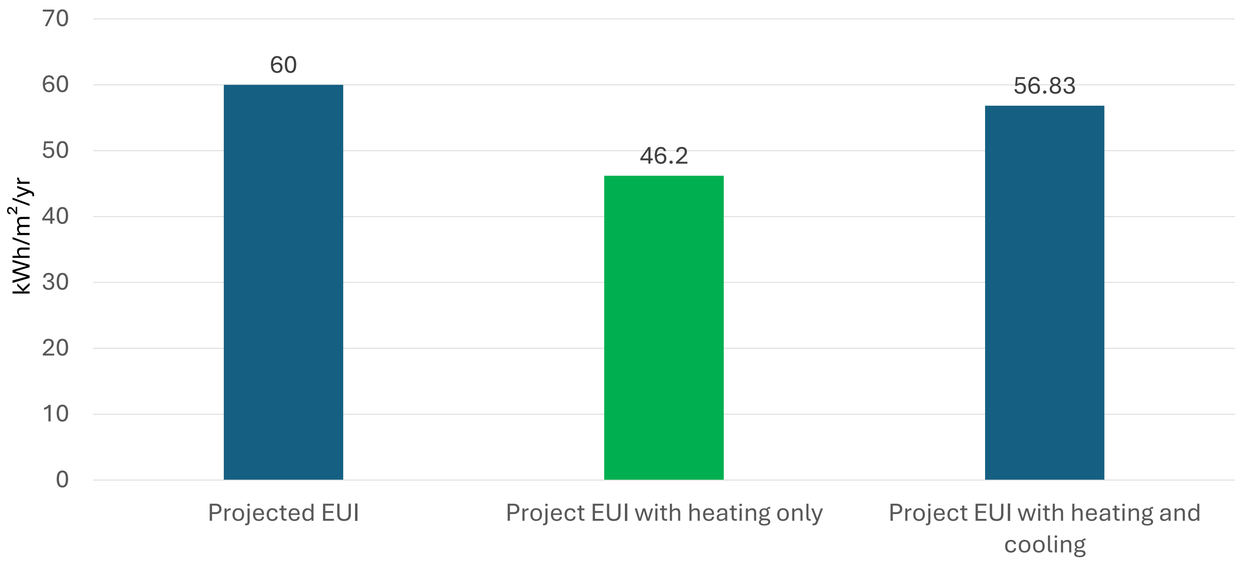



Figure 99: Projected EUI versus actual for project

## 8.4 RENEWABLE ENERGY

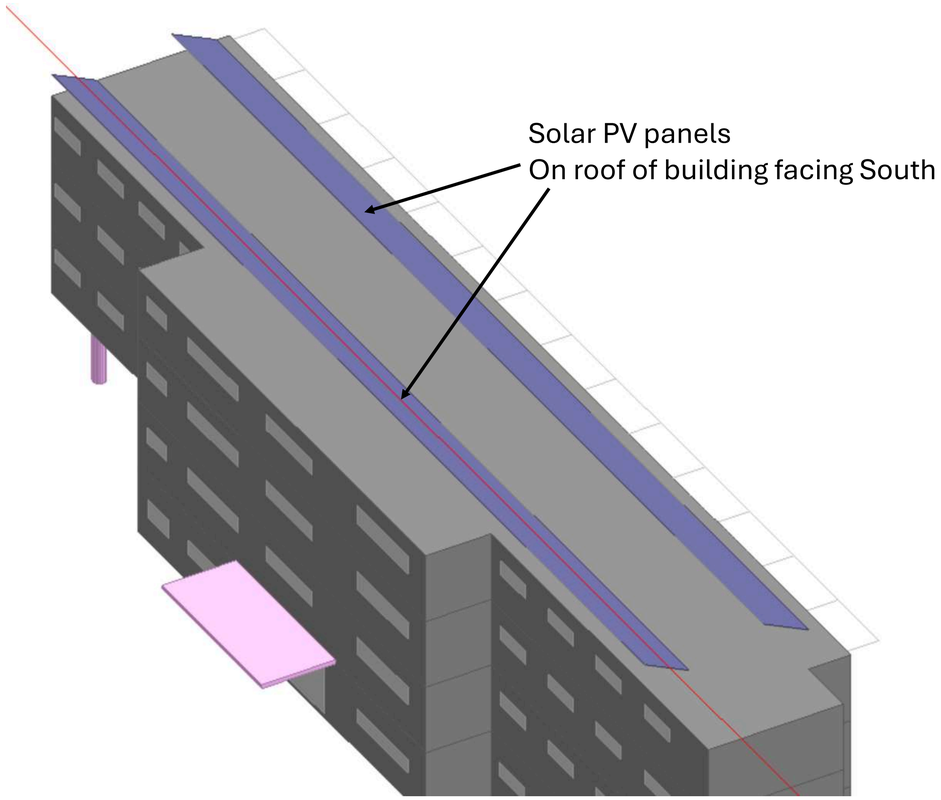



Figure 100: Solar PV on the roof

On-site renewable energy reduces the building's total reliance on the grid.

Examples common in Seattle are; Solar photovoltaics (PV), Solar Collectors for hot water, Building Integrated Photovoltaics(BIPV) and ground source heat pump.

Solar panels were placed on the roof to generate electricity for the project.

A total area of 226.45m2 of solar PV panels will generate 43096.89kWh of electricity for the building.

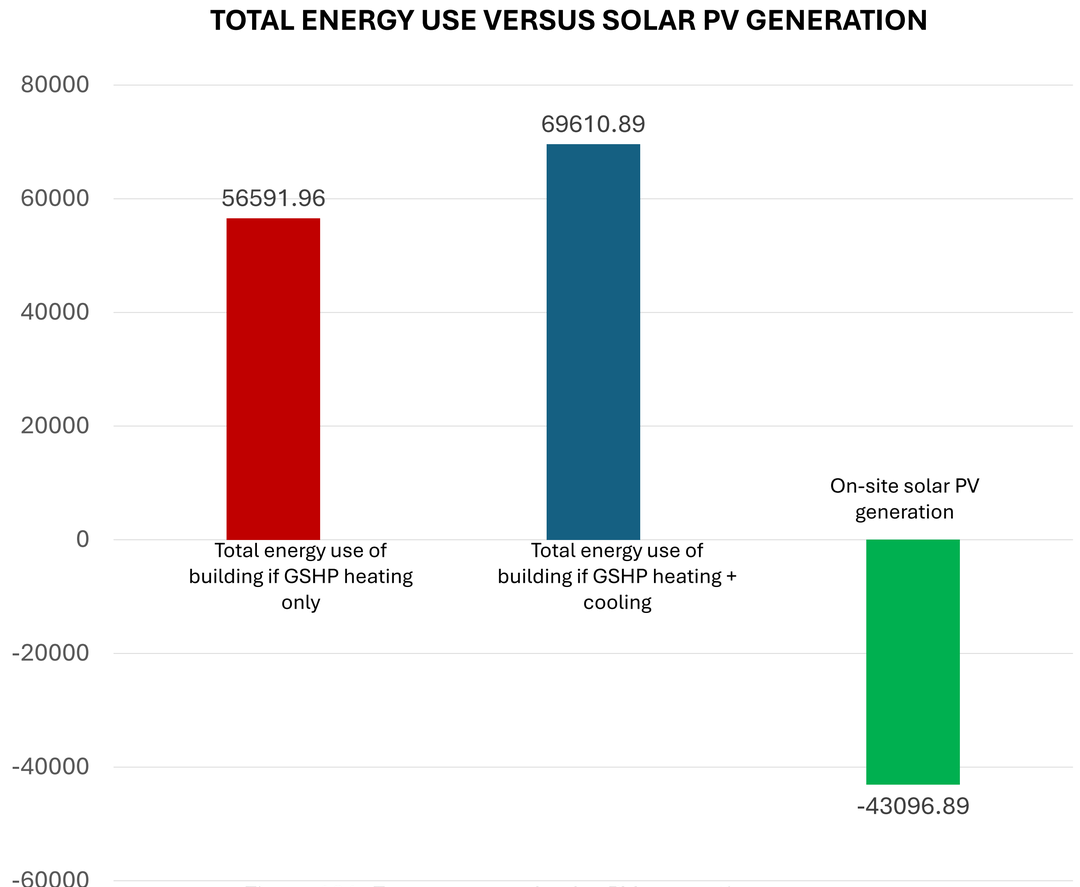



Figure 101: Energy use and solar PV generation

**Conclusion**: The solar PV system can offset 76.2% of the total energy use of the building if the GSHP provides heating only.

With both heating and cooling, the system will cover 61.9% of the total energy use of the building.

**Opting for heating only during the winter and natural ventilation during summer combined with the solar PV will help the project become energy efficient and reduce its reliance on the grid.**

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**END OF PRESENTATION**